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Machine learning in laser welding of electric vehicle batteries:

Process control and quality assurance

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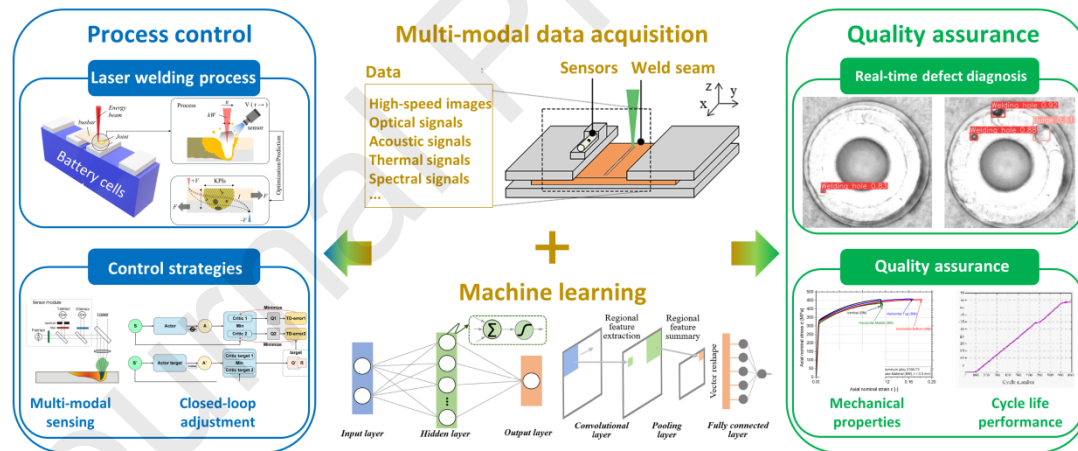
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Graphical abstract



Highlights

- The systematic review of machine learning in battery laser welding was presented.
- The effect of intelligent parameter optimization on process stability was discussed.

- The influence of multi-sensor fusion on real-time defect diagnosis was investigated.
- The welding quality consistency is enhanced by adaptive closed-loop control.

Abstract: Laser welding serves as a pivotal technology for ensuring the structural integrity and electrical connectivity of electric vehicle (EV) battery packs, which directly influences product safety and mass production consistency. However, the inherent non-linearity of process dynamics, compounded by the limitations of traditional open-loop control and lagging offline inspection, presents significant challenges to achieving zero defect manufacturing. Emerging machine learning technologies offer a transformative data-driven paradigm to address these bottlenecks by enabling real-time monitoring and intelligent decision-making. This paper presents a systematic review of machine learning applications in battery laser welding and categorizes them into two core domains, including intelligent process control and smart quality assurance. For process control, the review discusses the transition from passive monitoring to active adaptation through multimodal sensing, parameter optimization and reinforcement learning-based closed-loop adjustment. Regarding quality assurance, the paper analyzes strategies for real-time defect diagnosis and the quantitative prediction of key mechanical and electrical properties, and explores the prospective correlation between welding process signatures and battery lifecycle performance. Furthermore, the review critically examines current challenges, including industrial data scarcity, model generalization and system integration. Finally, it highlights future directions such as physics-informed neural networks and digital twin technologies to realize the vision of intelligent manufacturing of EV batteries.

Keywords: Machine learning; Laser welding; Electric vehicle battery; Process control; Quality assurance; Intelligent manufacturing

1. Introduction

1.1. Role of power batteries in electric vehicles

Driven by global strategies to achieve carbon neutrality and mitigate the severe challenges of climate change, the electrification of the transportation sector has emerged as an irreversible trend and a global consensus [1,2]. As the primary driver of the transition, the electric vehicle (EV) market is witnessing an unprecedented surge in penetration. Recent statistics indicate that global EV sales have exhibited an exponential growth trajectory over the past few years [3]. The rapid expansion imposes substantial market demands and rigorous technical challenges on power batteries, particularly lithium-ion batteries which serve as the core power source for EVs.

Power batteries represent not only the single most expensive component in an EV but also the most technology-intensive subsystem [4,5]. Their energy density, power density, and cycle life directly dictate the vehicle's driving range, acceleration capabilities, and service lifespan [6,7], while their safety remains the paramount concern for consumers [8]. Furthermore, the manufacturing cost and production efficiency of power batteries are critical determinants of whether EVs can economically displace traditional internal combustion engine vehicles (ICEVs). Consequently, the development of advanced battery manufacturing technologies aimed at achieving high-quality, highly consistent, efficient, and low-cost mass production has emerged as a focal point of competition within the global new energy industry and a prominent research hotspot in academia [9].

1.2. Laser welding applications in battery manufacturing

The manufacture of EV batteries is a complex, interdisciplinary endeavor spanning electrochemistry, materials science, and precision engineering [10]. Within the intricate manufacturing chain, joining technology plays a pivotal role. Achieving reliable electrical and mechanical connections among dissimilar-metal components at the cell, module, and pack levels is the foundation for constructing high-efficiency battery systems [11]. Among the myriad of joining technologies, although ultrasonic welding and resistance spot welding were extensively adopted in the early stages of battery manufacturing [12], they exhibit inherent limitations, such as insufficient capability for thick-plate joining, severe tool head wear, and difficulties in single-

sided access welding. In contrast, laser welding has gradually supplanted these traditional joining methods and established an indispensable core position in modern EV battery manufacturing [13]. The dominance is attributed to its significant advantages, including high energy density, superior processing precision, rapid welding speeds, a narrow heat affected zone (HAZ), and high adaptability for automated integration.

The application of laser welding technology deeply permeates the entire production chain, spanning from microscopic battery cells to macroscopic battery packs. The specific applications of laser welding in EV batteries are illustrated in **Fig. 1**. Specifically, in cell-level joining, the technology is employed for the precision welding of positive and negative tabs to caps or terminals, a process commonly known as tab-to-cap or tab-to-terminal welding, establishing a critical channel for the output of electrochemical energy to external circuits [14]. In the module-level joining stage, it is primarily responsible for the welding of series-parallel busbars between cells and the structural fixation of side plates, directly dictating the internal resistance consistency and structural strength of the module [15]. Extending further to pack-level joining, laser welding undertakes the tasks of electrical interconnection between modules and the large-scale sealing of the battery enclosure, aiming to ensure the system's electrical integrity and Ingress Protection rating in harsh environments [16]. Given these critical roles, the quality of laser welding joints fundamentally determines the conductive performance, mechanical reliability, and hermetic sealing of EV batteries, thereby influencing their ultimate service life and the overall safety of the vehicle.

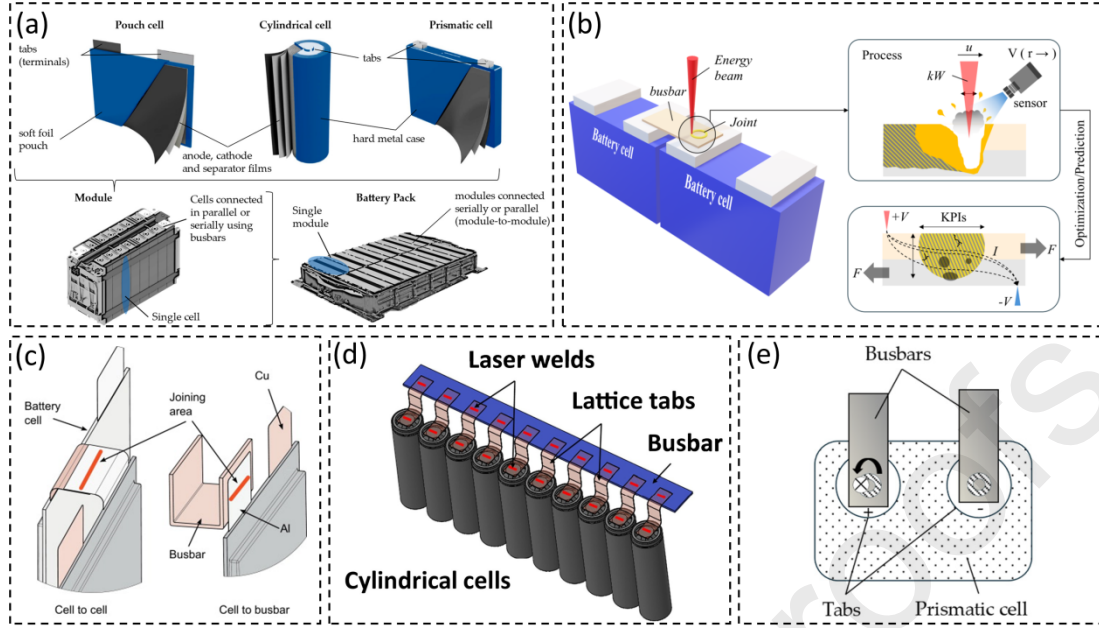


Fig. 1. Applications of laser welding in EV battery manufacturing: (a) schematic of an EV battery pack [17]; (b) schematic illustration of the laser welding process [17]; (c) cell-to-cell and cell-to-bus bar interconnections [18]; (d) tab-to-busbar joints made during cylindrical cell-based module manufacture [19]; (e) laser welding of busbars onto prismatic cell tabs [17].

1.3. Challenges in process control and quality assurance

Notwithstanding the significant advantages of laser welding in terms of efficiency and precision, achieving sustainable and stable high-quality welding in the mass production of EV batteries remains a formidable challenge. These challenges are primarily manifested in two dimensions, namely the stochastic nature of process control and the reactive nature of quality assurance.

In terms of process control, although the EV battery manufacturing system involves various metallic materials, including stainless steel for cylindrical casings and titanium alloys for specialized components, Cu-Al dissimilar metal welding has become a critical process for enabling series-parallel cell connections [20]. This is necessitated by the fundamental material system where aluminum and copper serve as the positive and negative current collectors, respectively. In the scenario, laser welding is intrinsically an extremely complex process characterized by transient, high-temperature, and strongly coupled multi-physics phenomena. It involves intricate mechanisms including laser-matter interaction, the dynamic evolution of the keyhole, the turbulent flow of the molten pool, and complex metallurgical reactions [21–23]. These physical processes exhibit hypersensitivity to variations in process parameters, such as laser power, welding speed, and defocusing distance, as well as minor

environmental disturbances [24]. Traditional open-loop control strategies, which rely on predetermined fixed process windows, fail to perceive and compensate for these inevitable process disturbances in real-time. Consequently, it leads to significant stochasticity and inconsistency in welding quality, making it difficult to meet the stringent requirements for high-yield production [25].

On the other hand, the quality assurance system also faces numerous bottlenecks. Since battery systems store high energy, even minor joint failures can trigger severe thermal runaway accidents. Therefore, the EV battery industry imposes extremely rigorous quality control requirements regarding welding defects, including spatter, porosity, cracking, spatters, etc. However, existing industrial-grade quality assurance systems remain heavily reliant on post-process inspection. The primary methods include metallographic cross-section analysis [26], destructive tensile testing [27], as well as off-line X-ray [28] and computed tomography (CT) inspection [29]. The intrinsic limitations of these traditional methods lie in their offline, destructive, and time-consuming nature. Constrained by cost and production efficiency, they are often restricted to small-ratio sampling inspection. The fundamental drawback of the mode is its inability to provide real-time quality feedback for process correction or to achieve 100% in-line full inspection for every product, thereby leaving non-negligible safety hazards for the long-term reliability of EV batteries [17].

1.4. Industrial state-of-practice in laser welding quality assurance

Beyond traditional post-weld inspection methods, numerous advanced commercial monitoring systems have been widely deployed across industrial laser welding applications [30]. Representative solutions include inline monitoring platforms provided by companies such as Precitec and IPG Photonics [31,32]. These systems integrate photodiodes, cameras, and proprietary signal processing algorithms to facilitate real-time process observation. For instance, technologies including penetration depth detection, which is primarily based on optical coherence tomography [33], and process monitoring modules embedded within fiber laser sources can closely track keyhole behavior, optical emissions, and reflected signals during the welding operation.

Despite their industrial maturity and robustness, these commercial configurations face specific challenges in fully addressing the complex demands of next-generation intelligent manufacturing. First, most existing solutions rely on predefined threshold-

based decision rules or simplified physical indicators calibrated under specific process conditions [34]. Consequently, adapting these rules to highly dynamic and nonlinear welding scenarios, particularly Cu-Al dissimilar metal joining characterized by strong reflectivity fluctuations, remains a considerable challenge [35]. Second, these systems are predominantly designed for passive monitoring and alarm triggering, leaving significant room for the development of autonomous, real-time closed-loop control to adjust process parameters in response to detected instabilities [36]. Finally, the proprietary nature of commercial systems frequently restricts user access to raw data and underlying model architectures, which can complicate custom algorithmic optimization and cross-scenario generalization by third-party developers.

1.5. Machine learning-driven intelligent welding technologies

Importantly, machine learning in laser welding should not be interpreted as a purely black-box tool for defect classification or data fitting. The monitored optical, acoustic, thermal, and visual signals are physical manifestations of laser-material interaction, and their temporal and spatial fluctuations reflect changes in keyhole geometry, vapor plume dynamics, melt-pool convection, and solidification behavior. For example, in vision-based monitoring, convolutional neural networks (CNNs) do not merely see images as abstract pixel arrays; rather, they can hierarchically extract physically relevant visual patterns, such as molten-pool contours, keyhole opening variations, plume morphology, brightness distribution, and spatter trajectories, which are closely associated with penetration evolution and process stability [37]. Therefore, the real value of machine learning lies not merely in mapping sensor inputs to output labels, but in learning physically meaningful correlations between multimodal process signatures and the underlying thermo-fluid-metallurgical phenomena that govern defect formation and joint performance.

Against this background, both academia and industry are actively investigating the deep integration of machine learning with laser welding processes to overcome the aforementioned physical complexities in process control and efficiency bottlenecks in quality assurance [38]. The integration has witnessed a significant evolutionary trajectory, transitioning from shallow feature mapping to deep perceptual decision-making.

In the initial application phase, research efforts were primarily concentrated on employing traditional algorithms such as support vector machines (SVM) and random

forests (RF). By utilizing hand-crafted feature engineering, these studies mapped single physical quantities, including molten pool width and optical radiation intensity, onto welding quality labels, thereby achieving the classification of basic defects like porosity and insufficient penetration in specific scenarios [39]. However, when confronted with the massive volume of unstructured data typical of EV battery manufacturing, these traditional methods exhibited inherent limitations characterized by time-consuming feature extraction processes and weak generalization capabilities.

Coinciding with the proliferation of deep learning, CNNs have been widely adopted to facilitate end-to-end feature extraction, directly mapping molten pool imagery to defect categories. The data-driven approach has significantly bolstered identification accuracy for intricate defects, particularly spatter and micro-cracks [40,41]. Furthermore, to address the inherent high-dynamic and time-varying characteristics of the laser welding process, long short-term memory (LSTM) networks and transformer architectures have been employed to process time-series signals derived from acoustic emission and thermal imagery. These models have successfully captured the long-range temporal dependencies between molten pool fluctuations and keyhole collapse [42].

Currently, frontier exploration is advancing towards deep reinforcement learning (DRL), which attempts to endow welding systems with self-learning capabilities akin to those of human experts. The approach enables the dynamic adjustment of laser power and focal length within milliseconds based on sensor feedback, thereby realizing true closed-loop adaptive control in environments characterized by uncertain disturbances [43]. Consequently, the paradigm offers a novel and revolutionary pathway for unraveling the complex physical processes of laser welding and driving the technological transformation of EV battery manufacturing from automation to intelligence.

Realizing the industrial potential of these advanced algorithms necessitates the integration of multi-modal sensors including high-frequency photodiodes, high-speed infrared cameras, acoustic emission sensors, and spectrometers into welding optical heads or fixtures for real-time process monitoring. By utilizing machine learning algorithms to instantaneously analyze massive process data, intelligent welding systems aim to achieve two core objectives. The first is to transition process control from passive, fixed-parameter open-loop strategies to active, closed-loop adaptive control based on real-time feedback, thereby significantly enhancing process

robustness and stability amidst external disturbances [44]. The second is to establish an intelligent quality assurance system that bridges the gap from lagging offline sampling to comprehensive real-time online inspection, utilizing process data signatures to perform predictive assessment and precise quality control for every individual weld [45,46].

Although a number of review studies have addressed laser welding technologies or data-driven manufacturing, their focus is typically limited to either general welding processes [47], specific sensing techniques [48], or isolated machine learning applications [38]. Therefore, systematic investigations of machine learning specifically tailored to EV battery laser welding remain limited. Existing reviews rarely provide a unified perspective on this domain, particularly considering the unique challenges of battery manufacturing, where stringent requirements on electrical conductivity, mechanical reliability, and safety impose significantly higher demands on process control and quality assurance.

Consequently, this paper presents a systematic review of the latest research progress, critical challenges, and future directions of machine learning in advancing the intelligence of laser welding for electric vehicle batteries. Aiming to provide a holistic view of the interdisciplinary field, the remainder of the paper is structured around two central pillars including intelligent process control and smart quality assurance. Section 2 establishes the theoretical framework by outlining key process principles and common defects while analyzing the limitations of traditional methods in mass production. Building upon the foundation, Sections 3 and 4 provide a comprehensive analysis of the two core themes. Specifically, Section 3 covers intelligent sensing, parameter optimization, and adaptive control strategies, whereas Section 4 explores defect diagnosis via multi-sensor fusion, online performance prediction, and the correlation between process signatures and battery lifecycle. Subsequently, Section 5 critically addresses current implementation hurdles, including data scarcity, model generalization, and computational constraints, while highlighting emerging technological paradigms such as digital twins and physics-informed machine learning. Finally, Section 6 offers concluding remarks and forward-looking insights to guide scholars and engineers, thereby facilitating the industrial deployment of intelligent welding technologies in the EV battery sector.

2. Laser welding process, defects, and traditional control methods

2.1. Laser welding process of EV batteries

The core of internal electrical interconnections in EV batteries lies in the realization of welding between dissimilar metals, specifically copper (Cu) and aluminum (Al). It represents one of the most technically challenging links in the entire battery manufacturing chain. The difficulty in dissimilar metal welding stems from the vast disparity in the thermophysical properties of the two materials. Cu possesses an extremely high melting point and heat capacity; in contrast, aluminum has a relatively low melting point. The discrepancy often leads to a scenario during welding where aluminum is already over-burned while Cu remains insufficiently melted. Furthermore, differences in thermal conductivity and thermal expansion coefficients between Cu and Al can induce cracks and voids within the weld seam [49].

Additionally, Cu exhibits a reflectivity of over 95% to conventional industrial lasers operating in the near-infrared spectrum, resulting in extremely low absorptivity [50]. Consequently, extremely high power density is required to initiate the melting process. Insufficient power density leads to a lack of penetration defects, whereas excessive power density may cause excessive penetration defects. Therefore, the operating window for laser welding is extremely narrow.

Moreover, as shown in **Fig. 2** (a), metallurgically, copper and aluminum are typical limited solid solutions with poor metallurgical affinity [51]. At high welding temperatures, a series of hard and brittle intermetallic compounds (IMCs) are formed within the weld as shown in **Fig. 2** (b), such as Al_2Cu , AlCu , Al_3Cu_4 , and Al_4Cu_9 as shown in **Fig. 2** (c). These IMCs possess significantly higher electrical resistivity and hardness compared to the base materials. An excessively thick IMC layer will drastically degrade joint performance. Solchenbach et al. [18] investigated the electrical performance of laser-welded Al-Cu joints. Their results indicated that when the IMC thickness exceeds 3–5 μm , it has a significant impact on contact resistance. Furthermore, mechanical strength was found to be more sensitive to IMC thickness than contact resistance. The relationship between contact resistance and IMC thickness is illustrated in **Fig. 2** (d). Although an extremely thin IMC layer can enhance the mechanical properties of the weld, once its thickness becomes uncontrolled, these compounds will severely impair weld performance. It leads to

increased resistivity and degraded mechanical properties, potentially causing brittle fracture of the joint. Elevated resistivity exacerbates heat generation during battery charging and discharging cycles, thereby accelerating the battery degradation process [52,53]. Therefore, the precise regulation of IMC formation and thickness remains a critical metallurgical issue that must be strictly controlled in welding quality assurance.

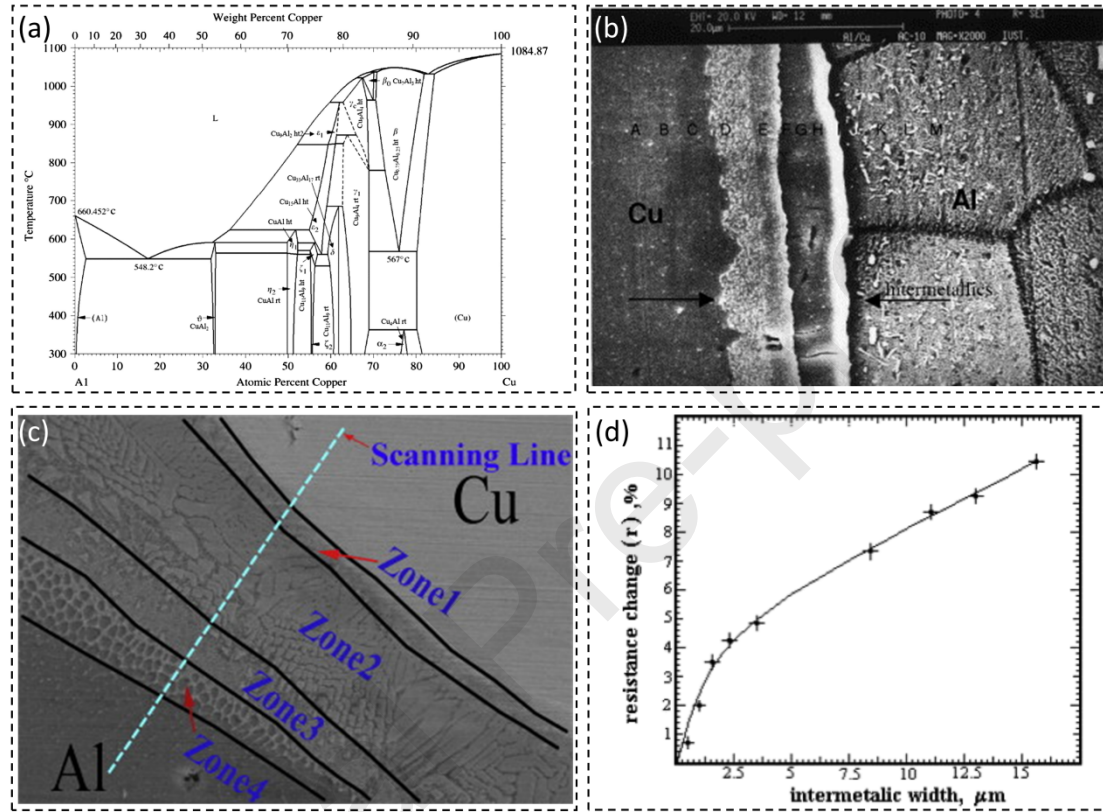


Fig. 2. Metallurgical challenges in Cu-Al dissimilar laser welding: (a) equilibrium phase diagram of the Al-Cu system [54]; (b) Al-line scan across the intermetallic width and layers [55]; (c) morphology of four zones in the intermediate layer [23]; (d) variation of resistance change with intermetallic width [55].

Addressing critical challenges in the welding of dissimilar metals such as copper and aluminum, specifically low energy absorption rates, melt pool instability, and the difficulty in controlling brittle IMCs, academia and industry have proposed a series of innovative solutions from multiple dimensions, including process optimization, beam shaping, and the development of novel light sources. These technical pathways aim to suppress defects such as porosity, cracks, and spatter by improving laser-matter interaction mechanisms, regulating the thermal fluid dynamics of the molten pool, and optimizing metallurgical reaction processes, ultimately achieving high-quality and highly reliable precision joining.

Near-infrared fiber lasers represent the most widely used laser sources in the industry, characterized by their ability to deliver high power output. However, as illustrated in **Fig. 3 (a)**, due to the extremely low absorptivity of solid-state copper to infrared light, excessively high power density is required to initiate the melting process. Infrared laser welding primarily operates in keyhole mode, where the molten pool is inherently unstable and prone to defects like porosity, cracks, and spatter. Huang et al. [56] combined experiments with CFD models to reveal the influence of laser oscillation on weld seam performance from a microstructural perspective. They utilized laser beam oscillating welding to stabilize the molten pool by expanding the HAZ, thereby obtaining weld seams with superior performance. Ali et al. [57] performed a comparative analysis between dual-beam and single-beam laser welding, concluding that dual-beam welding yields joints with higher performance, whereas single-beam welding results in numerous defects. Additionally, introducing a high-affinity filler metal between copper and aluminum to prevent direct contact can also suppress IMC formation and improve weld performance [58]. Ma et al. [59] investigated the effects of different pulse shape energy ratios in YAG laser welding on weld morphology, metal cross-section mixing, microstructure, failure modes, mechanical strength, and electrical conductivity of Al-Cu welds. Wu et al. [60] studied the impact of different welding speeds on weld formation, interface microstructure, mechanical properties, and fracture phenomena, further demonstrating that rational selection of welding parameters is essential for obtaining excellent welded joints.

The emergence of blue and green laser technologies has brought a revolutionary breakthrough to copper welding. Their fundamental advantage lies in the fact that copper exhibits an absorptivity of 60–70% for the blue light band, which is more than 10 times that of infrared lasers [61]. The qualitative change delivers fundamental benefits. A comparison of laser welding results at different wavelengths is shown in **Fig. 3 (c)** and **Fig. 3 (d)**. Sadeghian et al. [62] investigated the application of blue laser systems for joining low-thickness nickel-coated copper and mild steel, while Das et al. [63] studied the feasibility of laser welding multi-layer thin stainless steel foils using a 450 nm blue laser. Research indicates that the blue laser welding process proceeds primarily in a stable conduction mode, resulting in shallow and wide weld seams with a calm molten pool, gentle evaporation, and negligible spatter generation. Furthermore, due to the absence of a high-temperature deep-penetration keyhole, the

growth of brittle IMCs is effectively suppressed. As evidenced by the studies of Sadeghian et al. [62] and Iqbal et al. [64], the formation of IMCs is significantly reduced in blue laser welding compared to infrared laser welding. Consequently, blue laser-welded Cu-Al joints typically exhibit lower interfacial resistance and superior mechanical properties.

IR-blue hybrid laser technology offers a novel pathway for solving the welding difficulties of highly reflective materials. A schematic of the IR-blue hybrid laser welding system is presented in **Fig. 3** (b). The scheme integrates the high absorptivity and stability of blue lasers with the high energy density of infrared lasers, achieving a comprehensive enhancement in absorptivity, melt pool stability, and penetration capability through synergistic effects [65]. In the hybrid welding process, the blue laser first acts on the copper surface. By virtue of its significantly higher absorptivity, it forms a stable and low-reflectivity initial molten pool at a lower energy density, creating favorable conditions for subsequent energy coupling. Subsequently, the high-power near-infrared laser irradiates the pre-molten region, allowing laser energy to efficiently enter the material interior, thereby realizing stable deep penetration welding [66]. Compared to single-wavelength approaches, the hybrid scheme not only avoids the strong reflection and violent spatter associated with infrared lasers acting directly on solid copper but also overcomes the limitations of blue lasers in power scaling and penetration depth [67]. Leveraging the dual advantages of high absorption and high-power deep penetration, IR-blue hybrid lasers demonstrate outstanding performance in thick copper and demanding dissimilar material welding scenarios, and are regarded as a significant development direction in the field of laser welding.

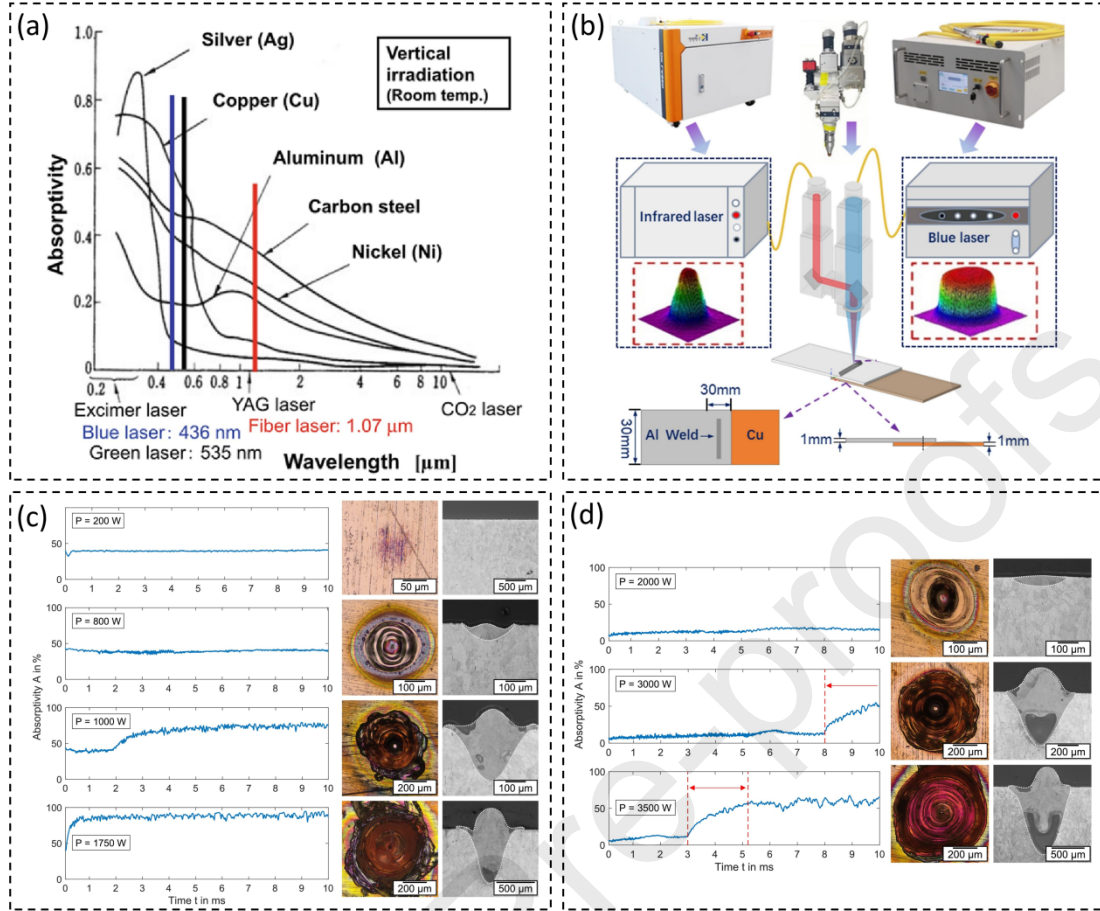


Fig. 3. Physical mechanisms and results at different wavelengths in laser welding: (a) laser absorption efficiency of different metals [61]; (b) schematic diagram of the NIR-blue hybrid laser welding system [68]; (c) measured absorptivity data, top view, and cross-sections of spot welds on Cu-ETP using $\lambda = 515 \text{ nm}$ laser radiation with different pulse powers [69]; (d) measured absorptivity data, top view, and cross-sections of spot welds on Cu-ETP using $\lambda = 1030 \text{ nm}$ laser radiation with different pulse powers [69].

2.2. Laser welding defects

Laser welding of EV batteries is a transient high-temperature process characterized by multi-physics coupling. The extremely high energy density and rapid heating-cooling cycles often result in a narrow process window, making the process highly susceptible to inducing various welding defects. These defects not only compromise the mechanical structural integrity of the joint but also pose direct threats to the electrical performance and long-term safety of the battery module. The section critically analyzes the six most common and detrimental defect forms illustrated in **Fig. 4**, namely porosity, cracking, spatter, insufficient penetration, and excessive penetration, and undercut. We will systematically explore the formation mechanisms of these defects and provide an in-depth analysis of how they pose potential threats to

the cycle life and vehicle-level safety of EV batteries by increasing contact resistance, compromising hermetic sealing, or triggering short circuits. A comprehensive summary of these typical defects, along with their formation mechanisms and impacts, is presented in **Table 1**.

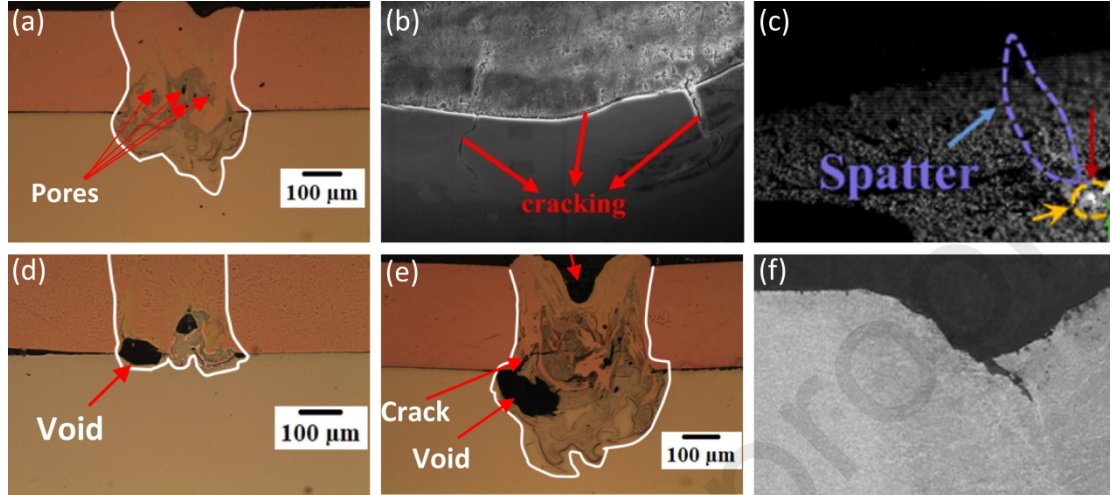


Fig. 4. Visual atlas of typical defects in laser welding: (a) porosity [19]; (b) cracking [70]; (c) spatter [66]; (d) insufficient penetration [19]; (e) excessive penetration [19]; (f) undercut [71].

2.2.1. Porosity

Under the equilibrium of surface tension, metal vapor recoil pressure, and shear stress, a keyhole is formed at the bottom of the molten pool [72,73]. Due to the poor fluidity of molten aluminum and excessively rapid cooling rates, the keyhole may collapse. Internal porosity is generated when the backfilling rate of the molten metal is significantly lower than the solidification rate. Upon solidification of the molten pool bottom, some pores remain trapped at the bottom, while others migrate along the molten pool towards the aluminum side or escape through the keyhole [74]. The presence of porosity reduces the effective load-bearing area of the weld, degrades the mechanical strength of the joint, and promotes the initiation of fatigue cracks. Furthermore, porosity compromises the airtightness of the joint, making porosity control imperative during the welding process [75]. In EV battery applications, welding porosity not only weakens the mechanical strength of the solder joint but also significantly increases local contact resistance, leading to elevated heat generation at the joint. During long-term charge-discharge cycles, it localized heating accelerates the aging of the joint and adjacent materials, causing further deterioration of contact performance and thereby shortening the battery's cycle life. For battery structures requiring high airtightness, internal porosity may also serve as leakage pathways,

affecting battery consistency and safety reliability. Porosity tolerance in EV battery welding is extremely strict, as porosity is known to significantly affect joint conductivity and mechanical reliability [76]. Pores on the order of tens of micrometers, especially when clustered, may already degrade performance and are therefore generally considered unacceptable in critical current-carrying connections.

2.2.2. Cracking

Kumar et al. [19] conducted an in-depth study on the influence of laser power on weld defects through joint visualization. They posited that high thermal input induces distortion and welding stresses. Under the stress action of thermal cycling, hot cracking will be generated. Additionally, the mismatch in thermal expansion coefficients between the two materials also contributes to crack initiation. They discovered that low laser power input or low heat input could mitigate crack formation. Cracking within the weld seam lead to stress concentration when the joint is subjected to loads, drastically reducing the fatigue life of the joint. During the long-term operation of EV batteries, cracking in the weld seam is highly prone to propagation under cyclic stress and temperature rise, ultimately leading to joint failure. Once a crack penetrates the welding interface, it causes an interruption in the conductive path or a sudden surge in resistance. It not only triggers battery performance degradation but may also lead to local overheating in extreme cases, increasing the risk of thermal runaway and posing a threat to the safety of the battery system. Accordingly, cracking is generally considered unacceptable in EV battery welding, since even micro-cracks can propagate under cyclic loading and compromise both mechanical integrity and electrical continuity [77,78].

2.2.3. Spatter

Hugger et al. [79] analyzed the causes of spatter formation during laser welding using high-speed imaging experiments and numerical simulations. They pointed out that due to metal evaporation at the keyhole surface and more intense evaporation at the keyhole bottom, the inclination angle of the keyhole increases when the welding speed exceeds 0.08 m/s, thereby facilitating spatter generation. Moreover, violent fluctuations in metal vapor pressure inside the keyhole can induce explosive spatter [80,81]. Molten metal droplets are ejected from the molten pool at high speeds. If these particles fall into the battery cell, they may pierce the separator and trigger an

internal short circuit, presenting an extremely severe safety hazard. Additionally, spatter adhering to the product surface requires extra cleaning processes. Such internal short circuits caused by spatter are often sudden and irreversible, serving as significant inducements for thermal runaway and safety accidents in EV batteries. Therefore, in the battery manufacturing process, spatter control is not only related to weld formation quality but is also directly correlated with the safety of the battery system. For EV battery applications, spatters are generally subject to near-zero tolerance, since ejected particles may damage the separator or induce internal short circuits, while severe spatter often also leads to part rejection in high-quality production [82].

2.2.4. Insufficient and excessive penetration

When laser power is too low or welding speed is too fast, insufficient heat input per unit length prevents the penetration depth from meeting design requirements. The phenomenon is technically referred to as insufficient penetration. The most typical manifestation of the defect is that the molten pool fails to penetrate the upper tab layer or reaches the interface without fusing with the underlying terminal. It directly leads to a reduction in the effective load-bearing area of the joint and substantially lowers mechanical connection strength. Furthermore, insufficient penetration is often accompanied by cold laps or voids [83], which drastically reduce the fatigue life of the joint. For components requiring high airtightness, such as battery casings, the defect implies hermetic sealing failure.

Conversely, when laser power is too high or welding speed is too slow, excessive heat input results in the molten pool existing for an extended period with overheated liquid metal. The condition leads to excessive penetration. The most severe consequence is excessive penetration, where molten metal collapses and is lost under the influence of gravity [84], destroying the integrity of the workpiece. In EV battery welding, the hazards of excessive penetration are particularly fatal. High-energy laser beams may penetrate the bottom tab layer or even burn through the underlying polymer separator, directly causing contact between the positive and negative electrodes and triggering severe internal short circuits.

For EV batteries, insufficient penetration and excessive penetration correspond to connection failure risks and catastrophic safety hazards, respectively. Insufficient penetration leads to premature fracture of the weld joint under high-frequency

vibration and thermal cycling, causing a sharp rise in contact resistance and accelerating battery capacity decay. In contrast, excessive penetration may instantly trigger thermal runaway by destroying the internal insulation structure of the cell. Therefore, strictly controlling the laser energy density within the optimal process window is key to ensuring both battery lifespan and high safety. In practical battery manufacturing, both insufficient and excessive penetration are tightly restricted. Literature on battery interconnect welding indicates that an ideal joint should ensure sufficient fusion without bottom-surface breakthrough, for example, approximately 63% penetration of the lower sheet with no bottom-surface penetration has been reported as suitable for battery applications [85].

2.2.5. Undercut

Undercut is typically caused by excessive laser energy density. The excessive heat generates a large volume of molten metal, and the liquid metal at the edge of the molten pool solidifies before it can spread back and refill under the action of surface tension. The results in the formation of a depressed groove along the weld direction at the junction of the weld toe and the base material [86,87]. The geometric discontinuity not only directly reduces the effective load-bearing cross-sectional area of the joint but also induces severe stress concentration effects at the groove. It easily becomes a source for fatigue crack initiation, thereby significantly reducing the fatigue life of the welded structure under dynamic loads. In EV battery modules and packs, welded structures are subjected to long-term vibration, shock, and thermal cycling loads. Stress concentration caused by undercut will significantly accelerate fatigue failure of the solder joint and reduce connection reliability. Once a fatigue fracture occurs at the solder joint, it not only leads to single-cell failure but may also cause current redistribution within the module, leading to local overheating and consequently affecting the safety and service life of the battery.

Table 1. Typical defects in EV battery welding and their impacts on safety and lifespan.

Defect type	Brief formation mechanism	Impact on weld seam/joint	Impact on battery lifespan	Impact on battery safety
Porosity [88]	Keyhole collapse, insufficient backfilling of molten metal, rapid solidification	Reduced effective load-bearing area, increased local resistance, decreased airtightness	Increased local Joule heating, accelerating aging of the joint and adjacent materials, shortening cycle life	Potential leakage pathways, affecting battery consistency and reliability
Cracking [89]	Excessive heat input, welding residual stress, and incompatible thermal strain caused by differences in thermal expansion coefficients	Stress concentration, significant decrease in fatigue strength	Crack propagation under cyclic loading, premature joint failure, performance degradation	Crack penetration may lead to local overheating, increasing the risk of thermal runaway
Spatter [90]	Keyhole instability, violent fluctuations in metal vapor pressure	Degraded surface quality, potential formation of secondary defects	Reduced joint formation quality and consistency, decreased fatigue life under cyclic loading, indirectly accelerating battery performance degradation	Spatter particles entering the cell may pierce the separator, triggering internal short circuits, a severe safety hazard
Insufficient penetration [91]	Insufficient heat input, excessive welding speed	Incomplete weld connection, insufficient load-bearing capacity	Increased contact resistance, premature joint failure during service, accelerated capacity decay	Hermetic failure, affecting the sealing reliability of the casing
Excessive penetration [91]	Excessive heat input, overheated molten pool	Destruction of workpiece integrity	Direct damage to cell structure, drastic reduction in lifespan	Laser may penetrate tabs or separators, triggering severe short circuits or even thermal runaway
Undercut [71]	Excessive energy density, failure of molten metal at the pool edge to backfill in time	Geometric discontinuity, stress concentration	Significantly reduced fatigue life, decreased connection reliability	Current redistribution following joint fracture, leading to local overheating

2.3. Traditional process control methods

Traditional laser welding process control predominantly adopts open-loop control strategies [92]. The core practice involves determining a fixed set of optimal process parameters, such as laser power, welding speed, and defocusing distance, offline through extensive trial-and-error experiments before production. Such parameter optimization generally relies on so-called offline optimization methods, which explore the relationship between process parameters and welding quality through limited experimental runs under laboratory or controlled conditions. Specifically, orthogonal experimental design utilizes orthogonal arrays with uniform representativeness to substitute full-factorial experiments with a reduced number of combinations, thereby identifying primary influencing factors and their trends at a lower experimental cost [93]. Meanwhile, response surface methodology (RSM) fits quadratic polynomial models to construct mathematical relationships between process parameters and welding responses, such as penetration depth and weld width, further facilitating parameter optimization and interaction effect analysis [94]. These methods are widely applied in traditional industrial processes to define the initial process window.

In recent years, accompanying the development of artificial intelligence, the application of machine learning prediction in optimizing laser welding process parameters has increased. Tsai et al. [95] constructed a high-precision three-dimensional CFD model and utilized artificial neural networks (ANNs) to predict the effects of welding speed and power combinations on molten pool depth, width, and cooling rate, finally determining the optimal parameter combination through screening. Chuang et al. [96] determined laser welding parameters, such as laser power and welding speed, based on finite element (FE) simulation, regression analysis, and ANN to minimize angular distortion and enhance the mechanical properties of the joint.

However, the open-loop control mode possesses fundamental limitations. First, it is essentially a static optimization based on the assumption that the production environment, material state, and equipment performance remain constant during the welding process. Consequently, it is incapable of coping with inevitable dynamic disturbances in actual production. Second, due to the absence of a real-time feedback mechanism, the welding system cannot perceive instantaneous process fluctuations,

including laser output power instability, variations in shielding gas flow, workpiece surface contamination, or assembly gap deviations, and thus cannot perform real-time online adjustments based on the welding state. Finally, it suffers from insufficient robustness. Once material batches change, environmental temperature and humidity fluctuate, or equipment status degrades, the optimal parameter combination obtained under offline conditions may rapidly become invalid. It leads to a decline in weld consistency or even fluctuations in product yield. Evidently, traditional open-loop control methods based on experience and offline optimization can no longer satisfy the demands of high-end manufacturing for welding quality stability, process consistency, and intelligent regulation.

2.4. Traditional quality assurance methods

Traditional laser welding quality assurance systems rely primarily on post-process inspection, which involves evaluating product quality after welding is completed. The approach is intrinsically a passive verification mode where quality issues are often detected only after defects have formed or even after mass production has occurred. Current post-process inspection methods mainly comprise two categories, namely destructive testing (DT) and non-destructive testing (NDT). Common inspection methods are illustrated in **Fig. 5**.

Regarding destructive testing, typical methods include metallographic analysis [26], tensile/shear testing [27], and peel testing [55]. These methods provide high-precision information on microstructural characteristics, penetration morphology, and joint mechanical properties, and are considered among the most reliable means of welding quality evaluation. However, the testing process is typically time-consuming, requiring multiple steps such as sample preparation, polishing, etching, and microscopic observation. It is costly and results in the complete scrapping of the sample, making it difficult to implement on a large scale on production lines.

Although non-destructive testing methods such as X-ray inspection [28], industrial CT [29], and ultrasonic testing [97] can identify internal weld defects without damaging the workpiece, they involve massive investment in inspection equipment and time-consuming imaging or analysis processes. Furthermore, constrained by material thickness, geometric structure, and surface conditions, achieving high-speed continuous detection on actual battery welding production lines remains difficult. Simultaneously, as EV battery structures become increasingly

complex and weld sizes trend toward miniaturization and high precision, the sensitivity and spatial resolution of traditional NDT also face challenges.

These traditional quality assurance methods have gradually exposed multiple core bottlenecks under modern high-end manufacturing demands. First, due to limitations in cost, cycle time, and equipment capability, both DT and NDT are mostly conducted via sampling inspection rather than covering the entire product volume, thereby introducing significant risks of missed detection. Second, post-process inspection exhibits obvious latency. When quality issues are detected during sampling, a large number of non-conforming products have often already been produced, leading to rework, scrapping, or recalls, which cause severe economic and time losses. Moreover, it cannot provide real-time feedback to the welding process for immediate correction. More critically, these methods are fundamentally incapable of achieving 100% in-line full inspection. The quality status of the massive volume of products manufactured between the two sampling inspections remains effectively in an unknown zone. For application scenarios like EV batteries that rely heavily on welding reliability, it introduces immense potential safety hazards and long-term reliability uncertainties.

In summary, the traditional quality assurance mode based on post-process inspection struggles to meet the requirements of current intelligent EV battery manufacturing for high consistency, high reliability, and high-tempo production. A transformation toward an intelligent inspection system featuring real-time monitoring, online identification, and closed-loop quality control is urgently needed.

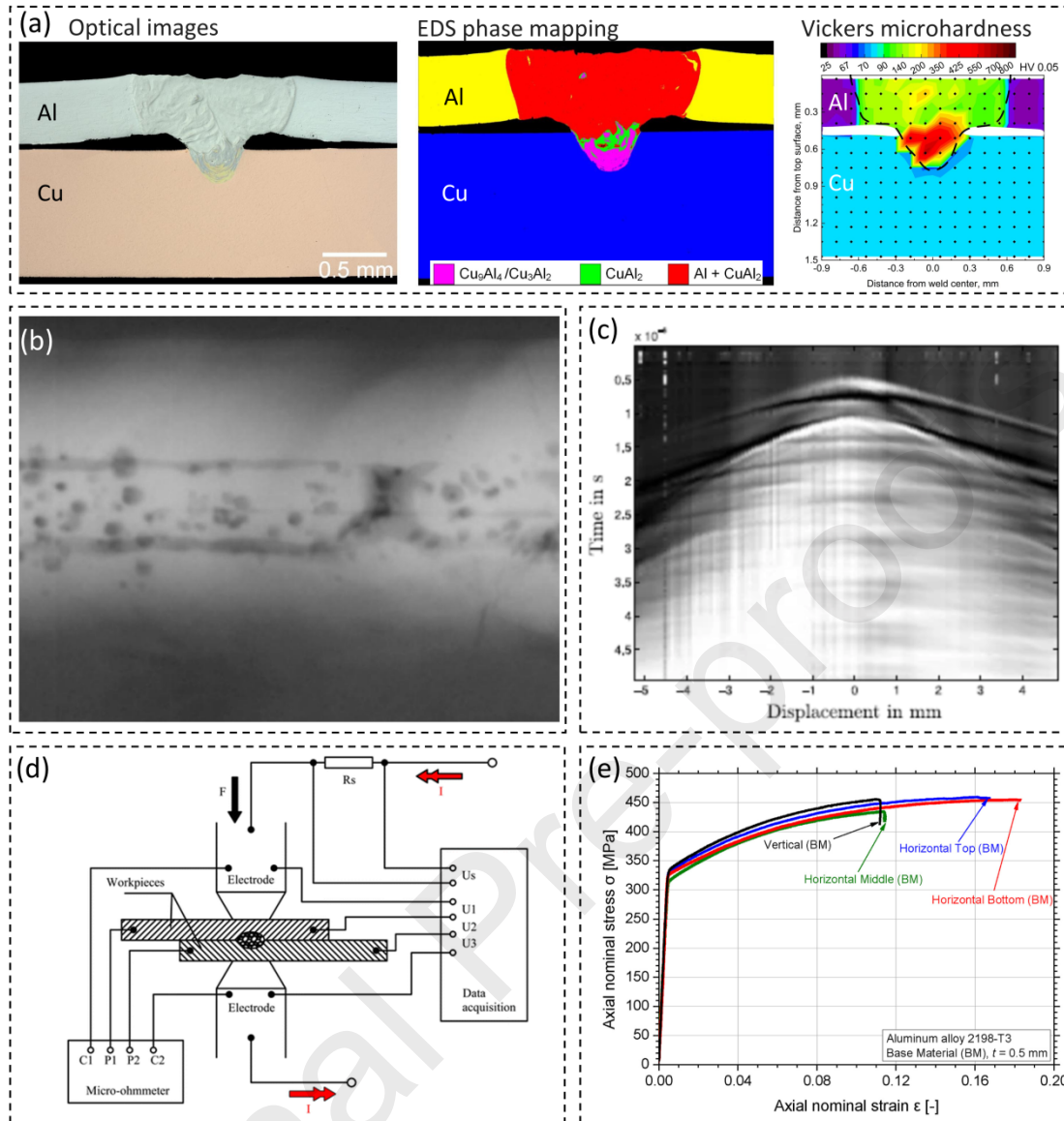


Fig. 5. Conventional post-process quality inspection methods: (a) microstructure analysis [98]; (b) weld X-ray image [99]; (c) B-scan of a 2-mm thick aluminum alloy plate [29]; (d) resistance testing [100]; (e) tensile testing [27].

3. Machine learning-enabled process control

3.1. Intelligent sensing and monitoring systems

Precise process control originates from a deep understanding of physical phenomena. Laser welding is an instantaneous, high-temperature, nonlinear, and complex process involving strong multiphysics coupling. During the process, laser-material interaction induces a series of dynamic phenomena including keyholes [101], molten pools [102], plasma [103], and spatter [104]. These phenomena provide insight into the internal state of the weld by emitting rich multimodal signals such as

light, sound, and heat. These signals effectively correlate with weld quality and serve as key indicators for evaluating weld seam integrity.

Traditional process control typically relies on offline inspection, resulting in significant delays. In-situ monitoring technology based on multimodal sensing combined with machine learning acting as an intelligent brain to interpret complex information enables the establishment of nonlinear mappings between signals and welding quality metrics like penetration depth, defects, and surface formation [105].

Fig. 6 illustrates the classification of commonly used welding process sensing technologies. This section systematically elaborates on the physical mechanisms, hardware architecture, and latest monitoring advancements enabled by machine learning in intelligent sensing for laser welding.

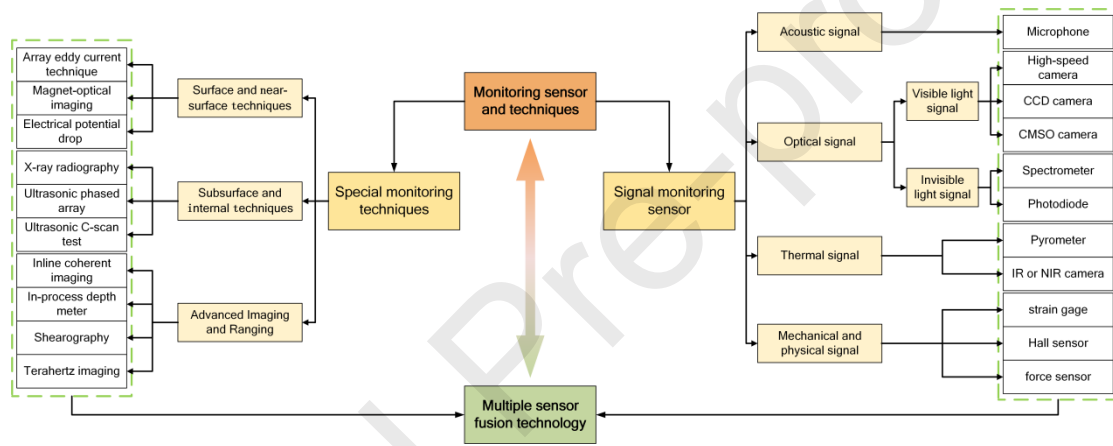


Fig. 6. Classification of monitoring sensor technologies.

3.1.1. Vision-based sensing

Visual sensing, characterized by its high resolution and intuitive nature, is currently the most widely adopted monitoring method. Based on optical path configuration, it can be categorized into coaxial and paraxial monitoring; according to imaging wavelength bands, it encompasses visible light, near-infrared, and ultraviolet (UV) bands.

Visual sensing is the core technology for real-time monitoring of laser welding. Through high-speed cameras, CCD/CMOS cameras, and other equipment, it directly observes the macroscopic morphology, dimensions, and dynamic behavior of phenomena such as the molten pool, keyhole, plasma plume, and spatter. **Fig. 7 (d)** illustrates a laser vision sensor. **Fig. 7 (b)** depicts a visual sensing system. Gao et al. [106] established fundamental correlations between the geometric characteristics of the molten pool and the stability of the laser welding process through quantitative

analysis. Zhang et al. [107] employed CCD cameras to capture dynamic images of the molten pool and keyhole, providing a data foundation for molten pool morphology analysis.

However, in practical applications, a single viewpoint often struggles to capture the three-dimensional fluctuations of the molten pool. Consequently, multi-view fusion has emerged as a trend to obtain more comprehensive information. **Fig. 7 (a)** illustrates a dual-camera system. Zhang et al. [108] constructed a dual-CCD system to simultaneously observe the upper and lower surfaces of the molten pool. Their findings revealed that the upper surface is shorter and more stable than the lower surface, uncovering the dynamic equilibrium mechanism of surface tension during penetration. You et al. [109] exploited spectral differences by simultaneously capturing spatter images with two high-speed cameras in the NIR and UV/VIS bands, demonstrating the short-wavelength band's advantage in isolating thermal radiation backgrounds and precisely capturing spatter behavior. To address wide-field monitoring needs, Chen et al. [110] developed a four-camera array monitoring system. Test results indicate the system holds significant potential for large-area measurements without compromising resolution or accuracy, offering new approaches for welding monitoring of large, complex components.

Optimizing imaging quality poses a significant challenge for visual sensing in the face of extreme light intensity and radiation interference during laser welding. In addition to traditional narrow-band filtering techniques [111,112], active illumination methods [113–116] have emerged as a practical solution. Luo et al. [102] employed an auxiliary green laser to illuminate the molten pool, capturing dynamic images with a CMOS camera to effectively extract the molten pool edges.

With the development of computer vision, data processing has evolved from traditional image processing techniques including edge detection and binarization to deep learning-driven semantic understanding. CNNs have become the mainstream tool for extracting high-dimensional visual features. For instance, object detection models like YOLO or Faster R-CNN enable real-time counting and trajectory tracking of spatter particles. Meanwhile, semantic segmentation networks such as U-Net can precisely delineate the contours of plasma feathers, quantifying their area, brightness, and center-of-mass fluctuations. These high-level semantic features exhibit strong correlations with penetration depth variations, providing critical insights for real-time process control. **Fig. 7 (c)** illustrates a visual model for detecting weld bead profiles.

Xue et al. [117] proposed a lightweight multi-output CNN architecture that enables simultaneous monitoring of the multi-state characteristics of AISI 304 stainless steel welding. Furthermore, Nam et al. [118] introduced a novel “end-to-end” visual inspection code framework that extends beyond mere monitoring. It directly generates executable control codes from a single weld seam image, achieving fully automated seam tracking and laying the foundation for vision-driven closed-loop adaptive welding.

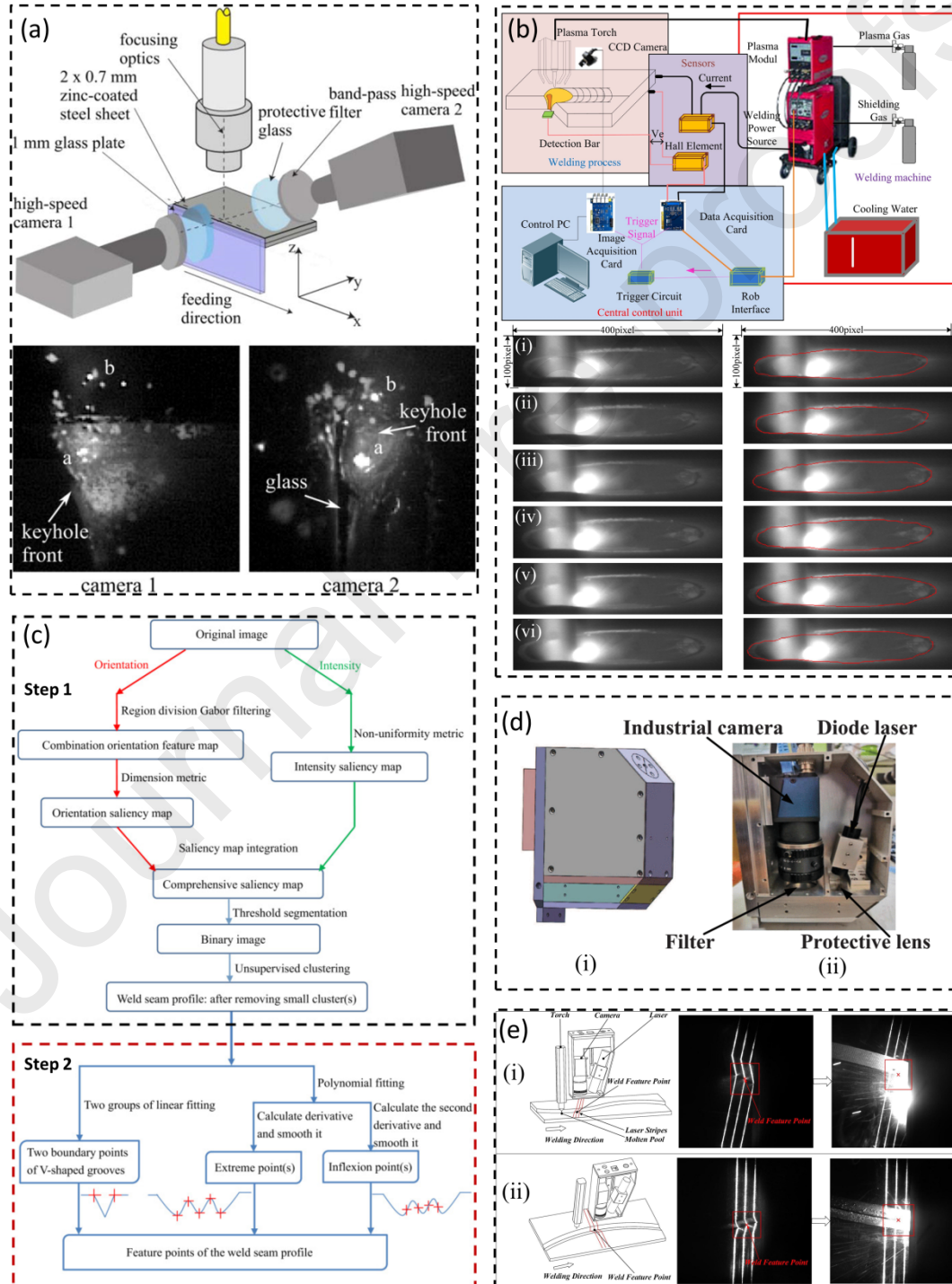


Fig. 7. Representative vision-based sensing systems and image processing frameworks for laser welding monitoring: (a) a dual-camera system, camera 1 measures steam plume velocity, camera 2 measures melt front velocity at the keyhole front wall [119]; (b) a top-layer weld pool visual sensing system designed for K-PAW along with the acquired weld pool and extraction boundary [120]; (c) a schematic of a visual model for detecting weld bead profiles [121]; (d) a self-developed laser vision sensor (i) 3D model (ii) physical apparatus [122]; (e) schematic of weld tracking using a laser vision sensor (comprising a camera and laser generator) for (i) lap welds and (ii) butt welds [123].

3.1.2. Photodiode-based optical sensing

In the manufacturing process of EV batteries, particularly during the welding and tab connection of highly reflective non-ferrous metals like Cu and Al, the nonlinear and unstable nature of the welding process readily leads to defects such as porosity, cracking, and insufficient penetration depth. Optical sensing technology, leveraging its non-contact nature, rapid response speed, and high information capacity, has emerged as the core method for capturing transient welding characteristics. Currently, mainstream optical sensing solutions primarily rely on spectrometers and photodiodes to decode optical signals from the weld zone across two dimensions: spectral composition and temporal fluctuations in light intensity. **Fig. 8** (a) shows the spectrometer signals for DP steel welding plate welding and Cu/Al re-soldering.

When lasers interact with metals, they generate high-temperature metal vapors and photo-induced plasmas. The positions, intensities, and broadening of characteristic spectral lines in their emission spectra directly reflect the plasma's electron temperature, electron density, and elemental composition. For Al/Cu dissimilar metal lap welds commonly found in EV batteries, spectral information serves as a critical basis for determining elemental mixing ratios within the weld and the formation of IMCs. Sibillano et al. [124–127] conducted in-depth investigations into the optical radiation mechanisms of plasma plumes. Based on the local thermal equilibrium (LTE) assumption, they established a correlation model between plasma electron temperature and weld penetration depth. They developed a spectral monitoring system utilizing a CCD detector array, enabling real-time calculation of electron temperature fluctuations to infer penetration depth changes. The technology has demonstrated significant practical value in industrial applications. Building upon the foundation, Kang et al. [128,129] further addressed the more complex heterogeneous material welding in the EV battery sector. Using a single-spectrometer sensor, they achieved quantitative characterization of Cu/Al lap welds and duplex

steel cladding processes. Their proposed multi-sensor model, integrating CCD images with spectral signals, significantly reduced monitoring costs and provided theoretical support for quality control in heterogeneous metal welding for EV batteries.

Although spectrometers offer high resolution, their substantial data processing demands make them unsuitable for real-time closed-loop control in ultra-high-speed welding. Photodiodes, with their extremely high sampling frequencies (typically $>10\text{kHz}$), can sensitively capture fluctuations in light intensity across specific bands (ultraviolet UV, visible VIS, infrared IR). The temporal characteristics of these signals correlate closely with keyhole stability and molten pool hydrodynamic behavior. In battery tab-to-busbar connections, highly reflective materials readily cause keyhole collapse or gap fluctuations. Eriksson et al. [130] utilized a three-band sensing architecture capturing the T signal for infrared thermal radiation, the P signal for plasma intensity in the ultraviolet and visible bands, and the R signal for laser reflection associated with the keyhole aperture. Through the collective analysis of these independent data dimensions, the thermodynamic state of the molten pool can be accurately reconstructed. Addressing common assembly gap issues in battery module assembly, Chianese et al. [131] discovered in their study of laser welding copper-steel battery connectors that even minor changes in workpiece gaps produce significant step responses in plasma signals. It demonstrates the sensitivity of photodiodes in diagnosing component alignment accuracy.

To ensure reliable electrical connections, Shu et al. [132] proposed a machine learning-driven method for diagnosing weak welds in busbar laser welding based on photoelectric signals. The authors also discussed the differences in photoelectric signals caused by various defects. For example, a reduction in laser power leads to a significant decrease in the overall intensity of the P signal. When the overlap gap is too large, this decrease in the P signal's intensity can reveal the underlying mechanism. This study establishes a physical-level correlation between the signals and these fundamental metallurgical phenomena, providing valuable insights into the model's interpretability.

With the penetration of artificial intelligence technology, simple signal threshold judgments are no longer sufficient to accommodate the complex process windows of EV battery welding. Research focus has now shifted toward extracting high-dimensional features from light intensity time-series signals. Typically, fast fourier transform (FFT) is employed to analyze periodicities in the frequency domain such as

the keyhole breathing frequency, while discrete wavelet transform (DWT) captures transient sudden changes. These feature vectors are fed into machine learning models such as SVM and ANN to achieve high-precision classification of states including stable deep penetration welding, thermal conduction welding, and keyhole instability. Lee et al. [133] specifically compared the performance of SVM, fully connected neural networks, and CNN for Al/Cu laser welding, which represents a typical scenario in battery manufacturing. The results demonstrate that deep learning models can uncover nonlinear correlations within photodiode signals that are difficult to discern visually, significantly improving prediction accuracy for copper-aluminum weld penetration states. It provides an intelligent solution for achieving zero-defect manufacturing on battery production lines.

3.1.3. Acoustic sensing

Acoustic sensing technology is based on the photothermal-to-acoustic energy conversion mechanism during laser welding, utilizing microphones or resonant sensors to non-contact capture acoustic signals or structural acoustic emission signals in the air [134]. Acoustic monitoring methods have garnered widespread attention due to their low cost and operational simplicity [135,136]. In the mass production of EV batteries, particularly during the welding of tabs and busbars made of highly reflective materials like copper and aluminum, the intense thermodynamic behavior within the keyhole generates a unique acoustic fingerprint. When a high-energy laser beam impacts a metal surface, the instantaneous vaporization of material generates high-pressure metal vapor ejected at high velocity. The resulting recoil pressure fluctuations not only maintain the dynamic equilibrium of the keyhole but also serve as the primary source of acoustic radiation. **Fig. 8 (c)** illustrates an acoustic signal detection method. Ao et al. [137,138] developed a detailed two-dimensional fluid dynamics simulation model, mechanistically revealing the acoustic signal characteristics induced by vapor flow during welding. Both simulations and experiments indicate that the vapor flow velocity at the keyhole center significantly exceeds that in peripheral regions, reaching a maximum of 280 m/s. The high-speed ejected vapor stream interacts with the surrounding air, inducing periodic compression and rarefaction of air density, thereby radiating high-frequency acoustic waves carrying information about the molten pool state.

Based on the aforementioned acoustic generation mechanism, researchers have focused on decoding the mapping relationship between acoustic signals and welding quality. For battery module welding, weld penetration depth is a critical indicator determining electrical connection reliability, where insufficient penetration increases contact resistance and causes overheating, whereas excessive penetration may puncture the battery casing and lead to electrolyte leakage. Huang et al. [139] conducted an in-depth study on the correlation between weld penetration depth and acoustic signals, defining and extracting time-frequency domain acoustic features such as sound pressure deviation and frequency band power. By incorporating neural network algorithms and multiple regression analysis, they successfully established a quantitative model linking acoustic features to penetration status. Results demonstrated the method's effectiveness in distinguishing between full penetration and partial penetration, offering a low-cost solution for defect interception during battery welding. Additionally, Lv et al. [140] designed an automated monitoring and control system specifically for gas-shielded tungsten inert gas welding (GTAW). Its control logic, based on acoustic signals and a linear fitting model of arc length (analogous to laser keyhole depth), provides valuable insights for implementing acoustic-based closed-loop control in laser welding.

However, the complex environment of EV battery production lines, where background noise generated by robotic arm movements and surrounding equipment operations easily drowns out faint welding sound signals, has become the primary bottleneck hindering the large-scale industrial application of acoustic monitoring. The waveforms from four sets of acoustic signal detection experiments are shown in **Fig. 8** (d). To overcome the signal-to-noise ratio challenge, Luo et al. [141] proposed a hardware-level solution: designing a planar array system comprising eight microphones. Utilizing time-delay recognition technology, the system enables spatial localization and amplification of sound sources, significantly suppressing background noise interference from non-welding areas. At the algorithmic level, He et al. [142] developed an advanced signal processing workflow tailored for stainless steel laser welding. They proposed an adaptive resonance demodulation method based on variational modal decomposition (VMD) parameter optimization, innovatively integrating sparrow search algorithm (SSA) for acoustic signal preprocessing. Experimental results demonstrate that the integrated hardware-software noise reduction strategy effectively isolates environmental noise and accurately restores

acoustic features characterizing weld formation quality. It validates the feasibility of acoustic sensing for monitoring the welding quality of highly reflective materials like copper and aluminum in complex industrial environments.

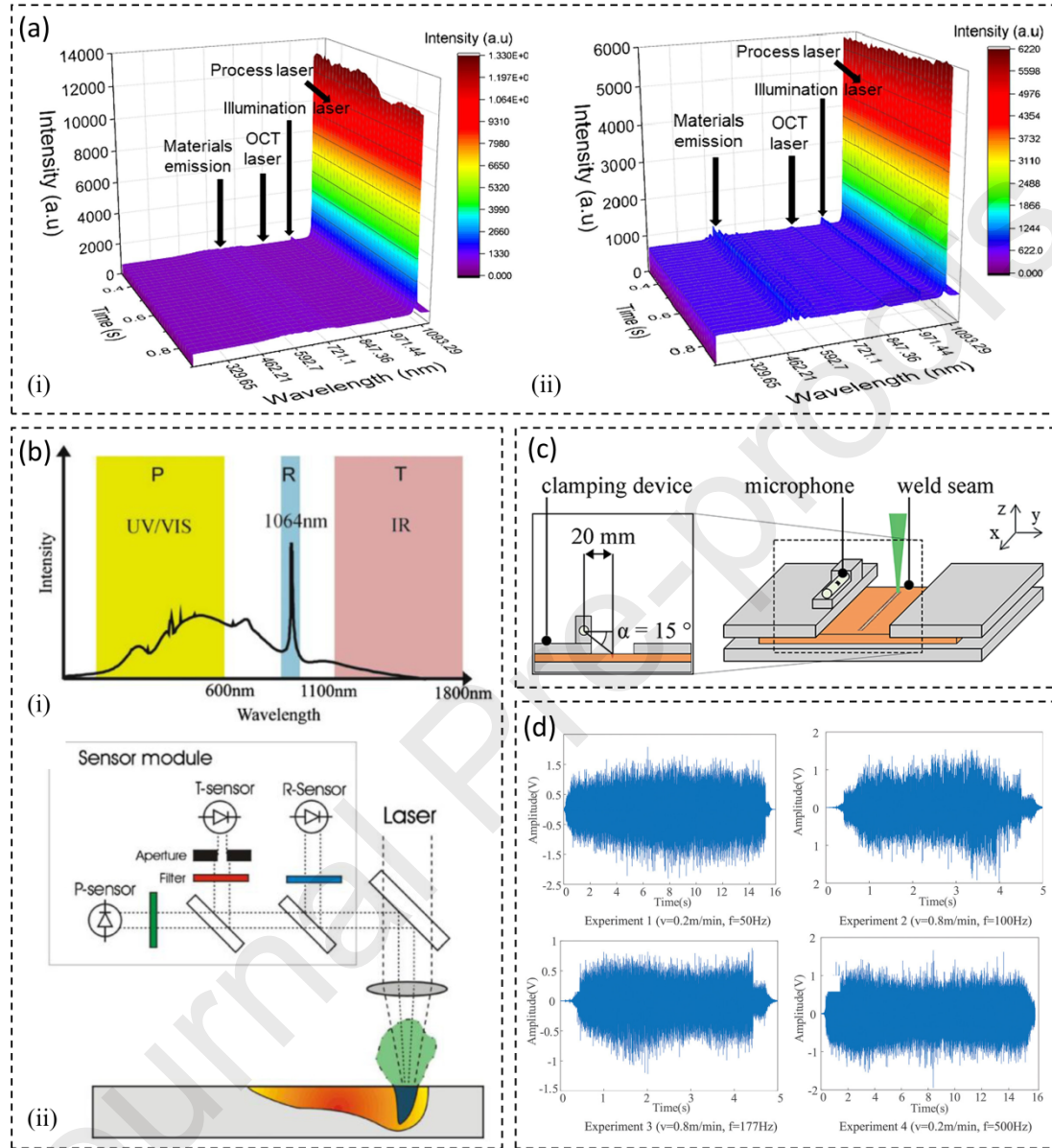


Fig. 8. Multi-modal sensing strategies for welding monitoring including optical, spectral, and acoustic signals: (a) spectrometer signals for (i) DP steel bead-on-plate welding and (ii) Al/Cu overlap welding [129]; (b) schematic of optical signal monitoring method for (ii) and spectral monitoring range for (i) [130]; (c) acoustic signal detection method [143]; (d) waveforms from four sets of acoustic signal detection experiments [142].

3.1.4. Multi-sensor fusion

In the mass production of EV batteries, the pursuit of rapid cycle times often forces laser welding processes into extremely short timeframes. The raw data collected by sensors, including high-frequency voltage values from photodiodes and

massive pixel arrays from high-speed cameras, is inherently high-dimensional, non-stationary, and saturated with redundant noise. Particularly during the laser cladding of dissimilar metals like copper and aluminum, dramatic fluctuations in material surface reflectivity introduce complex random disturbances that traditional fixed-threshold physical models fail to manage effectively enough to meet zero-defect quality control requirements. The core value of machine learning technology lies in its nonlinear mapping capabilities that automatically extract deep features intrinsically related to welding defects like cold joints, spatter, and porosity from the complex multi-source heterogeneous data. The process ultimately enables the precise classification and prediction of welding quality. **Fig. 9** illustrates the application of multiple AI-based technologies for data processing and analysis to achieve various monitoring objectives.

In the context of data dimensionality reduction and multi-source feature fusion, the prerequisite for establishing high-precision models lies in eliminating redundancy and extracting effective features from massive sensor data. Principal component analysis (PCA), as a classical statistical method, has been widely applied due to its outstanding performance in spectral feature extraction and data denoising [144–146]. For high-power welding scenarios, Liu et al. [147] developed a multi-sensor fusion system. They applied normalization and PCA techniques to perform orthogonal transformations on 15-dimensional time-domain features, selecting key principal components as input vectors for SVM. Experiments demonstrated the method effectively eliminates noise unrelated to the molten pool state. Although the research focused on high-power disc welding, its feature engineering approach holds significant reference value for suppressing environmental noise in battery busbar welding. Furthermore, addressing the issue of insufficient information from single sensors in battery manufacturing, Lei et al. [148] proposed a multi-information fusion neural network integrating laser process parameters, such as power and speed, with visual morphology features of the molten pool. The approach successfully achieved end-to-end prediction of weld geometric dimensions, such as melt width and penetration depth, validating the advantages of multi-source data fusion in enhancing model generalization capabilities.

In terms of model selection, SVM, ANN, Radial Basis function neural networks (RBFNN) [149], and LSTM [150,151] are widely applied in laser welding

monitoring. Two common application areas of data-driven laser welding are analyzed below.

Predicting keyhole stability and penetration depth based on time-series data represents one of the most critical research directions in data-driven welding processes. Keyhole stability directly determines the connection quality of battery tabs. Photodiodes and acoustic sensors capture typical time-series data, making LSTM networks and recurrent neural networks (RNNs) the preferred models for processing such data due to their inherent memory capabilities. Luo et al. [152] addressed keyhole collapse caused by sudden parameter changes during welding. They employed a coaxial vision system for real-time keyhole morphology monitoring, feeding the data into a state observer based on radial basis function neural networks (RBFNN). The approach successfully enabled early warning of potential porosity defects. For more precise penetration depth prediction, Li et al. [153] leveraged the unique gating mechanism of LSTM networks to deeply mine the time-dependent characteristics of plasma spectral signals. They established a dynamic mapping relationship between spectral intensity and welding depth, achieving a predicted average relative error of only 2.31%. The high-precision prediction holds significant engineering value for preventing excessive penetration or insufficient penetration in EV battery sealing welds. Additionally, Nunez et al. [154] employed a pre-trained CNN to extract keyhole depth features, using these as input for a regression model to achieve rapid prediction of operational outcomes with errors controlled within 3%. It demonstrates the effectiveness of deep learning in extracting physical features from unstructured data.

Computer vision-based spatter and feathering analysis is a current research hotspot. Spatter represents one of the most hazardous defects in EV battery welding. Once metal particles generated by spatter enter the battery module interior, they can readily cause high-voltage short circuits or even thermal runaway. Currently, CNNs have become the mainstream tool for analyzing high-speed camera images and quantifying spatter risks. Researchers widely employ object detection models like YOLO and Faster R-CNN for real-time counting of spatter particles, while segmentation networks such as U-Net are utilized to extract morphological features (area, height, brightness) of plasma arcs. Gao et al. [155] employed dual-band ultraviolet and visible light cameras to capture metal plumes. By integrating PCA dimensionality reduction with a particle swarm optimization (PSO)-enhanced SVM

model, they successfully achieved weld width estimation through plume morphology inversion. To meet the ultra-high online inspection speed demands of battery production lines, Cai et al. [156] established a high spatio-temporal resolution in-situ observation platform and innovatively proposed a lightweight spatter detection model. They first removed background metal vapor interference through point operations, then rapidly trained the model using an automatic labeling generation method. Validation results showed the lightweight model operates 5.11 times faster than the baseline model while achieving 96.71% spatter detection accuracy, providing a practical technical solution for real-time zero-tolerance spatter monitoring on EV battery production lines.

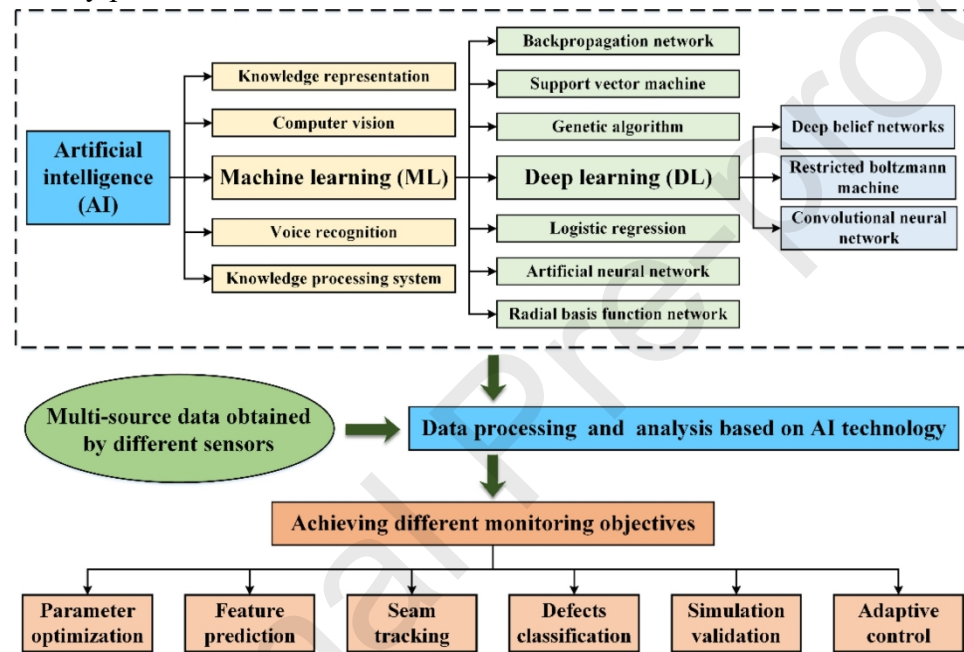


Fig. 9. A variety of AI-based technologies are used for data processing and analysis to achieve different monitoring objectives [48].

Despite the impressive predictive accuracies reported in these multi-sensor fusion studies, a critical gap remains in the physical interpretability of the learned features. Most current literature adopts a black-box paradigm where stochastic fluctuations in optical, acoustic, or thermal signals are directly mapped to quality labels such as good versus defective through deep learning architectures [103,133,157]. This approach often overlooks the causal chain rooted in welding physics. It must be emphasized that signal modulations are not merely data patterns but extrinsic manifestations of high-speed transient physical phenomena. For example, the synchronous attenuation in photodiode intensity and specific frequency shifts in acoustic emissions are physically coupled with keyhole instability and vapor

plume dynamics [130,132,158]. A sudden collapse of the keyhole, which serves as the primary precursor to stochastic porosity, triggers a distinct spatiotemporal signature across multiple sensing modalities. Treating these signals as purely statistical inputs without considering the underlying melt pool thermodynamics and fluid dynamics limits the scientific rigor and industrial reliability of the models. Future research should shift from simple feature-level concatenation toward physics-constrained fusion strategies where the fusion weights or latent spaces are regularized by known metallurgical mechanisms, such as the relationship between keyhole aspect ratio and signal back-reflection.

Furthermore, a critical hardware-engineering challenge often overlooked in current multi-sensor fusion literature is the strict requirement for spatiotemporal synchronization across heterogeneous data streams. While integrating acoustic, optical, and thermal signals offers theoretical superiority, it introduces severe physical alignment bottlenecks during highly transient laser welding processes [159]. A primary source of misalignment stems from the disparate propagation velocities of these physical phenomena: optical and thermal emissions propagate at the speed of light with negligible latency, whereas acoustic waves travel through air at approximately 340 m/s. To illustrate, if a microphone is positioned 20 cm from the melt pool, an inherent physical delay of roughly 0.6 ms occurs. In the context of high-speed imaging operating at 10,000 frames per second, this latency corresponds to a shift of six frames. Consequently, the camera captures the visual aftermath of an event long before the corresponding acoustic signature of a keyhole collapse reaches the sensor. Despite this, many existing studies rely exclusively on software-level timestamp alignment, often neglecting that operating system jitter can severely disrupt the causal correlation between these microsecond-scale physical events.

Through a systematic review of the core literature on multi-sensor fusion cited in this paper, we found that this oversight at the engineering level is prevalent in current research. In these studies, approximately 80% of the papers [104,135,142,149,152] merely demonstrate the high accuracy of the models after fusion, while completely avoiding the specific implementation mechanisms for spatiotemporal synchronization among sensors. Even in studies that partially mention “synchronous triggering” [153], most focus solely on software-level timestamp alignment within non-real-time operating systems, failing to quantitatively address the inconsistent delays in the physical propagation paths and analog-to-digital conversion (ADC) processes of

acoustic, optical, and thermal signals. This blind alignment may result in the model learning only statistical spurious correlations rather than true physical causality. Furthermore, the vast majority of the literature [95,96,99,104,142,150,152] fail to address the negative impact of computational overhead resulting from the fusion of multiple high-frequency data streams on real-time closed-loop control. Through this critical review, this paper aims to highlight that without rigorous hardware-level synchronization and computational performance evaluation, the industrial feasibility of existing multi-sensor fusion models in handling microsecond-scale transient processes such as laser welding remains highly questionable. To achieve scientifically rigorous multi-modal fusion, future systems must abandon simple array concatenation and implement nanosecond-level hardware-based triggering mechanisms, such as synchronized clock boards.

3.2. Data-driven parameter optimization

In the mass production of EV batteries, laser welding processes face extreme challenges. On one hand, the widespread use of highly reflective non-ferrous metals like copper and aluminum leads to highly unstable laser energy absorption. On the other hand, battery structural components including explosion-proof valves, electrolyte filling ports, and tabs demand zero-defect weld sealing and conductivity. The traditional trial-and-error approach to process development relies on extensive repetitive physical experiments to traverse the parameter space. The method is not only time-consuming and material-intensive but also struggles to precisely pinpoint the narrow process window within the multidimensional, coupled parameter space.

With the deep integration of Industry 4.0 and artificial intelligence technologies, machine learning offers a novel approach to addressing these challenges: a data-driven surrogate model optimization framework. The core concept involves constructing a mathematical surrogate model capable of approximating the input-output mapping relationship of the actual physical welding process. By transforming costly, time-consuming real-world welding experiments into low-cost, millisecond-response virtual computations, researchers can leverage global optimization strategies like genetic algorithms (GA) and PSO to execute tens of thousands of virtual iterations on computers. It enables the systematic revelation of complex coupling mechanisms among multiple variables, such as laser power, welding speed, and beam

oscillation patterns, achieving a leap from trial-and-error experimentation to intelligent optimization.

3.2.1. Surrogate modeling architectures

A surrogate model is essentially an empirical regression function based on statistical learning theory. Its mathematical objective is to construct an approximation function $\hat{f}(x)$ within the design space Ω that approximates the true physical response function $f(x)$, using a finite set of training data points $D = \{(x_i, y_i) | i = 1, \dots, n\}$. Its general mathematical expression is as follows [160]:

$$y = f(x) = \hat{f}(x) + \varepsilon(x) \quad (1)$$

where $x = [x_1, x_2, \dots, x_k]^T$ is the k-dimensional input process parameter vector, y is the welding quality response value, and $\varepsilon(x)$ is the error term comprising model deviation and random noise.

In the field of laser welding, the process of constructing high-fidelity surrogate models typically follows standard black-box modeling logic, with the specific steps and underlying mathematical principles as follows:

(i) Design of experiments (DoE): To maximize information gathering with minimal experimental runs, a scientific sampling strategy is essential. For multidimensional parameters involved in battery welding, such as beam oscillation trajectories featuring “ ∞ ” shaped or helical patterns, common approaches include orthogonal design, central composite design (CCD), and Latin hypercube sampling (LHS), which is better suited for high-dimensional spaces. LHS ensures uniform distribution of sample points along the edges and within the parameter space, capturing the highly nonlinear penetration depth variation trends observed in Cu/Al dissimilar metal welding.

(ii) Model selection and mathematical construction: Selecting a surrogate model requires balancing computational efficiency, predictive accuracy, and model interpretability. In the process optimization of laser welding for EV batteries, data often exhibits characteristics of small sample size, high dimensionality, and strong nonlinearity, particularly since destructive tensile testing is costly and typically restricts sample sizes to fewer than 50. Therefore, researchers must select appropriate algorithmic architectures based on specific data features.

(iii) Polynomial response surface methodology (PRSM): RSM is the most classical parametric model, assuming that the system's inputs and outputs follow a

smooth explicit functional relationship. In the preliminary process window screening for battery busbar welding, a second-order polynomial model is commonly used to approximate the local relationship between laser power, welding speed, and penetration depth [160–162]:

$$\hat{y}(x) = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{ii} x_i^2 + \sum_{i < j}^k \sum_{j=1}^k \beta_{ij} x_i x_j + \varepsilon \quad (2)$$

where β_i , β_{ii} and β_{ij} represent the coefficients for linear effects, quadratic curvature effects, and interaction effects, respectively. The model's strength lies in its exceptional interpretability, where the sign of β_{ij} directly reveals coupling mechanisms such as power-velocity interactions on heat input. However, its fitting capability is often limited when addressing highly nonlinear phenomena like complex molten pool instability or spatter.

(iv) Kriging/Gaussian process regression (GPR): Kriging models are considered the gold standard for deterministic simulation experiments, particularly suited for expensive experiments with sparse samples, such as long-term aging or fatigue testing of battery modules. They treat response values as realizations of a Gaussian stochastic process composed of a global trend model $F(\beta, x)$ and a local random offset $Z(x)$ [160,163,164]:

$$\hat{y}(x) = F(\beta, x) + Z(x) \quad (3)$$

The general form of the global trend model $F(\beta, x)$ is [163,164]:

$$F(\beta, x) = \beta f_i(x) \quad (4)$$

where $f_i(x)$ is the basis function of the regression model, and β is the regression coefficient vector. The primary advantage of the Kriging model is its capacity to yield both predicted values $\hat{y}(x)$ and their associated uncertainty, expressed as a confidence interval derived from the mean squared error. Its probabilistic nature renders it highly suitable for adaptive sampling, thereby facilitating the automatic identification of model uncertainty regions and guiding subsequent experiments to attain the optimal Cu/Al welding process point with minimal iterations.

(v) Support vector regression (SVR): By adhering to the principle of structural risk minimization (SRM), SVR exhibits remarkable resilience to process noise, notably fluctuations in photoelectric signals encountered during laser welding. SVR does not aim for zero training error but instead seeks a hyperplane where the distance deviation of all sample points to the plane falls within the tolerance ϵ . Its core optimization objective is to minimize the sum of the squared weight vector $\|w\|^2$ and the slack variable ξ [165]:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (5)$$

By introducing the Kernel Trick [165], defined as $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$, SVR can map low-dimensional nonlinear welding parameters into a high-dimensional feature space for linear regression. This enables it to effectively avoid the curse of dimensionality when handling high-dimensional inputs, specifically when simultaneously optimizing 5-6 variables including power, speed, defocusing distance, oscillation amplitude, and frequency, making it commonly used for predicting the formation quality of battery seal rivet welding.

(vi) ANNs: ANNs have emerged as the preferred architecture for handling massive process datasets, such as tens of thousands of historical weld joint records, owing to their robust nonlinear modeling capabilities underpinned by the Universal Approximation Theorem. A typical backpropagation (BP) network employs a multi-layer perceptron (MLP) structure, utilizing nonlinear activation functions, such as Sigmoid or ReLU, to capture extremely complex mapping relationships [166]:

$$y_k = \varphi(\sum_{j=0}^m w_{kj}^{(2)} \cdot \varphi(\sum_{i=0}^n w_{ji}^{(1)} x_i)) \quad (6)$$

Although ANNs achieve exceptional accuracy with sufficient data, precisely simulating microstructural changes caused by molten pool fluid dynamics, their strong black box nature and susceptibility to overfitting with small samples pose challenges. Therefore, during battery R&D, regularization techniques or ensemble learning are often integrated to enhance their generalization capabilities.

(vii) Model evaluation metrics: To verify whether proxy models can accurately replace physical experiments, evaluation is typically conducted using leave-one-out cross-validation (LOOCV) or independent test sets. Core metrics include the coefficient of determination (R^2) and root mean square error (RMSE) [163]:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

The closer R^2 is to 1, the stronger the model's ability to explain performance variation caused by process parameters.

(viii) Key input-output parameter system: In the specific application of laser welding for EV batteries, the selection of variables directly determines the success or failure of optimization. Input variables [38,167] primarily encompass key process parameters, material property parameters, and environmental parameters. Key process

parameters include laser power, welding speed, defocus amount, oscillation amplitude, and oscillation frequency for oscillation welding. Material property parameters encompass material type, thickness, thermal conductivity, and specific heat capacity. Environmental parameters consist of welding position, such as horizontal, vertical, or inclined, shielding gas type, and flow rate. These parameters collectively determine the energy density distribution, melt pool dynamics, thermal cycling path, and the resulting microstructural evolution.

Key performance indicators (KPIs) vary across different application scenarios. In the welding of precision electronic components such as EV batteries, output response primarily encompasses three dimensions: mechanical, electrical, and metallurgical. Regarding geometric morphology, both penetration depth D and width W , especially the former, must be strictly controlled to prevent battery casing breakdown or damage to the coil core beneath the tab. It also includes heat-affected zone depth, porosity, and defect grade. Mechanical/electrical properties primarily include residual stress, deformation, joint shear strength, and contact resistance. For metallurgical quality in Cu/Al connections, the average thickness and morphology of the IMC layer represent a highly challenging optimization target, typically requiring control below $10\text{ }\mu\text{m}$ to ensure joint ductility.

It is worth noting that Pareto conflicts often exist among these KPIs. For instance, increasing laser power density can enhance penetration depth and tensile strength, but simultaneously heightens the risk of spatter caused by metal vaporization and may result in excessive IMC layer thickness due to excessive heat input, thereby increasing contact resistance. While boosting welding speed improves efficiency, it tends to cause lack-of-fusion defects. Therefore, a single mathematical model often proves inadequate. A multi-objective optimization framework must be established to identify the optimal process balance point that maximizes overall performance.

The significance of surrogate models manifests in three key aspects: First, they can capture nonlinear relationships between process parameters and performance metrics using limited experimental data, thereby circumventing the prediction inaccuracies caused by parameter interactions inherent in traditional methods. Second, surrogate models exhibit exceptional computational efficiency, completing simulations in seconds that would require hours or even days using conventional approaches. Finally, they provide uncertainty estimates, which hold substantial value for risk management and decision support in process parameter optimization.

3.2.2. Machine learning algorithms for surrogate modeling

In the manufacturing process of EV batteries, laser welding involves multi-physics coupling of light, heat, flow, and force, exhibiting strong nonlinear and transient characteristics. Particularly in the connection between battery tabs and busbars, dissimilar welding of highly reflective materials like copper and aluminum is prone to generating porosity, cracking, and brittle IMCs. To establish a mapping relationship between process parameters and weld quality, including penetration depth, tensile strength, and electrical resistivity, in a cost-effective and efficient manner, various machine learning algorithms are widely applied in proxy model development. **Table 2.** presents a comparison of common algorithms and their application characteristics.

Table 2. Feature comparison of algorithms for building surrogate models.

Algorithm name	Model category	Core advantages	Main limitations	Data sample efficiency	Hyperparameter sensitivity	Typical applications for battery welding
Gaussian process regression	Probability	Provides confidence intervals alongside predicted means (quantifying uncertainty);	Computational complexity increases cubically with sample size ($O(N^3)$),	Extremely high (suitable	Medium (requires optimal kernel	Small-sample process design; fundamental Bayesian
(GPR) [168]	model	performs exceptionally well with small samples; highly mathematically interpretable.	making it unsuitable for large datasets; high-dimensional features are difficult	for expensive	function)	optimization models; process window exploration requiring risk
			to handle.	experimental data)		assessment
Artificial neural network	Neural network	Exceptionally strong nonlinear mapping capability; versatile function approximator;	Prone to local optima; requires large datasets to prevent overfitting; “black	Low (requires large	High (extremely sensitive to the	Establish complex process parameter-melt depth/melt width
(ANN/BP) [169,170]		excels at handling complex coupling issues in multiple-input multiple-output (MIMO)	box” model lacking physical interpretability; slow training.	amounts of data)	number of layers, nodes, and	mapping; multi-objective process parameter inversion
		systems.			learning rate)	
Convolutional neural network	Neural network	Specializes in processing grid-based data (images); capable of automatically extracting	Requires extremely large amounts of data; consumes significant computational	Extremely low (requires	High	Prediction of penetration status based on high-speed camera/melt
(CNN) [171,172]		spatial/texture features from molten pool images or spectra without requiring manual	resources; features complex model structures.	massive amounts of		pool images; analysis of spatter and slag morphology; end-to-end
		feature engineering.		annotated images)		weld seam tracking
Long short-term memory	Neural network	Specifically designed for time series; its unique gating mechanism captures long-range	Long training times; poor parallel computing capabilities; gradient propagation	Medium (requires long	Medium	Keyhole stability prediction based on photodiode/acoustic signal
network (LSTM) [172,173]		dependencies and dynamic time-varying characteristics during the welding process.	issues have improved but persist.	sequence data)		sequence data; dynamic penetration depth monitoring
Support vector regression	Kernel trick	Minimizes structural risk for strong generalization capabilities; performs stably in small-	Difficult to implement with large-scale training samples; sensitive to missing	High	High (sensitive to penalty coefficient	Regression analysis of medium-scale experimental data;
(SVR) [174,175]		to-medium sample sizes and high-dimensional spaces; demonstrates robustness to noise.	data; the choice of kernel function determines the final outcome.		C and kernel parameters)	classification and recognition of multi-sensor fusion signals
Random forest (RF)	Ensemble	Strong anti-overfitting capability; capable of evaluating feature importance; fast parallel	In regression problems, it cannot predict values beyond the training set (weak	Medium	Low (relatively robust)	Key influence factor screening for welding defects; preliminary
[176,177]	learning	processing speed; insensitive to outliers.	extrapolation capability); the prediction results are not sufficiently smooth.			process modeling and rapid validation
XGBoost [178,179]	Ensemble	Prediction accuracy typically outperforms RF; supports custom loss functions; state-of-	Prone to overfitting on noisy data; features a large number of parameters,	Medium-high	Medium	High-precision prediction of joint tensile strength; online quality
	learning	the-art model for structured tabular data; high training efficiency.	making tuning complex.			grade classification; rapid modeling of complex nonlinear
						relationships
Response surface	Traditional	Simple models (typically quadratic polynomials); minimal computational cost; easy	Can only fit smooth, low-order nonlinear relationships; poor at fitting complex	High (based on	Extremely low	Preliminary determination of the initial process window;
methodology (RSM)	statistics	analysis of factor interactions; high acceptance in industry.	local fluctuations; experimental volume explodes as variables increase.	orthogonal/CCD design)		parameter optimization when fewer variables are involved (<5).
[180,181]						

GPR stands as a representative probabilistic method occupying a significant position in process optimization with limited data due to its ability to not only provide predictions but also output confidence intervals for uncertainty quantification [38]. For battery cell welding, destructive testing is costly and sample acquisition is challenging. GPR effectively identifies model blind spots and guides subsequent experimental design through active learning, enabling rapid convergence to the optimal process window. Dong et al. [168] employed GPR to process complex welding data, enabling real-time prediction of the weld pool morphology. In copper-aluminum welding, melt pool fluctuations directly influence IMCs formation, and the robustness of the method provides theoretical support for monitoring melt pool stability. Sterling et al. [182] combined GPR as a surrogate model with Bayesian optimization (BO), an approach successfully adapted to laser welding for automatically identifying optimal parameter combinations that suppress spatter. To address the lack of physical interpretability in purely data-driven models, Wang et al. [183] not only employed GPR for regression but also integrated it with a thermo-fluid-solid coupled FEM model. By introducing a physical information kernel (PIK) function, they embedded stress distribution patterns into the predictive model. The dual-driven “physics + data” approach holds significant reference value for predicting thermal stress concentration and optical power loss during battery module welding.

ANNs, particularly the BP neural network, represent a prominent approach for modeling surrogate models. Leveraging their robust nonlinear mapping capabilities, these models have become the preferred tool for addressing multi-input multi-output or MIMO challenges in laser welding. In EV battery welding, it is often necessary to simultaneously optimize both weld morphology including penetration depth and width as well as mechanical properties such as shear strength. ANNs provide the foundational architecture for solving such multi-objective optimization problems. However, traditional BP networks exhibit high sensitivity to hyperparameters such as hidden layer nodes and learning rate, and are prone to getting stuck in local minima. Consequently, the hybrid strategy of “meta-heuristic algorithms + neural networks” has emerged as the current mainstream approach. Li et al. [184] proposed a BP neural network optimized by the dung beetle optimizer (DBO) for fiber laser welding, combined with the NSGA-II algorithm for multi-objective optimization. The research not only improved welding quality but also reduced carbon emissions from an energy efficiency perspective, aligning with the green manufacturing philosophy of EVs.

Zhang et al. [185] introduced the whale optimization algorithm (WOA) and differential evolution (DE) to optimize BP network weights. Experiments demonstrated that the hybrid model accurately predicts laser welding-induced thermal deformation (error ± 0.1 mm). Controlling welding deformation is critical in battery assembly for ensuring module dimensional accuracy and battery safety.

As a kernel-based method, SVR is widely used for modeling under conditions of small sample sizes. Based on the principle of minimizing structural risk, SVR demonstrates excellent generalization capabilities when handling high-dimensional, small-sample regression problems. For copper-aluminum dissimilar metal laser welding, experimental data often exhibit high noise characteristics due to complex metallurgical reactions. Through the nonlinear mapping of kernel functions, SVR can effectively capture key patterns. Zhang et al. [186] developed a sample-sorting support vector regression (SS-SVR) meta-modeling approach. The study demonstrated that SVR can accurately fit the relationship between process parameters and weld geometric features, including the depth-to-width ratio, even with limited experimental data, providing process guidance for scenarios demanding extreme airtightness such as battery sealing welds. Liu et al. [147] integrated PCA with SVM to achieve multi-sensor signal fusion classification during high-power laser welding. The approach enables real-time identification of cold welds or excessive penetration defects in battery welding, serving as a key technology for achieving “zero-defect” production line targets.

Ensemble learning algorithms enhance overall prediction accuracy by constructing multiple weak learners and inherently excel in feature importance assessment. In complex battery welding processes, determining the relative impact weights of laser power, welding speed, defocus amount, and oscillation amplitude on final quality is central to process development. Representative ensemble methods include RF and gradient boosting trees (XGBoost). RF can quantify the contribution of each input parameter. For instance, in laser cladding, RF models successfully identified laser power as the most critical factor controlling dilution rate. Zhang et al. [187] further applied Bayesian optimization (BO) to RF hyperparameter tuning, incorporating it as the objective function within a multi-objective jellyfish search algorithm (MOJS). When applied to copper-aluminum welding, the process effectively balances penetration depth with intermetallic compound layer thickness, significantly enhancing joint quality. As a representative of boosting algorithms,

XGBoost demonstrates robust performance on structured industrial data. Zhang et al. [188] employed XGBoost for online prediction of tensile strength in Al-Li alloy laser-welded joints ($R^2 = 0.9383$). Given the similarity in thermophysical properties between aluminum-lithium alloys and common aluminum alloys used in battery casings, the approach demonstrates the significant potential of gradient-boosted tree models for predicting the mechanical properties of battery structural components.

3.2.3. Surrogate-based optimization strategies

In the manufacturing of EV batteries, the process window for laser welding is often extremely narrow. Particularly in critical processes such as copper-aluminum dissimilar metal bonding, even minor parameter fluctuations can lead to incomplete welds or excessive penetration. Traditional trial-and-error methods are not only time-consuming and wasteful of materials but also struggle to cover the high-dimensional parameter space. Once a high-precision surrogate model is constructed, it serves as a digital twin, enabling engineers to conduct tens of thousands of virtual experiments on a computer. Combined with intelligent optimization algorithms, the system can traverse the entire parameter space at minimal computational cost, precisely identifying the optimal combination of process parameters that balances both mechanical and electrical performance.

(i) Single-objective optimization: When engineering objectives are clear and singular, such as maximizing penetration depth to ensure structural integrity in battery module side plate welding or minimizing contact resistance in busbar connections, classical heuristic algorithms prove highly effective for solving such nonlinear problems. GA are based on natural selection and genetic principles, performing random searches within the solution space through selection, crossover, and mutation operations. Due to their gradient-free nature, GAs demonstrate exceptional global search capabilities when handling non-convex, multi-peak functions common in laser welding, rarely getting stuck in local optima. PSO mimics the foraging behavior of bird flocks, updating individual velocities based on both personal and collective historical optimal positions. Compared to GA, PSO requires fewer parameters and converges faster, making it highly suitable for rapid sensitivity analysis of welding parameters. **Fig. 10** (a) illustrates a schematic of a process parameter optimization procedure. In the context of battery manufacturing, Ai et al. [189] first constructed a genetic algorithm-optimized backpropagation neural network (GA-BPNN) as a

surrogate model for laser welding processes. Subsequently, they employed a PSO algorithm to perform secondary optimization on the model's prediction outputs. Experimental results demonstrate that the strategy successfully identified parameter combinations that optimized weld integrity. Compared to unoptimized welded joints, the optimized joints exhibited significantly enhanced shear strength and markedly reduced quality dispersion caused by process fluctuations.

(ii) Multi-objective optimization: In practical engineering applications of laser welding for EV batteries, optimizing a single parameter often fails to meet all requirements, and inherent physical conflicts may exist between different parameters. A quintessential example of the trade-off is found in Cu-Al dissimilar metal laser welding. While higher heat input is required to promote element diffusion for mechanical strength, excessive thermal energy induces the growth of brittle intermetallic compounds, which significantly compromise joint conductivity and ductility [190].

To address trade-offs such as strength against conductivity or penetration depth against low spatter, multi-objective optimization algorithms aim to identify a set of Pareto optimal solutions, also known as the Pareto front. The Pareto front represents the collection of all solutions where no further improvement in one objective can be achieved without sacrificing another. **Fig. 10 (b)** illustrates the Pareto frontier for four objectives, while **Fig. 10 (d)** shows one Pareto optimal solution set. For the multi-objective optimization problem, the non-dominated sorting genetic algorithm II (NSGA-II) is currently the most widely applied multi-objective algorithm. It incorporates fast non-dominated sorting and a crowding distance calculation mechanism, ensuring that while approximating the true Pareto frontier, the solution set maintains good distribution within the objective space. Through non-dominated sorting and crowding distance calculation, NSGA-II generates a Pareto front representing a set of optimal solutions in a single run. This front comprises all solution sets where no other solution can improve one objective without worsening another. Engineers can then select the optimal trade-off solution based on specific application requirements, such as prioritizing strength over conductivity. Aghaei et al. [191] employed NSGA-II to simultaneously optimize four primary geometric features and four performance objectives during the welding process. By analyzing the Pareto front, they revealed the competitive mechanisms of different parameters on welding quality. Addressing the stringent dimensional accuracy requirements for battery pack

assembly, Tan et al. [192] utilized eight welding parameters as inputs for an ANN, employing both NSGA-II and the multi-objective particle swarm optimization (MOPSO) algorithm for optimization. Results showed that process parameters based on Pareto optimal solutions reduced residual stress by approximately 1.88% and weld deformation by over 20%, effectively ensuring module assembly tolerances. The Pareto frontier provided visual decision support for process engineers. For instance, in battery casing welding, engineers can use the frontier curve to clearly define the trade-off boundaries among strength, toughness, and softened zone width [193]. Combined with weighted methods or expected hypervolume improvement (EHVI) strategies [194], it enables rapid screening of final production process parameters from solution sets based on specific vehicle safety standards, such as prioritizing deformation or contact resistance.

(iii) Active learning and Bayesian optimization: Although surrogate models can replace physical experiments, training high-quality surrogate models still requires a certain amount of labeled experimental data. For objects like lithium batteries where destructive testing such as tensile peel tests and metallographic sectioning is costly and materials are expensive, achieving rapid optimization with minimal samples in a small data regime has become a research hotspot. Bayesian optimization is a framework for the global optimization of black-box functions, particularly suited for complex objective functions lacking explicit expressions. The method is often combined with active learning strategies using acquisition functions to balance exploitation and exploration. In GPR-based surrogate models, the variance of predicted values quantifies uncertainty. Higher variance indicates less model knowledge in that region and greater potential information gain, guiding the algorithm to prioritize sampling in unknown areas to avoid local optima. Active learning and Bayesian optimization represent the state-of-the-art integrated framework for experiment design, modeling, and optimization [195,196]. The core philosophy shifts from passively fitting existing data to enabling algorithms to actively query the design space. It is achieved using acquisition functions that balance exploration by sampling in unknown regions and exploitation by refining known high-performance areas. By leveraging variance information provided by probabilistic models such as GPR, the algorithm automatically identifies regions of high uncertainty and recommends the next physical experimental parameter yielding the greatest information gain.

Consequently, the approach enables the approximation of the global optimum with minimal experimental iterations.

Haas et al. [197] developed a comprehensive cost function for complex welding quality control. Using a Bayesian optimizer, the research team identified the optimal process window satisfying the objectives within a vast parameter space after only 29 physical experiments, thereby achieving several-fold efficiency gains over traditional orthogonal experimental design. Beyond process parameter optimization, Zhang et al. [198] applied Bayesian optimization methods to the hyperparameter tuning of deep learning models including CNN, addressing the inefficiency of manual parameter adjustment and indirectly enhancing prediction accuracy for molten pool dynamics. The closed-loop strategy characterized by simultaneous experimentation, learning, and optimization holds revolutionary significance for developing laser welding processes for novel battery materials such as coated current collectors or complex joint configurations.

Emerging research is integrating Bayesian optimization with automated welding workstations to establish closed-loop automated optimization systems. In the framework, algorithms automatically generate the next set of parameters based on current experimental results to directly control the laser welding process, followed by automatic feedback acquisition via sensors. The autonomous approach minimizing human intervention represents the ultimate direction for intelligent and adaptive production on future EV assembly lines.

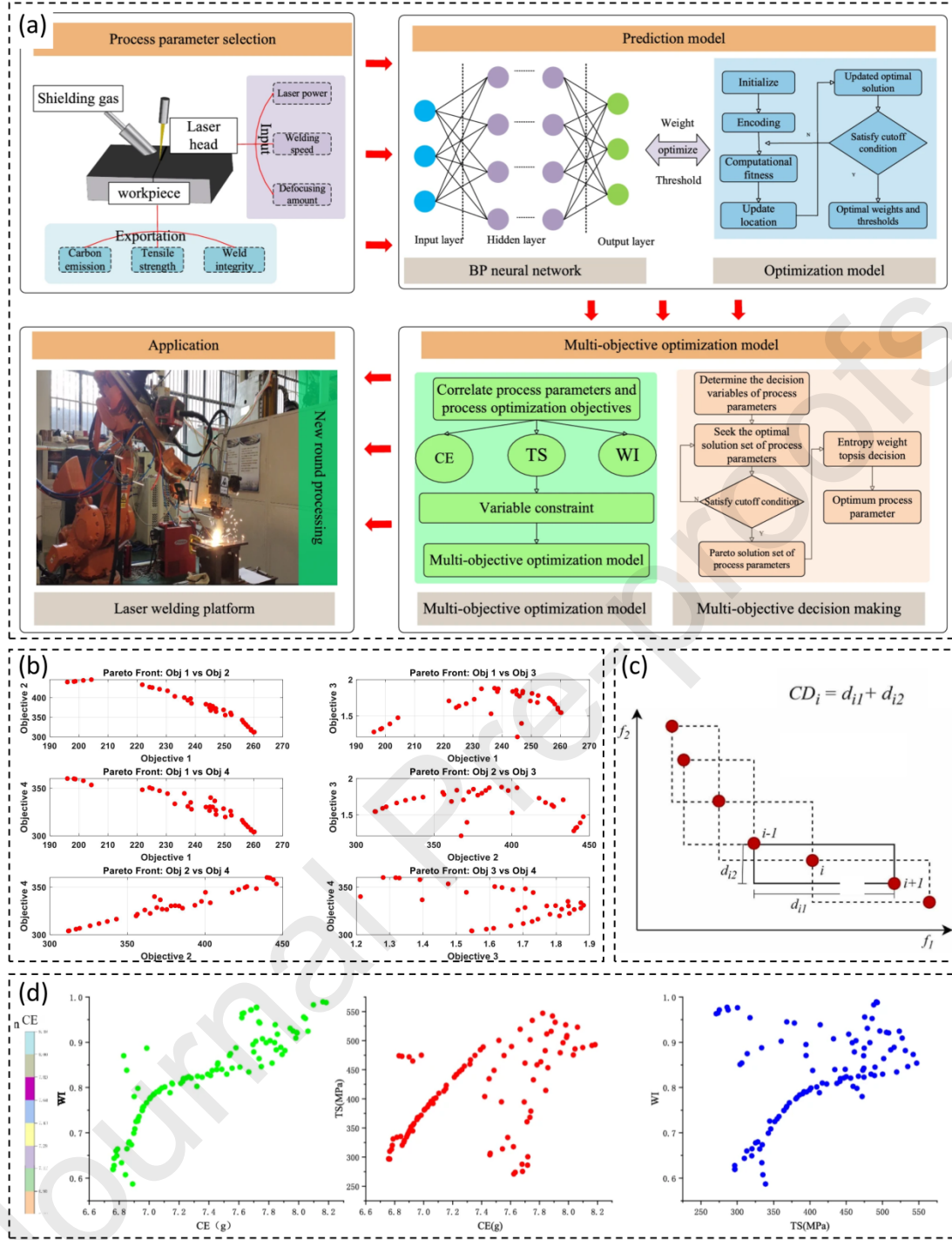


Fig. 10. Schematic representations of machine learning-based process parameter optimization frameworks and multi-objective evolutionary algorithms: (a) schematic diagram of the process parameter optimization procedure [184]; (b) Pareto frontier for four objectives [191]; (c) crowding distance for two objectives [191]; (d) set of Pareto optimal solutions [184].

3.3. Real-time adaptive closed-loop control

With the widespread adoption of laser welding in high-end manufacturing, demands for consistency, stability, and predictability of quality in the welding process continue to rise. However, the laser welding process itself exhibits significant multiphysics strong coupling involving thermal, fluid, phase transition, and plasma dynamics, alongside highly nonlinear, time-varying, and disturbance-sensitive characteristics. This makes it challenging to address uncertainties in actual operating conditions, such as joint assembly deviations, thermal accumulation effects, fixture instability, material variations, and external disturbances, by relying solely on offline parameter settings or open-loop control. Consequently, online adaptive closed-loop welding control integrating multi-source sensing, machine learning, and control theory has emerged as a core research direction in intelligent welding. The essence of adaptive welding lies in the transition from passive monitoring to active intervention, utilizing real-time sensing data to perceive welding conditions and dynamically adjusting process parameters via control algorithms to maintain consistent weld quality. In recent years, the introduction of machine learning and deep learning technologies has provided new pathways for establishing end-to-end intelligent closed-loop control systems, significantly enhancing the adaptability of the system to nonlinear and time-varying welding processes.

From a technological evolution perspective, existing research has generally progressed from rule-based control and classical feedback control to machine learning-assisted control, gradually advancing toward reinforcement learning-driven intelligent decision-making closed-loop systems. The following provides a systematic review at two levels: traditional machine learning-driven online closed-loop control methods and emerging reinforcement learning-based adaptive welding strategies.

3.3.1 Supervised learning for online control

Traditional adaptive control typically relies on simple linear models or empirical rules, whereas the incorporation of machine learning techniques endows controllers with enhanced feature extraction and state prediction capabilities.

(i) Supervised learning-assisted feedback control: Supervised learning represents the most mature and engineering-feasible approach in current online closed-loop welding control systems. Such methods typically employ a perception-

prediction-regulation architecture. First, supervised learning models such as CNN and MLP extract key geometric features including weld width and penetration depth from molten pool images or sensor signals. Subsequently, the deviation between predicted values and target setpoints is fed into traditional controllers such as PID or optimization algorithms for parameter adjustment. The schematic diagrams of the MLP and CNN architectures are shown in **Fig. 11** (a), while **Fig. 11** (d) illustrates an example of hybrid machine learning incorporating MLP and CNN applied to adaptive welding speed adjustment.

In laser welding scenarios, extensive research employs optical signals, electrical signals, acoustic signals, and visual information for real-time monitoring of welding conditions. Regression or classification models are then utilized to identify and predict welding states. For instance, online estimation of molten pool width, keyhole morphology, and plasma radiation intensity via CNNs or regression models effectively reflects welding energy coupling states and process stability. Kershaw et al. [199] proposed a real-time machine learning framework for gas tungsten arc welding (GTAW) robots. It utilizes CNN to extract melt width features from molten pool images, while integrating MLP to establish the mapping relationship between welding speed and melt width, enabling precise estimation of the back-side melt width. Cai et al. [200] proposed a penetration depth assessment method based on U-Net and CNN approaches. Experimental validation demonstrated strong robustness and generalization capabilities, achieving prediction accuracy up to 98.68%.

These predictive results are subsequently used as feedback signals, integrated with gradient descent methods, PID control logic, gain scheduling control, or robust controllers to achieve real-time adjustment of laser power, welding speed, or focal position. Bollig et al. [201] developed a predictive control scheme that uses neural networks to forecast the penetration depth in laser welding and dynamically adjusts laser power based on the predictions to maintain penetration depth consistency.

Typical control objectives for the method encompass the stable regulation of molten pool geometry, with a primary focus on pool width, length, and fluctuation amplitude. Concurrently, the approach ensures consistent weld bead formation, effectively suppressing uneven welds caused by gap variations or thermal input fluctuations. Furthermore, by stabilizing keyhole formation, it fundamentally reduces critical defects such as porosity and collapse.

The advantages of supervised learning-assisted feedback control lie in its clear implementation path, strong interpretability of control structures, and ease of integration into existing industrial control systems. However, its performance is highly dependent on the coverage of training data. When operating conditions deviate significantly from the distribution of training samples, the model's generalization capability remains markedly insufficient.

(ii) Learning model-driven model predictive control (MPC): To further enhance control capabilities over the multivariable coupling characteristics of welding processes, researchers have begun integrating machine learning models into the MPC framework, forming a synergistic strategy of learning models combined with optimization control. In the approach, machine learning models such as GPR, deep neural networks (DNNs), or ensemble learning models serve as surrogate models for the welding process. These models approximate the dynamic relationship between welding input parameters and key state variables, such as molten pool size, temperature distribution, or defect probability. Based on the predictive model, MPC solves a constrained optimization problem within each control cycle to achieve coordinated regulation of multiple input variables such as laser power and scanning speed. Zhang et al. [202] proposed an adaptive filler model based on BPNN. Inputting assembly gaps and misalignment quantities, it employs GA inverse optimization to output optimal combinations of laser power, wire feed rate, and welding speed in real time, addressing complex joint geometric variations. Zhang et al. [188] introduced a novel ensemble learning model, RFPCA-XGBoost, for predicting the strength of laser-welded joints in aluminum-lithium alloys used in rocket tanks. Comparisons with other feature selection methods, tree-based ensemble learning models, and grid search parameter optimization techniques demonstrated that the proposed model achieved the highest R^2 value of 0.9383.

Compared to traditional feedback control, MPC driven by learning models demonstrates significant advantages in multiple aspects. The approach not only explicitly handles multivariable coupling and constraints such as power limits and energy input smoothness, but also enables proactive regulation within a specific prediction time horizon, thereby achieving early suppression of process instability. Furthermore, it provides a unified control framework for complex welding tasks like variable-gap welding or joining dissimilar materials, effectively enhancing system adaptability. However, such methods impose stringent demands on model accuracy

and computational efficiency, leading to persistent trade-offs between real-time performance and stability in high-frequency online control scenarios.

(iii) Intelligent control and hybrid methods: Beyond supervised learning and MPC, the research extensively explores various intelligent control and hybrid optimization methods. To enhance control system robustness, researchers have integrated intelligent algorithms such as fuzzy logic and neural inference systems with machine learning. Fuzzy control and neural network control address the limitations of traditional linear control in complex welding conditions by introducing nonlinear mapping capabilities. Neural-fuzzy systems, meanwhile, strike a balance between rule-based interpretability and self-learning capabilities. In the field of resistance spot welding (RSW), neural networks are frequently combined with fuzzy inference systems (ANFIS). By monitoring electrode displacement and velocity signals, ANFIS controllers can evaluate the growth state of the weld nugget in real time and adjust the welding current, effectively suppressing spatter and ensuring nugget size. Zaharuddin et al. [169] employed ANFIS to predict the weld strength of CR780 high-strength steel plate spot welds, comparing results with ANN. Findings indicated superior predictive performance for ANFIS. Safari et al. [203] developed an adaptive neuro-fuzzy inference system (ANFIS) to model and estimate the tensile shear strength (TSS) of dissimilar RSW joints. The optimal parameters for the ANFIS system were obtained using a teaching-learning-based optimization (TLBO) algorithm. Experimental results demonstrated that the ANFIS network achieved excellent predictive performance for TSS.

Additionally, methods such as Bayesian optimization, genetic algorithms, or particle swarm optimization are frequently employed for online or semi-online control parameter self-tuning. For instance, Bayesian optimization can search for optimal laser power adjustment strategies within a limited number of trials, thereby reducing trial-and-error costs while ensuring welding quality. To address the limited generalization capability of single models, some studies have introduced online updating mechanisms. These enable control models to continuously fine-tune their weights during the welding process, adapting to environmental drifts caused by electrode wear or thermal accumulation in resistance spot welding. Wu et al. [204] proposed a novel dynamic model based on online sequential extreme learning machines (OS-ELM) to predict penetration depth by measuring the width of the weld bead on the back side in real time. Extensive experimental results confirmed that the

proposed dynamic penetration method significantly outperformed other comparison algorithms in terms of prediction accuracy, efficiency, and robustness.

Overall, traditional machine learning-driven online closed-loop control methods have made significant progress in engineering applications. However, their control logic remains largely dependent on human-defined objectives and structures, and their autonomous decision-making capabilities in complex dynamic environments remain limited.

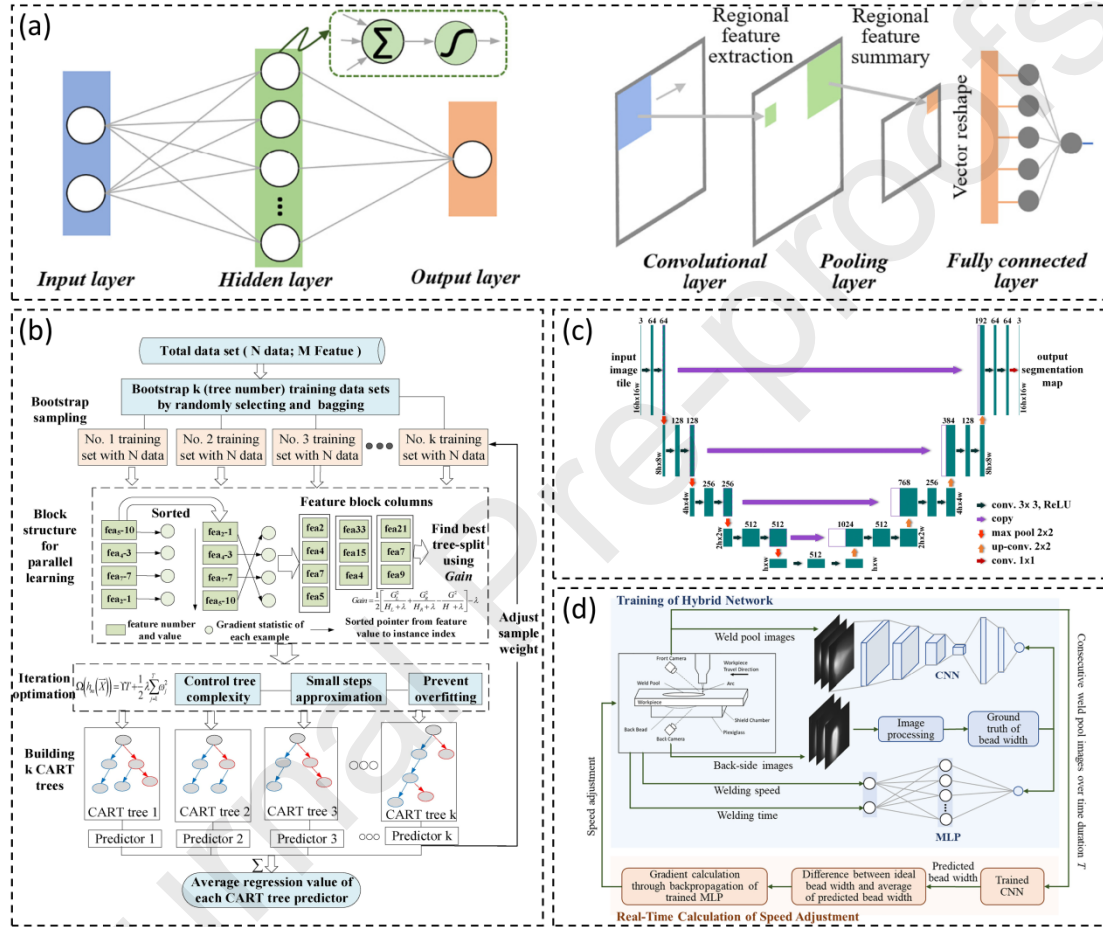


Fig. 11. Architectures of representative machine learning models and their application: (a) illustration of MLP and CNN architectures [199]; (b) schematic diagram of the XGBoost regression model [188]; (c) Detailed structure of the U-Net model [200]; (d) illustration of hybrid MACHINE LEARNING for adaptive welding speed adjustment, including training of hybrid CNN and MLP network as well as real-time calculation of speed adjustment [199].

Furthermore, a critical time-scale mismatch often remains unaddressed in the current literature regarding these closed-loop systems. In high-speed battery manufacturing, laser welding typically operates at speeds between 200 and 500 mm/s [193]. However, sophisticated deep learning models, such as deep CNNs or

Transformers, often exhibit inference latencies ranging from 20 to 50 ms on standard industrial controllers. To put this into perspective, at a welding speed of 200 mm/s, a 50 ms latency results in a 10 mm spatial offset. This displacement far exceeds the dimensions of typical battery tabs or the localized melt pool itself. It implies that by the time an algorithm completes its inference and prescribes a corrective action, the laser beam has already moved significantly past the detected anomaly, rendering the 'real-time' claim physically untenable for capturing microsecond-level transients.

Through a systematic real-time latency audit of the core literature cited in this paper regarding online control or closed-loop regulation, it indicates that this engineering-level time discrepancy is prevalent in current research yet has not been adequately addressed. In these publications, approximately 50% of the studies [195–197,199] emphasized only the models' high classification accuracy in validation, yet completely lacked quantitative reports on end-to-end inference latency. In studies that did report latency [193,194,198,200–202] models based on Transformer or deep CNN architectures generally had a single-frame processing time in the range of 20–100 ms. For microsecond-scale transient physical processes such as battery tab formation, this implies that by the time the controller issues an adjustment command, the laser head has already traversed a 10-mm control blind zone. Furthermore, even in studies that attempt model pruning or lightweight processing [189, 195], while inference speed has been improved to the millisecond range, they still fail to account for the cumulative latency introduced by industrial communication buses like EtherCAT and servo mechanism responses. This widespread neglect of control timing rigor means that the effectiveness of most current intelligent closed-loop models in high-speed industrial welding scenarios still faces significant theoretical limitations. Consequently, achieving true deterministic closed-loop control necessitates a paradigm shift from heavy, high-latency architectures toward model lightweighting techniques such as pruning and quantization, Edge AI deployment, and hardware-level acceleration via FPGAs or ASICs to bridge the gap between algorithmic processing and high-speed welding physics.

3.3.2 Reinforcement learning for autonomous control

To overcome the limitations of traditional model-based control, data-driven model-free control has gradually emerged as a research hotspot. Reinforcement learning (RL), a significant branch of machine learning, leverages its unique trial-and-

error learning mechanism and robust exploration capabilities in unknown environments. The approach offers a novel solution to the high uncertainty inherent in welding processes, shifting the paradigm from predict-and-adjust to decide-and-learn. The classification of classical reinforcement learning methods is shown in **Fig. 12 (b)**.

Unlike supervised learning, which relies on large amounts of labeled data, RL does not depend on fixed “input-output” labeled data. Instead, it treats the welding process as a sequential decision-making problem. The reinforcement learning agent continuously interacts with the physical welding environment and optimizes its control policy based on feedback reward signals. In welding modeling, the welding system can be abstracted as a MDP, where the welding state evolves over time, and control actions directly influence subsequent states and welding quality feedback. The objective of reinforcement learning is not to minimize single-step error but to maximize long-term cumulative reward. This makes it inherently suited for handling dynamic uncertainties and delay effects in welding where the agent can autonomously learn the optimal strategy for maintaining stable weld quality by continuously adjusting process parameters and observing outcomes, such as weld bead formation and arc sound changes, without requiring prior knowledge of complex molten pool physics. Consequently, reinforcement learning demonstrates superior autonomous decision-making capabilities in process control and is recognized as a key technology for realizing next-generation intelligent welding systems.

In recent years, the integration of deep learning, with reinforcement learning, known as DRL, has significantly expanded its application boundaries. DNNs empower agents to process high-dimensional perceptual data including molten pool video streams, spectrograms, and thermal imaging, enabling end-to-end intelligent welding control. From single-robot weld seam tracking and parameter fine-tuning to collaborative operations and obstacle avoidance in multi-robot systems, DRL is progressively reshaping the technological framework of welding automation. **Fig. 12 (c)** and **Fig. 12 (d)** illustrate two distinct intelligent welding sequence optimization (WSO) frameworks.

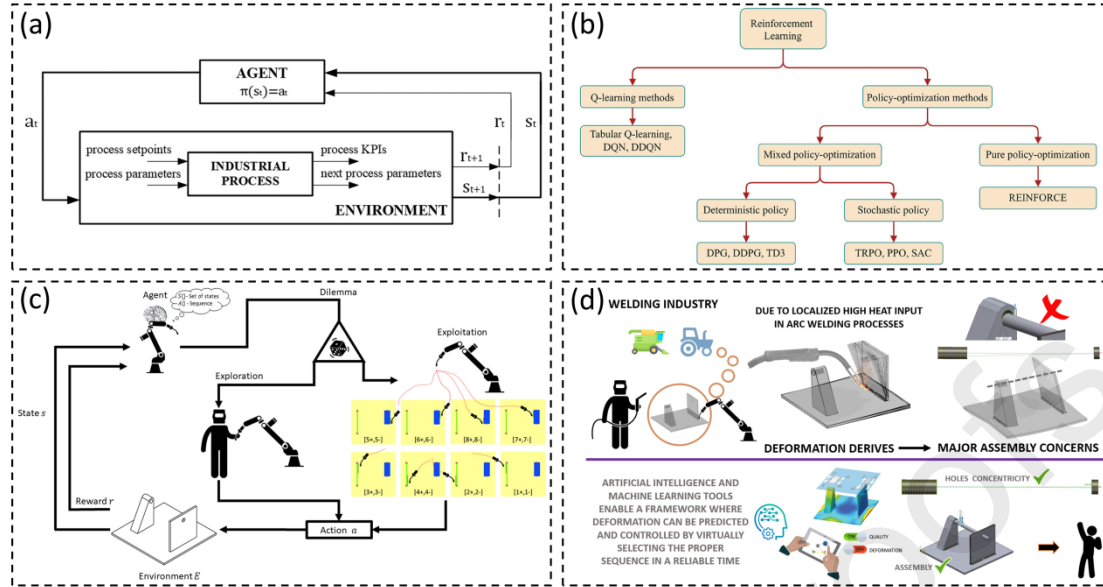


Fig. 12. Schematic framework of reinforcement learning algorithms and their application in welding sequence optimization: (a) schematic overview of reinforcement learning in industrial processes [205] (b) classification of classical reinforcement learning methods [205] (c) domain knowledge framework for reinforcement learning based on WSO [206] (d) intelligent welding sequence optimization (WSO) framework [206].

Transforming the continuous, dynamic physical process of welding into a mathematical problem that reinforcement learning algorithms can handle hinges on formalizing it as a MDP. A schematic diagram of the Markov decision process is shown in **Fig. 13** (a). The process is typically described by the quintuple $\langle S, A, P, R, \gamma \rangle$, where S represents the state space, A represents the action space, P denotes the state transition probability, R is the reward function, and γ is the discount factor. In welding applications, the precise definition of these five elements directly determines the algorithm's convergence speed and ultimate control effectiveness, forming the foundational framework for the agent's understanding and interaction with the environment.

The state space S serves as the sole window through which the intelligent agent perceives the welding environment. Given the welding process's pronounced black box characteristics, coupled with intense arc light, spatter, high temperatures, and electromagnetic interference, constructing a high-dimensional and robust state vector is crucial. Existing research typically employs multi-sensor fusion strategies, with visual information playing a dominant role. For instance, the agent must capture the geometric characteristics of the molten pool, including width, length, area, and keyhole features in laser deep penetration welding, using cameras equipped with

narrow-band filters. These visual features directly correlate with quality metrics such as penetration depth. Gao et al. [106] designed an active vision system to capture morphological information of the molten pool during high-power disk laser welding of Type 304 austenitic stainless steel plates. A diode laser was employed to illuminate the molten pool and generate its shadow. A high-speed camera equipped with a narrow-bandpass filter recorded the morphology of the molten pool and its shadow during laser welding. An algorithm was developed to segment the molten pool shadow, quantitatively characterizing the relationship between molten pool morphology and the stability of the high-power disk laser welding process. Guo et al. [207] employed a CCD camera with a narrow-band filter to capture clear images of the melt hole top surface during single-sided welding with double-sided formation. An image processing algorithm based on image features was developed to extract the edges of the melt hole and gap. Results indicated that the ratio of melt hole width to gap width increases with rising welding current. Nguyen et al. [208] developed a vision system for industrial laser welding robotic arms, comprising a laser projector and industrial camera. Based on a discriminative convolutional tracker and deep neural networks, it achieves three-dimensional object localization. Reinforcement learning was employed to enhance the tracker's accuracy and processing speed. Experimental results demonstrate the developed system's significant performance in ensuring tracking precision and welding quality.

In addition to visual information, to compensate for detail loss under intense arc light, the state space typically incorporates acoustic and electrical parameter signals. These include the arc acoustic frequency band energy ratio processed via fast Fourier transform (FFT), as well as real-time waveform statistical characteristics of welding current and arc voltage. Gokmen et al. [209] proposed a method utilizing acoustic techniques, FFT, and ANN to identify welding machine types. Experimental results demonstrated that ANN could effectively classify welding machine types based on recorded sounds after training, validation, and testing phases. Sheng et al. [210] proposed an adaptive signal denoising method for welding defect detection to overcome challenges such as multimode mixing and low signal-to-noise ratio (SNR) in laser-generated guided wave (LGGW) signals. First, the LGGW signal undergoes synchronous squeezing wavelet transform (SSWT). followed by extraction of the A0 guide wave using a two-dimensional Hann window. Subsequently, a PSO algorithm is applied to adaptively select window parameters. Finally, a scanning imaging method

termed PSO-SSWT-FFT is proposed to generate images by exploiting the difference in guide wave energy between defective and non-defective regions. Experimental results demonstrate that PSO-SSWT significantly outperforms traditional methods in extracting the A0 guide wave under high-noise conditions. Masinelli et al. [211] designed a deep CNN-based encoder to compress real-time optical and acoustic signals into low-dimensional state vectors as inputs for RL agents. Furthermore, system state is an indispensable component, particularly in multi-robot collaborative scenarios. The state vector s_t must incorporate kinematic information such as end-effector position $p_i(t)$, velocity $v_i(t)$, and the relative state of neighboring robots to ensure the safety and coherence of collaborative operations.

The action space A defines the specific means by which the agent controls the welding process. Its design depends on the requirements of the control hierarchy, encompassing multiple dimensions ranging from low-level process parameter adjustments to high-level path planning. In most adaptive welding tasks, the action space is designed as continuous, requiring the agent to output specific numerical values to finely adjust key variables such as welding speed, wire feed speed, and arc voltage. For instance, in laser welding control, the action variable might be the normalized laser power or its corresponding analog voltage signal. In tasks involving robotic motion control, actions manifest as joint angular velocities or the velocity vector of the end-effector in Cartesian space. While the continuous action space design increases training complexity, it avoids the control accuracy loss associated with discretization, meeting the stringent requirements for smoothness and responsiveness in precision welding. Cai et al. [212] proposed a control method for the joint of a deep-sea manned spherical shell. Specifically, they designed a control scheme based on dynamic equilibrium, employing a TD3 algorithm with a dual evaluator structure to enhance control accuracy within predefined state and action spaces. A composite reward mechanism was introduced to mitigate potential motor instability, and the CP-MPA algorithm was utilized to optimize the performance of the proposed m-TD3 composite controller. Simulation results validated the effectiveness and superiority of the proposed optimization strategy, demonstrating its applicability and potential for achieving precise robust control in complex nonlinear defect detection systems. Yu et al. [213] proposed a Dynamic Kuhn-Munkres Model-Free Policy Gradient (DKM-MFPG) method to achieve the dual objectives of optimal inertia and energy consumption for robots. They simulated DKM-MFPG performance

under various configurations. Results show that across various welding scenarios, DKM-MFPG exhibits a slight advantage in energy consumption (within 1%) compared to four baseline methods, while reducing the average inert robot ratio by 11.30%, 10.99%, 8.27%, and 10.39%, respectively. Günther et al. [214] proposed an integrated intelligent machine architecture that addresses complex control challenges in laser welding by fusing deep representations, real-time prediction, and autonomous control. The study applied autoencoder neural networks to laser welding data, enabling feature extraction from high-dimensional sensor information. Building upon the foundation, the researchers employed temporal-difference learning to construct general value functions (GVFs), achieving real-time prediction of weld seam quality. Finally, by integrating an actor-critic reinforcement learning algorithm, they realized closed-loop autonomous regulation of welding power.

The reward function R serves as the guiding principle in reinforcement learning, and its design quality directly determines whether the agent can learn the desired behavior. The reward function and reward values during training are shown in **Fig. 13** (c). It is particularly critical in welding, a typical multi-objective optimization problem. A mature reward function is typically composed of weighted combinations of multiple sub-components, such as quality-driven metrics, task efficiency, and safety constraints. Quality rewards are often constructed based on the deviation between actual measured values and target values. For example, when the target penetration depth is D^* and the actual penetration depth is D_t , the reward term can be set as [43,211,214]:

$$r_{quality} = -|D_t - D^*| \quad (9)$$

Or [43,211,214]:

$$r_{quality} = \exp\left(-\frac{(D_t - D^*)^2}{2\sigma^2}\right) \quad (10)$$

It guides the agent to minimize error. In complex scenarios such as multi-robot collaboration, reward function design becomes more refined, typically expressed as [43,215]:

$$R_{total} = k_1 r_{iw}(t) + k_2 r_{id}(t) + k_3 r_{ic}(t) \quad (11)$$

Where $r_{iw}(t)$ encourages the robot to maintain the welding state, $r_{id}(t)$ guides idle robots to move toward task points to enhance efficiency, while $r_{ic}(t)$ serves as a critical collision penalty term. When the distance between a robot and an obstacle or another robot falls below the safety threshold D_{safe} , this term imposes a

substantial negative reward, thereby compelling the agent to learn safe obstacle avoidance strategies. Through the fine-tuned adjustment of multi-objective weights, reinforcement learning algorithms can simultaneously ensure welding quality while balancing production efficiency and system safety. For instance, the Q-learning controller proposed by Quang et al. [216] calculates rewards using quality feedback signals, guiding agents to autonomously explore power strategies that maintain target quality.

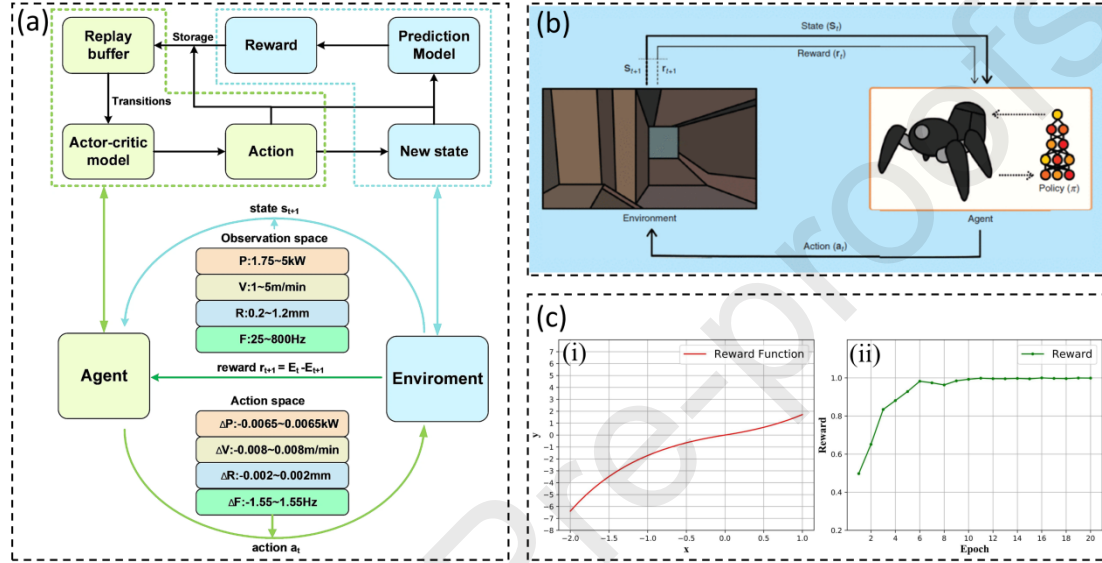


Fig. 13. Framework and application of reinforcement learning for process parameter optimization: (a) Markov decision processes [217]; (b) perception-action-learning cycle [215] (c) (i) reward function (ii) rewards during training [123].

With the integration of deep learning and reinforcement learning, laser welding control strategies have evolved from simple rule-based logic to high-dimensional state awareness and continuous action decision-making. Current research primarily focuses on two core dimensions: adaptive control of penetration depth and morphology, and complex trajectory weld seam tracking.

In adaptive control of penetration depth and morphology, the stability of the penetration state directly determines the mechanical properties and sealing integrity of the welded joint, serving as a core indicator for laser welding quality control. However, the laser deep penetration welding process involves intense thermal-fluid-solid coupling, accompanied by significant heat accumulation effects and time-varying characteristics. Traditional PID controllers often rely on linearized system models, making them prone to control lag or overshoot when encountering sudden operational changes including focal drift caused by thermal lensing effects or

variations in plate gaps. In contrast, RL demonstrates exceptional potential for fitting and adaptive control of nonlinear, strongly coupled systems through its end-to-end perception-decision learning capability.

During the early exploratory phase of intelligent welding control, research predominantly employed value-based algorithms such as Q-Learning or DQN (Deep Q-Network). Such studies typically constructed closed-loop systems based on coaxial vision, extracting geometric features like melt pool width and length as the state space. Discrete action commands such as adjusting laser power to +100 W, -100 W, or maintaining it unchanged were employed to maintain a constant melt pool morphology. Quang et al. [216] proposed a Q-learning-based controller that interacts with the laser based on feedback signals from the sensing system, adjusting processing parameters to achieve or maintain target weld quality. The controller's performance was validated in both controlled laboratory environments and more unpredictable industrial settings. Alzorgan et al. [218] proposed a motion planning model for aerial robotic arm systems, employing a Q-learning-based method to control end-effector trajectories. It enables the robotic arm to navigate around obstacles in operational environments, making aerial welding feasible. Romero-Hdz et al. [206,219] employed finite element simulations to compute welding deformation as reward signals, training a Q-learning agent to optimize welding sequences. They accelerated convergence by incorporating domain knowledge.

Although the feasibility of RL has been demonstrated, the discrete action space inherently resembles bang-bang control [220–222], inevitably causing high-frequency oscillations in process parameters around target values. Such fluctuations in power input not only affect the smoothness of weld bead appearance but may also lead to inconsistencies in microstructure thermal history, making it difficult to meet the stringent requirements for high-precision welding in fields such as aerospace.

To overcome the limitations of discrete control, actor-critic architecture algorithms based on policy gradients, particularly deep deterministic policy gradient (DDPG) and soft actor-critic (SAC), have increasingly dominated research in the field in recent years. These algorithms can directly output smooth control signals within high-dimensional continuous spaces, perfectly aligning with the continuous adjustability of physical parameters such as laser power, welding speed, and wire feed rate. The structure of the actor-critic network is illustrated in **Fig. 14** (a). Gao et al. [223] developed a multi-objective optimization framework based on reinforcement

learning, utilizing DDPG to generate efficient grinding strategies for grinding rail welded joints. The contour groove recovery rate improved from -0.41 mm to -0.18 mm. The framework of the DDPG method is shown in **Fig. 14 (c)**. Jin et al. [43] proposed an intelligent weld control strategy to regulate weld pool width based on the actor-critic reinforcement learning (ACRL) method. They selected gas tungsten arc welding (GTAW) and gas metal arc welding (GMAW) models for simulation experiments on welding process control, validating the controller's feasibility. Addressing the challenge that traditional genetic algorithms struggle to handle dynamic production environments in multi-robot path planning, He et al. [224] proposed an innovative solution based on reinforced pointer networks (RPNs). The study employs an Actor-Critic model to construct a path planning network targeting minimization of the maximum completion time, known as makespan, while integrating K-means clustering with genetic algorithms to optimize task allocation. Compared to conventional two-layer chromosome encoding methods, the approach demonstrates faster convergence and higher search efficiency, enabling autonomous dynamic planning under abnormal operating conditions. Addressing reward feedback acquisition delays in laser oscillation welding parameter optimization, Tian et al. [217] proposed a deep predictive reward reinforcement learning network (DPRRL-net). By constructing a differential evolution-optimized backpropagation neural network (DE-BPNN) as a predictive environment, the study achieved low-error predictions (relative error <2.7%) for weld porosity and penetration depth. The reward function constructed using the prediction model, combined with the double-delay deep deterministic policy gradient (TD3) algorithm, effectively avoids local optima and significantly enhances the efficiency and stability of parameter optimization. The structure of TD3 is shown in **Fig. 14 (b)**.

In the field of intelligent weld seam tracking under complex operating conditions, achieving precise alignment between the laser focus and the weld seam trajectory within unstructured and dynamically changing industrial environments presents another major challenge in ensuring welding quality. Traditional weld tracking technologies typically adopt a layered perception-planning-control architecture, heavily reliant on rule-based image processing algorithms. However, such approaches require complex manual feature engineering and are highly sensitive to intense arc light, spatter noise, and high reflectivity on workpiece surfaces. In contrast, RL offers a novel solution to complex path tracking problems by learning

through trial-and-error interactions with the environment. To enhance the robustness of weld feature extraction during weld tracking under intense arc light and spatter interference, Zou et al. [123] proposed a two-stage localization method combining convolution filtering with deep reinforcement learning (CF-DRL). The approach first employs convolutional filters for coarse target tracking, followed by a policy network to perform fine-grained corrections on feature point coordinates, effectively bridging the gap between general visual tracking and precise weld seam localization. Experimental results demonstrate an average tracking error of only 0.189 mm under complex operating conditions, significantly enhancing the visual system's interference resistance.

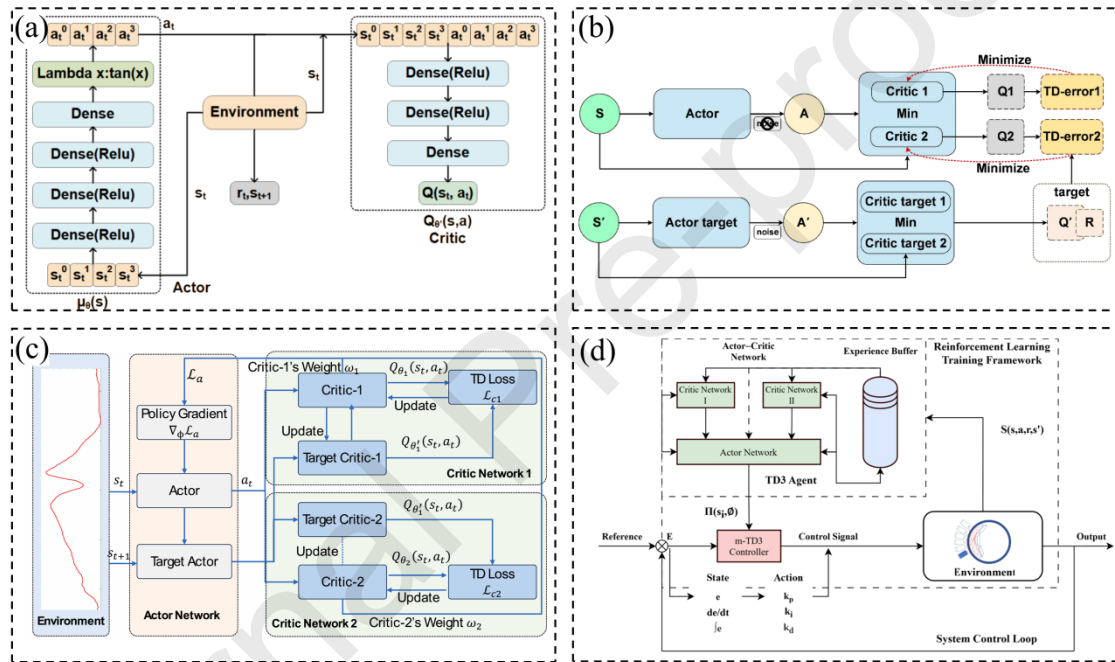


Fig. 14. Schematic diagrams of typical actor-critic reinforcement learning architectures: (a) Structure of the actor-critic network [217]; (b) Structure of TD3 [217]; (c) framework of the DDPG method [223]; (d) defect detection control framework based on TD3 [212].

Table 3. Key characteristics comparison of common reinforcement learning algorithms in welding applications.

Algorithm name	Algorithm type	Action space	Sample efficiency	Hyperparameter sensitivity	Complexity of implementation	Stability	Typical application scenarios for battery welding
Q-Learning [206,219]	Value-based	Discrete	Low (requires significant interaction)	Low (fewer parameters, easier to debug)	Extremely low	Medium	Simple laser power level adjustment; rule-based simple logic substitution
DQN / Double DQN [225]	Value-based	Discrete	Medium (introducing experience replay)	Medium	Medium	Medium	Rough state classification based on molten pool images (e.g., switching between deep penetration and heat conduction modes); preliminary processing of high-dimensional visual inputs
Deep deterministic policy gradient (DDPG) [213,217]	Actor-critic	Continuous	High (reusing historical data)	Extremely high (extremely sensitive to noise and learning rate, prone to divergence)	Medium	Low	Simple single-robot continuous trajectory control; process parameter optimization in laboratory environments
Twin delayed DDPG (TD3) [212,217]	Actor-critic	Continuous	High	Medium (sensitivity was mitigated through delayed updates.)	Medium-high	High	Precision tracking of complex geometric welds; multi-parameter (power, speed, wire feed) coordinated adjustment; defect detection control
Soft actor-critic (SAC) [224,225]	Actor-critic	Continuous	Extremely high (maximum entropy exploration)	Low (demonstrates remarkable robustness to hyperparameters)	High	Extremely high	Robust control in high-noise, strong disturbance environments; the preferred algorithm for sim-to-real transfer; stochastic exploration of welding processes
Proximal policy optimization (PPO) [226]	Strategy gradient	Continuous / discrete	Low (requires real-time data, non-reusable)	Low (easy to optimize)	Medium	Extremely high	Strategy validation phase; experimental welding with stringent safety requirements that preclude aggressive exploration; benchmarking in simulation environments.
MADDPG [227]	Multi-agent	Continuous	Medium	High	Extremely high	Medium	Multi-robot collaborative welding (e.g., dual-head butt welding); dynamic obstacle avoidance and task allocation coordination
Conservative Q-Learning (CQL) / BCQ [228,229]	Offline RL	Continuous / discrete	Extremely high (requires historical data only)	Medium	High	High	Policy learning based on historical welding data (PID control logs); risk-free “cold start” policy training

Reinforcement learning is profoundly reshaping the technological landscape of welding automation, propelling welding robots from simple trajectory-following actuators to intelligent agents capable of deep perception, autonomous decision-making, and adaptive behavior. By modeling the nonlinear, strongly coupled physical processes of welding as MDPs and integrating cutting-edge algorithms like DDPG, TD3, SAC, and MADDPG with multimodal deep perception mechanisms, the technology demonstrates performance far surpassing traditional control theories in critical areas such as high-precision weld seam tracking, dynamic adaptive process parameter tuning, real-time defect suppression, and multi-robot collaborative operations. Concurrently, the maturing digital twin and sim-to-real transfer technologies effectively overcome reinforcement learning's inherent limitations, specifically inefficient sample utilization and insufficient training safety in physical environments, establishing a virtual training and real-world execution implementation paradigm. Although RL has demonstrated immense potential for advancing toward fully autonomous welding in fields like laser welding, achieving large-scale industrial deployment still requires overcoming three-dimensional challenges including data and feedback mechanisms, the safety and real-time performance of physical systems, and transfer bias.

Although reinforcement learning offers a highly promising technical pathway for adaptive welding process control, several fundamental technical obstacles must still be overcome in the process of advancing it from theoretical models to industrial-scale applications.

The primary challenge lies in the conflict between the low sampling efficiency of algorithms and the high cost of trial-and-error in industrial settings. Existing model-free algorithms typically require tens of thousands or even millions of state interactions to converge. Training them directly from scratch on physical welding systems would result in unbearable material consumption, equipment wear, and time costs. More intractable is the sparsity and lag in reward functions where final quality metrics for welds, including internal porosity and tensile strength, often depend on post-weld destructive testing or CT scans, preventing agents from receiving real-time quality feedback during welding. Although indirect inference methods based on molten pool surface features currently exist, the semantic gap between surface characteristics and internal metallurgical quality severely limits the precision and reliability of reinforcement learning strategies.

Beyond data-level challenges, the safety and real-time constraints of physical systems form the second barrier to RL implementation. The core mechanism of reinforcement learning relies on optimizing strategies through the exploration of unknown actions. However, in high-energy beam processing such as laser welding, random exploratory actions involving extreme power or speed settings can easily trigger catastrophic consequences, including workpiece excessive penetration, optical component damage, or even structural damage to the robot. It makes algorithms lacking safety constraints difficult to apply directly. Simultaneously, the transient physical changes during welding demand control loop frequencies typically reaching 100 Hz to 1 kHz. Complex deep reinforcement learning models require substantial inference computation, making it difficult for standard CPUs to meet millisecond-level real-time response requirements. Deploying high-performance GPUs in industrial settings faces dual constraints of cost control and environmental adaptability.

Additionally, the simulation-to-reality (Sim-to-Real) transfer bias constitutes the third major bottleneck hindering the practical implementation of RL technology. Although digital twin technology has been introduced to mitigate the tension between sample scarcity and exploration risks, the reality gap remains significantly pronounced. Laser welding involves extremely nonlinear and transient thermo-fluid-solid coupled physical processes. Existing simulation models struggle to accurately reproduce complex features such as melt pool fluid dynamics, spatter generation, and micro-metallurgical phase transformations while maintaining computational real-time performance. The mismatch between models and physical entities leads to strategies trained to converge in virtual environments often suffering severe performance degradation or even failure when transferred to real physical systems due to subtle environmental dynamic differences. Simultaneously, there remains a lack of mature and universal theoretical paradigms for effectively embedding physical mechanism constraints into purely data-driven reinforcement learning frameworks to compensate for the model's deficiencies in generalization and robustness. It hinders agents from truly adapting to the variable operating conditions encountered in industrial settings.

To address the high cost and low efficiency of training reinforcement learning on physical entities, the future mainstream technological paradigm will establish a hybrid model characterized by virtual training followed by real-world fine-tuning. The approach relies on constructing high-fidelity multiphysics digital twin environments

that precisely replicate thermo-fluid-solid coupling behaviors during welding using CFD or FEM models. However, given the substantial computational overhead of high-fidelity simulations, research focus will shift toward developing lightweight, real-time thermal-fluid coupling surrogate models. It enables agents to complete the majority of training tasks cost-effectively within simulated environments. Building upon the foundation, domain randomization techniques will be introduced to artificially introduce perturbations into simulations, thereby covering real-world physical uncertainties. The approach breaks down the barrier between simulation and reality, enabling seamless strategy transfer to actual physical welding systems.

At the algorithmic mechanism level, to overcome the limitations of purely data-driven models that may violate physical intuition and exhibit weak generalization capabilities, physics-informed reinforcement learning (PIRL) will emerge as a core research direction. Future research will focus on establishing a dual-driven physics and data model, explicitly embedding classical welding physics equations, including the Rosenthal equation, energy conservation laws, and fluid control equations, as constraints or prior knowledge into the design of reinforcement learning loss functions and reward functions. The integration not only significantly reduces the exploration space and accelerates model convergence but also ensures control strategies strictly adhere to physical laws in unseen operating conditions, thereby enhancing the system's physical interpretability and safety. Simultaneously, offline reinforcement learning (Offline RL) provides a critical pathway to activate vast amounts of "dormant data" in industrial settings. The technology enables agents to directly extract features and learn optimal strategies from historically accumulated, non-optimal datasets based on traditional PID control. It eliminates the need for high-risk real-time interactions with the environment during early stages, thereby achieving low-cost intelligent upgrades.

Although the direction of technological evolution is clear, substantial progress in adaptability and system interpretability is still required to bridge the gap from laboratory validation to large-scale industrial applications. On one hand, to address the frequent changes in material composition, plate thickness, and joint configurations encountered in welding production, integrating meta-learning and few-shot learning technologies will be crucial. These approaches empower robots with meta-capabilities to rapidly adapt to new tasks using minimal trial samples, thereby overcoming sample efficiency bottlenecks. On the other hand, breaking the black box nature of deep

reinforcement learning and advancing explainable artificial intelligence (XAI) is essential for building technological trust. Particularly in high-safety-critical sectors like aerospace and nuclear power, engineers must clearly understand the underlying logic and decision boundaries when intelligent agents adjust process parameters. Establishing trust mechanisms in human-machine collaboration is vital to truly drive the industry-wide adoption of intelligent welding technology.

In summary, reinforcement learning-driven adaptive welding systems represent the future direction of intelligent manufacturing. With the deep integration of algorithms, computational power, and process mechanism research, they will inevitably propel welding manufacturing toward a new phase characterized by higher quality, greater efficiency, and enhanced flexibility.

4. Machine learning-driven quality assurance

Traditional quality assurance methods, due to their offline, destructive, and sampling nature, can no longer meet the extreme demands for quality, safety, and cost control in the mass production of EV batteries. This section provides an in-depth review of how machine learning drives a fundamental transformation in the quality assurance system. It aims to realize the leap from post-process inspection to in-process prediction, providing a viable technical pathway for 100% in-line full inspection.

4.1. Automated defect diagnosis

Automated defect diagnosis constitutes the foundational component of the machine learning-driven quality assurance system. Its core objective lies in replacing manual experience and rule-based judgment with data-driven approaches to achieve rapid, objective, and repeatable identification of welding defects. Depending on the sources of accessible information, relevant research focuses primarily on two technical pathways, namely the recognition of weld surface defects and the intelligent inference of internal defects.

4.1.1. Surface defect recognition via deep learning

Weld surface morphology serves not only as an intuitive representation of welding quality but also as an external mapping of process stability and energy input status. Defects such as spatter, collapse, undercut, and cracking are essentially caused

by welding parameter fluctuations or multi-physics coupling instability [230].

Therefore, surface defect recognition based on visual information acts not only as a basis for quality judgment but also as a critical entry point for achieving in-process monitoring.

Compared to traditional machine vision methods relying on manual feature extraction [231], deep learning, particularly CNN, has triggered a paradigm shift in weld surface inspection not simply because of higher classification accuracy, but because it can automatically learn multi-level visual descriptors that are more closely related to weld-surface evolution [83]. Traditional methods are often sensitive to lighting variation, texture disturbance, and background noise, and the hand-crafted features they use are difficult to generalize across complex industrial conditions. In contrast, CNN-based models can hierarchically capture physically relevant visual patterns, such as defect boundaries, molten-pool collapse traces, spatter morphology, brightness distribution, and surface profile discontinuities. These features are not merely abstract image representations; rather, they can be interpreted as external manifestations of unstable energy input, melt-pool fluctuation, insufficient backfilling, or excessive local heating. Accordingly, deep learning-based surface defect recognition provides a more effective way to relate weld appearance to process-state evolution, which explains its superior robustness in handling multi-scale and multi-morphological defects [232].

The deployment of the technology typically utilizes image streams collected by industrial cameras as data sources to complete automated defect detection via supervised learning. Notably, facing the practical challenges of defect sample scarcity and extreme class imbalance [233], as well as stringent requirements for detection speed on production lines, the introduction of lightweight architectures and few-shot learning strategies has become an inevitable trend [234]. By conquering these bottlenecks, visual recognition is breaking through the limitations of post-event quality inspection and transforming into an online process perception means, thereby providing support for establishing data-driven quality prediction and closed-loop control systems.

As the foundation of the intelligent welding quality monitoring system, image-level classification aims to construct a mapping function from high-dimensional raw image space to discrete defect labels. Driven by deep learning technologies, the field has transitioned from manual feature extraction combined with conventional

classifiers, specifically SVM and RF, to end-to-end adaptive feature learning. CNNs have become the mainstream architecture for processing welding texture features [235]. The dominance is attributed to their ability to maintain recognition robustness against field-of-view shifts caused by jitter or trajectory deviations, as well as their capacity to layer-wise map low-level edge gradient information into high-order defect topological structures, thereby achieving precise analysis of complex surface morphologies. Building upon CNNs, numerous classic models have emerged. Deep networks represented by VGG, ResNet, and DenseNet have enhanced the capability to capture subtle features by stacking convolutional layers and introducing residual connections. Meanwhile, lightweight models represented by MobileNet and ShuffleNet have optimized computational efficiency through depthwise separable convolutions. These classic models not only established the algorithmic cornerstone for defect recognition but also provided diverse options for addressing specific challenges in industrial scenarios.

In industrial welding scenarios, acquiring large-scale annotated data is extremely difficult. Addressing the challenge of defect sample scarcity, transfer learning strategies are widely applied. By fine-tuning classic models such as VGG, ResNet, or DenseNet, weld texture features can be effectively extracted. Seo et al. [236] constructed a small-scale dataset containing seven types of typical defects, including pores, overlaps, and craters, and applied transfer learning based on ResNet and AlexNet models. The results demonstrated that the models achieved high-precision classification of complex appearance defects without requiring training from scratch using massive welding data. Jiao et al. [237] developed a transfer learning method based on ResNet. The model was pre-trained on the ImageNet dataset, and its fully connected layer was modified according to their specific dataset. Their results indicated that the transfer learning method could shorten training time and increase prediction accuracy to 96.35%, proving that transfer learning significantly reduces the training difficulty of deep learning and meets training requirements even with a small amount of data.

In addition to transfer learning strategies, generative adversarial networks (GANs) are frequently employed to cope with challenges such as restricted welding data acquisition, scarcity of defect samples, and severe sample imbalance in industrial environments. Wang et al. [238] proposed an innovative GAN model and utilized transfer learning strategies to transfer welding knowledge obtained under laboratory

conditions to industrial environments. Experimental results showed that the LAB_IMP1 model, developed based on ATGAN and transfer learning, achieved an average recognition accuracy of 96.23%, effectively solving the problems of dataset scarcity and imbalance in industrial settings.

To overcome the inherent limitations of CNNs in capturing global context information, recent research exhibits a trend evolving towards transformer architectures. The self-attention mechanism of transformer enables the establishment of global contextual associations within images [239], which is vital for identifying large-scale distributed defects such as insufficient penetration and cracking. Compared to CNNs, transformer possess superior efficiency, feature robustness, and learning capabilities. However, since the computational complexity of the multi-head self-attention (MHSA) mechanism increases drastically with the increase of input sequence length and image pixel count [240], the training and inference speeds of the model decrease significantly when processing large-scale datasets and high-resolution images. Consequently, they fail to meet the practical application requirements for real-time detection.

The latest research tends to adopt CNN-Transformer hybrid architectures to synergize the local feature extraction capability of CNNs with the global information capturing capability of transformer. Tang et al. [42] designed a typical CNN-Transformer hybrid architecture named MCFNet to address the strong noise and interference fringe disturbances present in TOFD weld images. It utilizes a multi-scale feature fusion (MSFF) module constructed by CNNs to precisely capture local texture details of the weld. Simultaneously, it introduces an efficient and fast transformer module (EFTM) based on cascaded group attention (CGA) to establish global semantic associations. The deep architectural fusion design not only effectively circumvents the issue of excessive computational load associated with pure transformer models, but also realizes industrial-grade real-time inference while maintaining high classification accuracy. Similarly, to address the problems of low detection accuracy and high miss rates caused by low contrast and small-sized defects in welding X-ray images, Liu et al. [241] proposed an MFE-IDD model fusing CNN and transformer architectures as illustrated in **Fig. 15**. The model effectively captures long-range dependencies and spatial correlations, thereby efficiently extracting multi-scale defect features. Results indicated that the MFE-IDD model demonstrated

outstanding performance in welding defect detection on digital radiography (DR) images.

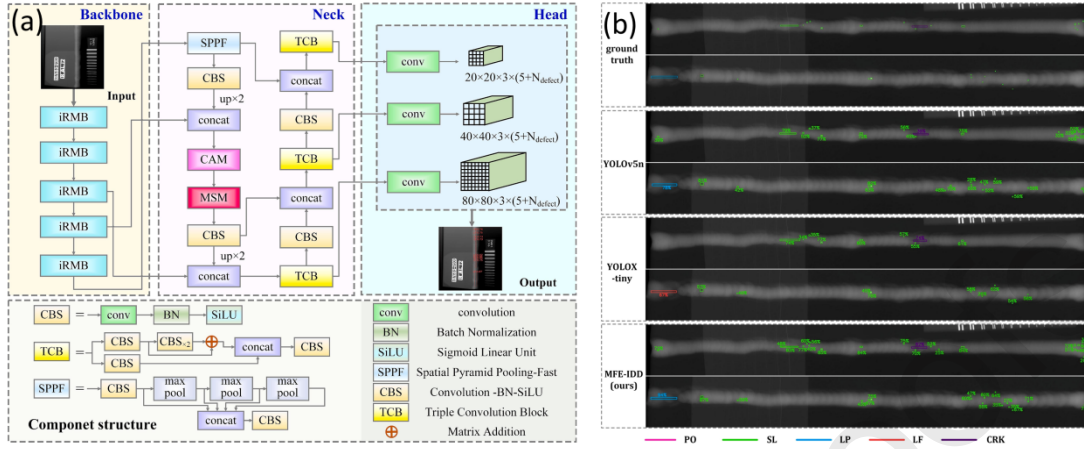


Fig. 15. A case of intelligent defect detection using CNN-Transformer [241]: (a) framework of the MFE-IDD; (b) defect detection results in high-resolution DR images using different algorithms.

With the acceleration of welding production rhythms, model response latency has become a core bottleneck for industrial deployment. Although deep networks excel in recognition accuracy, they often struggle to meet standard requirements for real-time monitoring on embedded terminals with limited computational capability [242]. Therefore, engineering practice tends to adopt lightweight networks characterized by lower computational loads [243].

With the increasing use of power batteries in the automotive industry, the inspection of laser welding defects in safety vents has become critical for ensuring safety and quality. In this context, Yang et al. [244] developed a lightweight laser welding defect detection model based on SqueezeNet and transfer learning. The model achieved high accuracy, reaching 99.57% for binary classification and 95.58% for seven-class classification on a dataset of 34,537 images, while maintaining a compact size of 1.2 MB and a fast inference speed of 4.9 ms. Deployment on a Raspberry Pi platform further demonstrated its suitability for real-time industrial applications.

As shown in **Fig. 16**, Zhang et al. [245] proposed a real-time lightweight defect detection algorithm, named Welding Defect Classification Network, based on a deep compression convolutional neural network, for detecting laser welding defects in the safety vent of electric vehicle batteries. The method incorporates an improved MobileNetV2 backbone combined with an Efficient Channel Attention mechanism and a teacher-free knowledge distillation strategy. Experimental results showed that

the model achieved an accuracy of 96.50% on a self-constructed welding defect dataset, outperforming several existing lightweight networks while maintaining low computational complexity. Furthermore, the model was deployed on an embedded platform consisting of a Raspberry Pi 4B and an Intel Neural Computing Stick, achieving an inference time of 29.89 ms per sample, thereby meeting real-time industrial requirements.

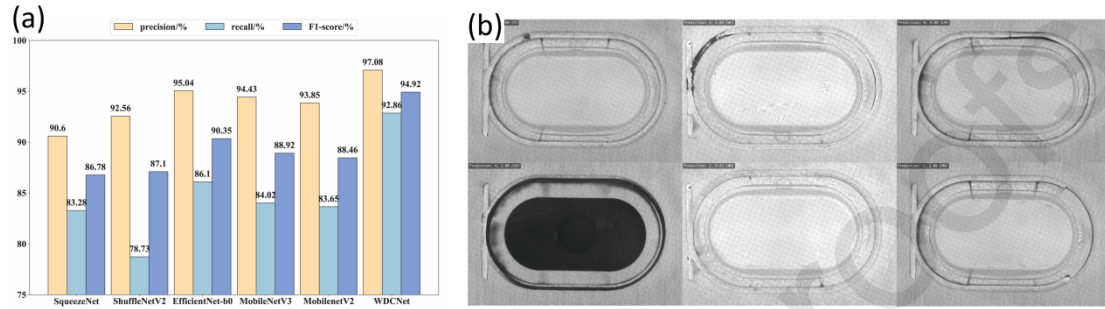


Fig. 16. A case of lightweight model [245]: (a) the comparisons of classification indexes for six lightweight models; (b) the detection results of the proposed algorithm for five types of defects (P, WC, LM, SU, MW) and normal product (NMs).

Despite the superior performance of deep learning models, their lack of interpretability remains a major trust barrier in the safety-critical field of welding. To enhance model interpretability, visualization techniques such as Grad-CAM++ [246] have been introduced. By overlaying heatmaps onto original images, these techniques assist in identifying which regions of the weld image influence the model's predictions, which is crucial for implementing effective interventions within maintenance workflows. López et al. [247] proposed WeldVGG, a convolutional model inspired by VGG for automatically classifying weld defects in radiographic images. By integrating Grad-CAM++, the model generates class-discriminative visual explanations for each prediction result, thereby improving transparency and supporting post hoc validation. To enhance the interpretability of deep learning-based welding defect classification, Hwang and Lee [34] employed Grad-CAM to visualize the decision-making process of the model as illustrated in **Fig. 17**. Their approach enables intuitive localization of discriminative regions in welding images, thereby improving the interpretability of CNNs. The results indicate that the integration of explainability techniques not only improves model transparency but also facilitates more reliable and trustworthy quality assessment in industrial scenarios.

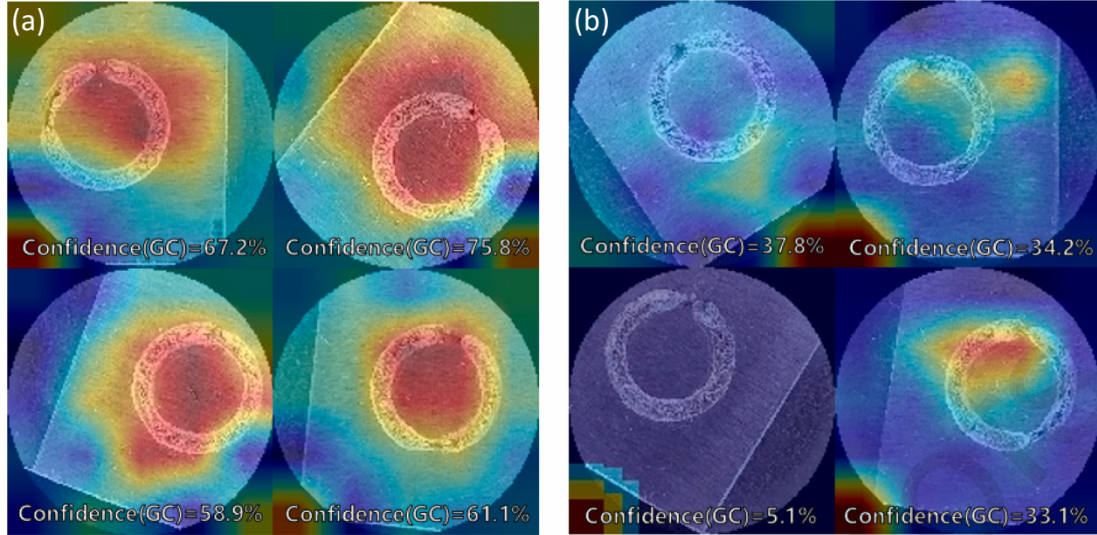


Fig. 17. Grad-CAM heatmap result image and confidence value [34]: (a) heatmap image and confidence values for correct classification; (b) heatmap image and confidence value when misclassified.

A comparison of mainstream architectures is presented in **Table 4**. Welding image classification technology is shifting from solely pursuing accuracy to multi-dimensional performance trade-offs, specifically the synergistic optimization of global feature understanding, few-shot learning, response speed enhancement, and industrial decision credibility. The evolutionary trend not only improves the reliability of online monitoring but also lays the data and theoretical foundation for subsequent refined defect localization and quantitative assessment.

Table 4. Comparative analysis of characteristics and applicability of mainstream architectures.

Architectures	Representative Models	Pros	Cons	Typical Scenarios
Deep CNN [237]	VGG, ResNet, DenseNet,	1. Excels at capturing local features.	1. Massive parameter count and high computational load	1. Offline precision quality
	AlexNet	2. Abundant pre-trained weights facilitate easy transfer learning.	make it difficult to meet industrial real-time requirements.	inspection.
		3. Residual connections effectively resolve the gradient vanishing problem in deep networks.	2. Limited by convolution kernel size, making it difficult to capture global information.	2. Few-shot transfer learning tasks.
Lightweight CNN [245]	MobileNet, ShuffleNet,	1. Extremely fast inference speed; significantly reduces computational	1. Feature extraction capability is weaker than that of deep	1. Online real-time monitoring.
	SqueezeNet	latency via depthwise separable convolutions.	networks; prone to missing minute defects.	2. Embedded and mobile
Transformer [239]	Vision Transformer (ViT), Swin Transformer	2. Deployment-friendly with small model size; suitable for integration into embedded terminals or robotic systems.	2. Robustness is limited in complex, high-interference industrial environments.	deployment.
		1. Self-attention mechanism enables the establishment of long-range spatial dependencies, thereby creating global associations.	1. Computational cost increases drastically with image resolution.	1. Detection of complex large-scale defects.
		2. Capable of recognizing large-scale defects; demonstrates excellent performance on distributed defects such as insufficient penetration and cracking.	2. Requires massive datasets for training.	
Hybrid CNN-Transformer [241]	MCFNet, MFE-IDD	1. Balances the local detail extraction of CNNs with the global semantic modeling of transformer.	1. Complex model topology makes hyperparameter tuning difficult.	1. Precision detection under complex operating conditions.
		2. Significantly improves efficiency compared to pure ViT, capable of meeting industrial real-time standards.	2. High-performance results still rely on specific computational support.	2. Simultaneous recognition of multi-scale defects.
Generative models [238]	GAN, CycleGAN,	1. Capable of generating high-quality virtual samples to ameliorate class imbalance.	1. Training is unstable and prone to mode collapse; generated images may not conform to physical laws.	1. Data augmentation.
	Diffusion Models	2. Achieves data style transfer between different environments or sensors.	2. Physical authenticity of generated data remains questionable.	2. Balancing long-tailed distribution datasets.

In the automated monitoring system for EV battery welding quality, compared to image-level classification, which only provides qualitative results, object detection technology enables the output of bounding boxes in the image coordinate system while identifying defect categories. The capability allows the quality assurance system to precisely quantify the spatial distribution and clustering degree of defects, providing critical positional feedback for the closed-loop adjustment of upstream welding process parameters. Deep learning-based welding defect detection algorithms are primarily categorized into two-stage networks represented by the R-CNN series, which pursue ultimate precision, and single-stage networks represented by the YOLO and SSD series, which prioritize real-time performance [47].

The core characteristic of two-stage detection algorithms lies in the introduction of a region proposal network (RPN) as an intermediate step. It first generates a series of potential regions of interest (ROIs) and subsequently performs refined classification and positional regression on these regions. Algorithms represented by Faster R-CNN [248], leveraging their deep feature extraction capabilities and multi-stage refinement strategies, exhibit outstanding detection accuracy when processing minute and extremely low-contrast welding defects such as tiny pores or micro-cracks [249]. Targeting weld defect recognition, Yang et al. [250] proposed an improved Faster R-CNN algorithm. By introducing a feature pyramid network (FPN), a convolutional block attention module (CBAM), and a K-means anchor strategy, they aimed to enhance the accuracy of weld defect detection and experimentally verified the effectiveness and robustness of the method. However, although the introduction of the RPN ensures high recall rates, it inevitably increases computational load and inference latency. Consequently, the FPS of two-stage algorithms typically struggles to meet the real-time requirements for high-speed full inspection on EV battery production lines.

To break through the real-time bottleneck, single-stage detection algorithms discard the explicit candidate region generation step and reconstruct the object detection task as an end-to-end regression problem directly from input images to class probabilities and bounding box coordinates [251]. Among them, the you only look once (YOLO) series and single shot multibox detector (SSD) algorithms constitute the mainstream for current industrial deployment [252,253]. Liu et al. [254] proposed an improved YOLOv9-based welding defect detection algorithm to address challenges such as large defect scale variation and complex background interference in industrial

scenarios. The model enhances feature extraction through a multi-scale fusion module in the backbone and introduces an adaptive mechanism in the neck network to better capture fine and slender defects. In addition, an attention-based detection module is employed to strengthen the model's focus on critical defect regions. Experimental results showed that the improved model achieved superior performance in both convergence and average precision compared with the baseline, while maintaining the real-time advantages of single-stage detectors.

As shown in **Fig. 18**, to improve the detection of small defects in lithium battery pole welding, Yang et al. [255] proposed an enhanced YOLOv5-based model by redesigning the backbone and head networks and introducing an improved loss function. The model achieved a mean average precision of 97% and a detection speed of 270 frames per second, outperforming conventional methods. Further validation on public datasets demonstrated consistent performance improvements and enhanced capability for small target defect detection.

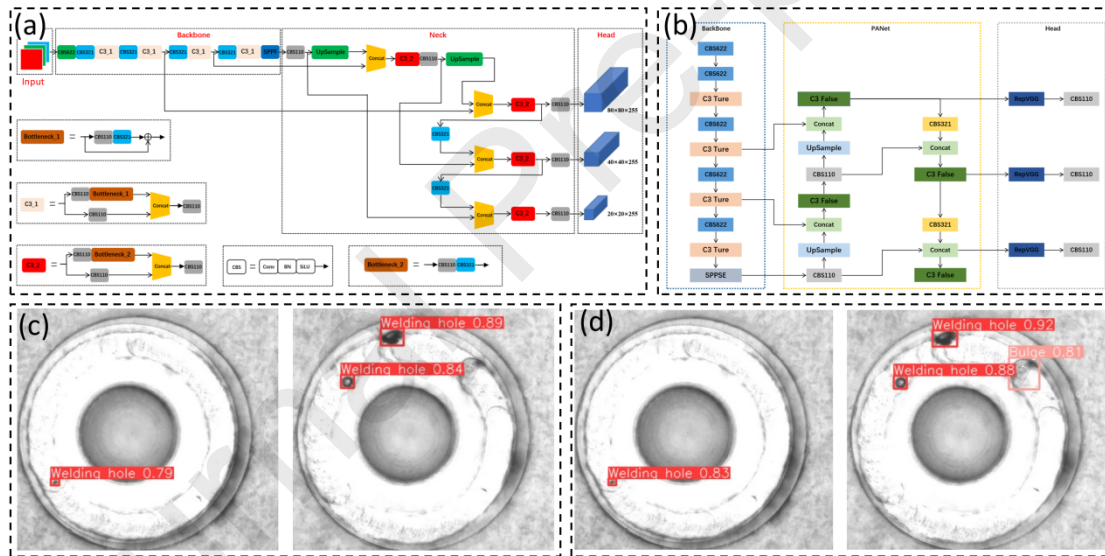


Fig. 18. A case of algorithm based on YOLO [255]: (a) the network structure of the YOLOv5 model; (b) the architecture of the improved YOLOv5 model; (c) the detection results of YOLOv5 model; (d) the detection results of the proposed improved YOLOv5 model.

In summary, although two-stage networks represented by Faster R-CNN still hold an irreplaceable precision advantage in offline precision analysis, single-stage networks that balance high precision with high real-time performance have established their dominant position in online quality monitoring under the stringent real-time requirements of large-scale EV battery mass production. With the deep integration of attention mechanisms, multi-scale feature fusion, and advanced data

augmentation strategies, modern single-stage networks have effectively overcome challenges regarding background noise interference and the missed detection of minute targets in complex industrial environments. They have achieved a substantial leap in detection accuracy while maintaining extremely fast inference speeds, thereby providing solid technical support for constructing an intelligent welding quality assurance system.

Although object detection algorithms can achieve defect localization with high real-time performance, the rectangular bounding boxes they output inevitably contain significant background information from non-defect areas, making it difficult to precisely describe the complex geometric morphologies of defects such as pores, cracks, or insufficient penetration. To meet the demand for precise quantization of welding quality in EV battery manufacturing, segmentation technology has been introduced into the quality assurance system. The technology primarily encompasses two technical routes, namely semantic segmentation and instance segmentation. The former aims to classify every pixel in an image as weld, base material, or defect, thereby generating a region mask consistent with the original image resolution. The latter further differentiates discrete instances within the same category, including dense spatter particles, thereby enabling independent counting and measurement [256]. The leap in representation capability from box to pixel enables the quality assurance system to transcend simple yield judgment. Consequently, it allows for the calculation of weld width consistency, molten pool edge roughness, and precise defect area ratios, providing high-confidence geometric data support for establishing physical correlation models linking weld morphology to mechanical properties.

In the architectural evolution of semantic segmentation, fully convolutional networks (FCN) and their derivative models constitute the mainstream technical framework. In particular, the U-Net architecture, by virtue of its unique symmetric encoder-decoder structure, has become the dominant model in the field of industrial defect segmentation [257]. The encoder of the U-Net is responsible for extracting high-level semantic features through continuous convolution and pooling operations, while the decoder progressively restores the spatial resolution of the image through upsampling. Its core innovation lies in the introduction of skip connections, which directly concatenate shallow high-resolution features from the encoding path to the corresponding levels in the decoding path, effectively solving the problem of edge detail loss during the downsampling process in deep networks [258].

To enable precise segmentation of welding defects in battery safety vents, Yang et al. [259] proposed a dual-branch deep learning model for pixel-level detection. The architecture integrates a spatial branch for detailed feature extraction and a context branch for semantic representation, followed by a feature fusion module. The method achieved 86.704% of mIoU and high detection accuracy, demonstrating its effectiveness in precise defect segmentation for industrial applications.

To address the challenges of scarce training samples and subtle defects in power battery laser welding, Wang et al. [260] proposed a dual-branch framework as illustrated in **Fig. 19**, integrating pixel-level segmentation with quality classification. The architecture shares encoder features to leverage multi-task learning and incorporates a coordinate attention module to enhance the capture of fine-grained spatial details without losing positional information. Additionally, the method employs Hough transform post-processing to extract precise geometric parameters from the segmentation masks. Experimental results on real-world production datasets demonstrated robust performance with high classification accuracy and superior segmentation precision, confirming its effectiveness for high-precision industrial inspection.

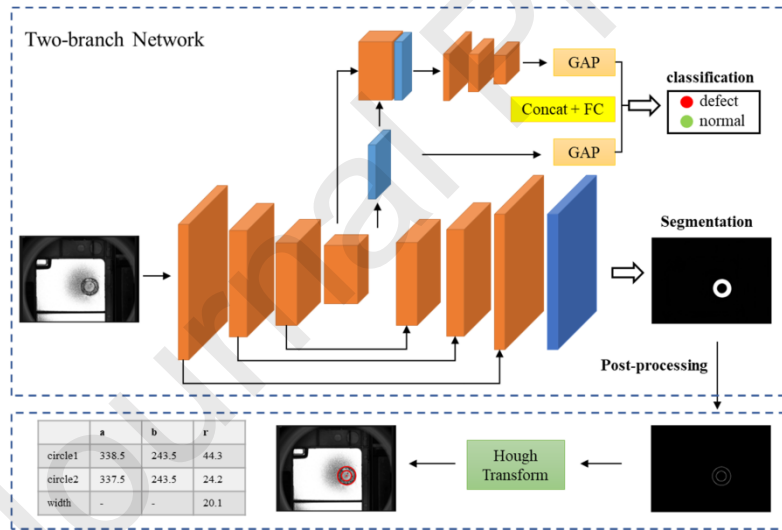


Fig. 19. Framework combining the two-branch network and conventional image processing [260].

For problems involving independent counting and individual analysis of dense defects, instance segmentation technology offers a more refined solution. Algorithms represented by Mask R-CNN extend Faster R-CNN by adding a parallel branch for predicting object masks based on the existing bounding box recognition branch [261]. The architecture not only inherits the high localization precision of object detection

but also realizes refined pixel-level boundary representation and independent identification of individuals within the same class, enabling the precise separation of every individual defect in densely occluded scenarios. In EV battery welding, the capability is particularly critical as it allows the system to perform independent individual segmentation and statistics on spatter particles. It enables not only the calculation of the total spatter area but also the analysis of spatter particle size distribution histograms, which facilitates the inverse inference of laser power density.

Kim et al. [262] developed an automated system presented in **Fig. 20**, integrating laser thermography and Mask R-CNN for the detection and precise quantization of weld cracks. The study utilized the pixel-level masks generated by Mask R-CNN to automatically calculate key geometric parameters such as crack length, maximum width, and inclination angle. The quantitative approach, based on instance segmentation, effectively resolved the subjectivity issues of human interpretation in traditional non-destructive testing, providing high-precision geometric data support for assessing the mechanical properties of welded structures.

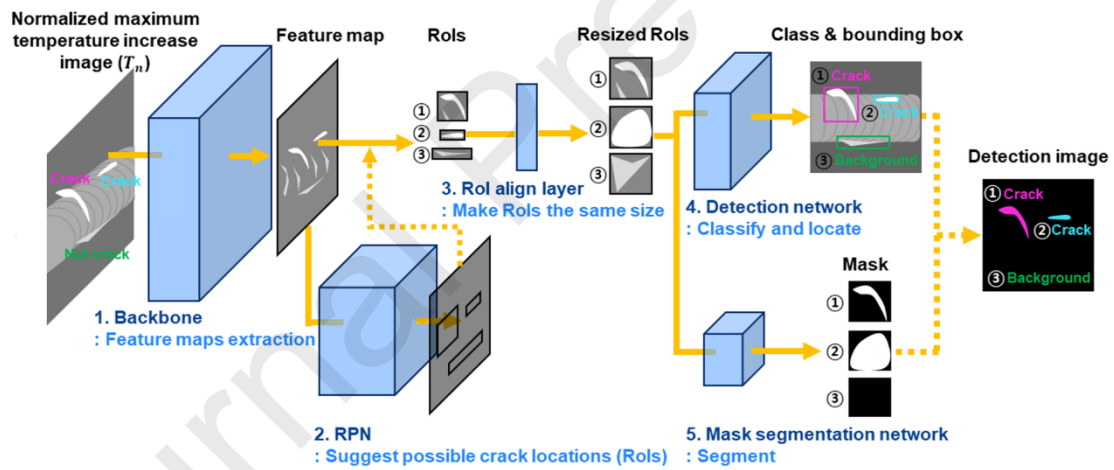


Fig. 20. Overview of crack detection and segmentation using Mask R-CNN [262].

However, the high cost of pixel-level annotation remains a bottleneck constraining the deployment of segmentation algorithms. Therefore, weakly supervised learning has become a current research hotspot, where researchers attempt to utilize inexact supervision, employing only image-level labels or bounding box labels to train segmentation networks [263]. Wu et al. [264] addressed the difficulty of low efficiency and high cost of pixel-level annotation in industrial scenarios by proposing a weakly supervised defect segmentation network based on classification activation maps (CAM). The study used only image-level class labels for training. By

designing a siamese network and improved modules, it effectively narrowed the gap between image-level and pixel-level supervision. Experimental results demonstrated that even with only image-level weak supervision signals, the model could still achieve high-precision segmentation on the KolektorSDD dataset.

Benefiting from the lightweight design of deep learning models, segmentation algorithms now possess millisecond-level real-time morphology quantification capabilities. It enables welding robots to dynamically adjust welding speed or laser power in real time based on current fluctuations in melt width, thereby providing a solid foundation for the transition from post-process inspection to in-process closed-loop control.

Zhang and Zhang [265] addressed phased array ultrasonic testing (PAUT) data for hybrid laser-arc welding in shipbuilding by proposing a lightweight U-Net defect segmentation model incorporating skip connections and residual blocks. The design successfully compressed the model parameter count to 1.25 M. Experimental results indicated that the model required only 0.11 s to segment PAUT data with dimensions of $64 \times 128 \times 128$ pixels on an RTX 6000 GPU. While achieving high-precision segmentation, this effectively verified the feasibility of using semantic segmentation models for real-time defect assessment in industrial sites.

As shown in **Fig. 21**, Zhu et al. [266] developed a lightweight multiscale attention-based semantic segmentation model for detecting welding defects in battery safety vents. The model integrates an improved Res2Net backbone with attention mechanisms, a localization module, and a boundary refinement strategy to achieve accurate segmentation of defects with varying sizes and shapes. Experimental results showed that the proposed method achieved a mean intersection over union of 84.67% and a high validation accuracy of 99.4%, while maintaining a compact model size of 3.8 MB and a real-time processing speed of 132.3 frames per second, demonstrating its effectiveness for industrial laser welding quality inspection.

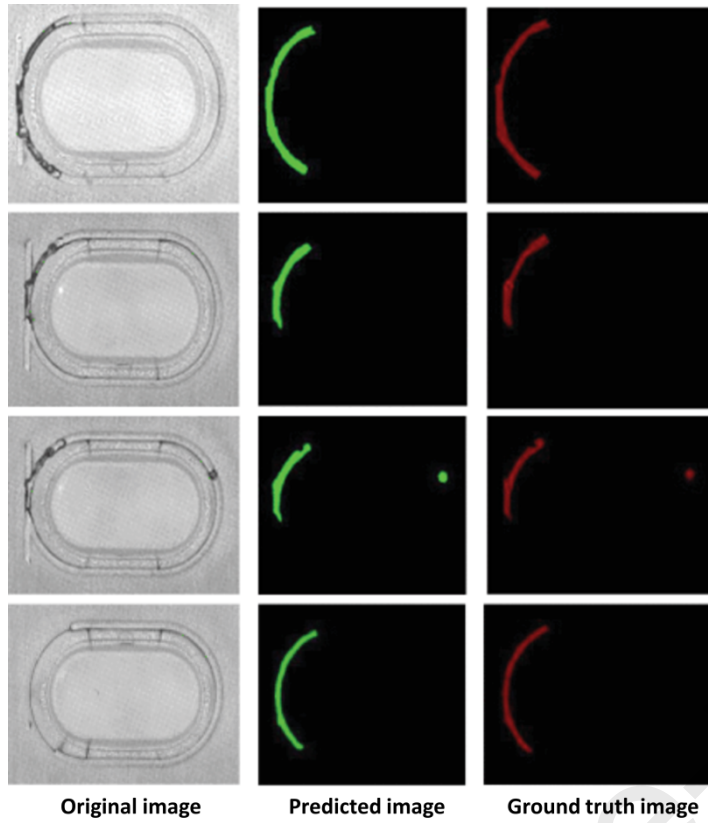


Fig. 21. Comparison between the predicted segmentation results and the ground truth images [266].

Looking ahead, incorporating the dynamic evolutionary trends of morphological features into statistical process control (SPC) systems will become a core strategy for predicting potential quality issues and ensuring the full lifecycle reliability of batteries.

4.1.2. Internal defect detection using multi-sensor fusion

Internal defects such as deep porosity, insufficient penetration, and micro-cracks are typically concealed beneath the weld surface and cannot be directly captured by traditional machine vision systems. However, these concealed defects pose a fatal threat to the electrical connection stability and long-term fatigue life of EV batteries. Traditional solutions rely on offline non-destructive testing methods like X-ray or CT, but their prohibitive equipment costs and slow inspection efficiency have become bottlenecks constraining mass production efficiency. Consequently, indirect inference based on in-process signal analysis has emerged as a promising alternative. Its underlying physical rationale is that thermodynamic instabilities inside the molten pool, such as keyhole collapse, metal vapor eruption, abnormal molten-pool flow, or

unstable heat accumulation, inevitably leave characteristic signatures in measurable acoustic, optical, electrical, and thermal fields. In this context, the role of machine learning is to enable online inference of internal weld quality by identifying physically meaningful correlations between multimodal process signatures and the underlying defect-generation mechanisms, thereby serving as an alternative to conventional offline inspection methods [267].

To comprehensively capture physical fluctuations during the transient welding process, constructing a multi-modal sensor array is a prerequisite for intelligent inference [268]. The multi-sensor system is illustrated in **Fig. 22**. Commonly employed sensors encompass photodiodes capable of capturing plasma light intensity fluctuations at extremely high sampling rates, acoustic sensors for monitoring arc sound signals, electrical signal sensors for recording welding voltage and current waveforms, and infrared thermal cameras capable of capturing the temperature distribution of the molten pool and HAZ.

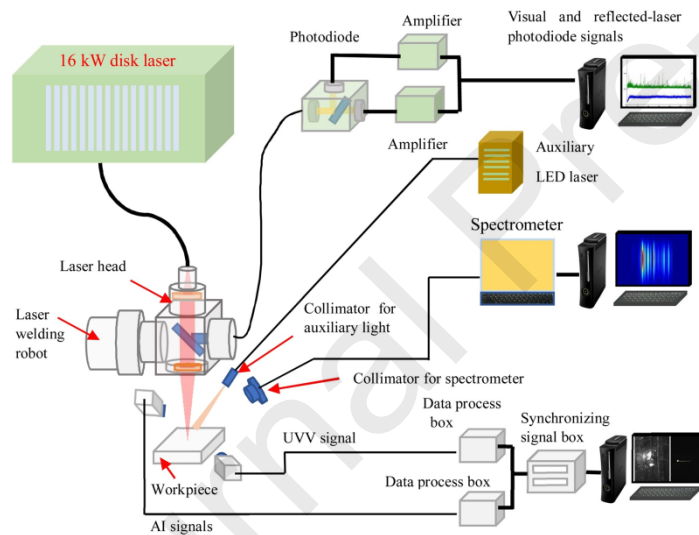


Fig. 22. A multi-sensor system for in-process monitoring of laser welding [46].

In early research, researchers primarily relied on manual feature engineering combined with traditional machine learning algorithms for modeling [269]. By performing time-domain and frequency-domain analyses on raw signals, these studies extracted features such as intensity fluctuation, dominant frequency components, and statistical descriptors, which can often be physically associated with keyhole instability, plasma plume fluctuations, or transient perturbations in molten-pool dynamics. These features were subsequently input into classifiers such as SVM, Random Forest, or XGBoost for defect detection [270–272].

The paradigm has been extensively validated in feature extraction for single physical fields. Wang et al. [273] targeted high-power disk laser welding. They extracted geometric and grayscale features, such as the area and centroid of metal vapor images, and utilized the sequential floating forward selection (SFFS) algorithm to screen for the optimal feature subset. These features were finally input into an SVM classifier, achieving real-time monitoring of welding quality. Zhang et al. [274] proposed an intelligent method based on arc sound sensing for the real-time assessment of weld penetration defect. They developed two feature selection methods based on Fisher distance and PCA to select frequency components correlated with weld defects. By combining these with a SVM optimized by grid search and cross-validation (SVM-GSCV), they established a novel classification model. Results demonstrated that the method possesses both accuracy and robustness.

As shown in **Fig. 23**, Caprio et al. [275] developed a multi-sensor monitoring framework combined with machine learning algorithms for laser welding of battery busbar connections. Multiple sensors, including photodiodes and a spectroscope, were employed to capture process emissions across different spectral ranges. Based on the extracted signal features, supervised learning models were trained to identify process deviations such as defocusing and gap variations. Experimental results demonstrated that photodiodes monitoring laser back-reflected signals provided effective indicators of process instability, and the trained models achieved high classification accuracy even with limited data.

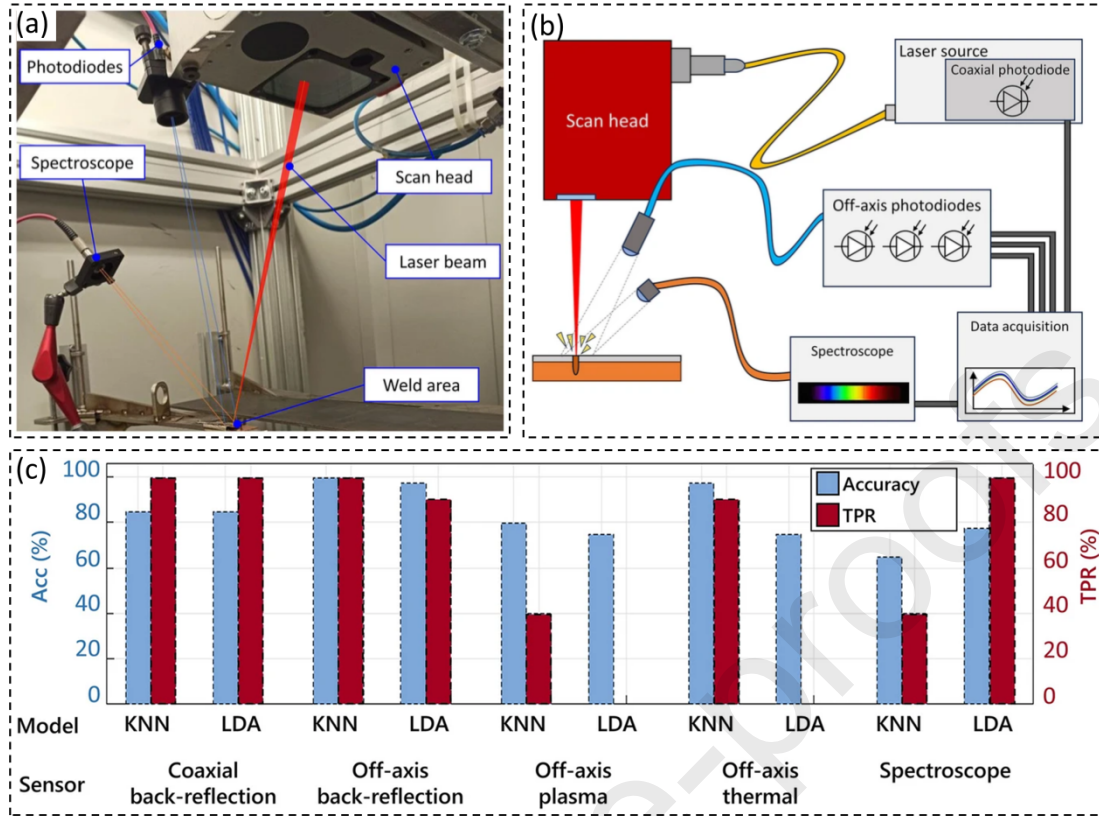


Fig. 23. The multi-sensor system [275]: (a) experimental set up of the monitoring sensors and laser welding system developed for the battery pack manufacturing; (b) schematic representation of the system with the monitoring acquisition system; (c) algorithm performance in terms of Acc and TPR obtained with KNN and linear discriminant analysis with data from different sensors and configurations.

Furthermore, to leverage complementary information across different sensors, Liu et al. [147] constructed a multi-sensor fusion system. The system utilized five sensors to extract features characterizing the keyhole, plasma, and spatter. Through normalization and PCA, fifteen features were analyzed, and the principal components served as input parameters for an SVM classifier. Experimental results indicated that the overall accuracy for training data reached 98%, while that for testing data reached 91%, thereby verifying the effectiveness of the system.

Although such methods possess good interpretability and low computational resource requirements, their performance is heavily reliant on manual feature engineering. They are prone to overlooking subtle and complex dynamic changes within raw signals. Consequently, their generalization capability is often constrained when facing complex operating conditions.

With the evolution of deep learning technologies, end-to-end modeling centered on CNNs has gradually replaced traditional methods to become mainstream. Unlike

traditional workflows relying on manual experience, deep learning models can operate directly on raw process signals or their transformed time-frequency spectrograms, utilizing multi-layer convolutional operations to automatically extract hierarchical abstract features.

As presented in **Fig. 24**, Deng et al. [45] proposed a multi-sensor data fusion network named IDDNet based on CNNs for process defect detection. Experimental results demonstrated that the multi-classification performance of IDDNet was superior to that of SVM, achieving an overall accuracy of 97.57%. Furthermore, IDDNet attained an average processing speed of 0.79 ms per sample on a single GTX 1080Ti GPU, fully satisfying the real-time requirements of industrial production.

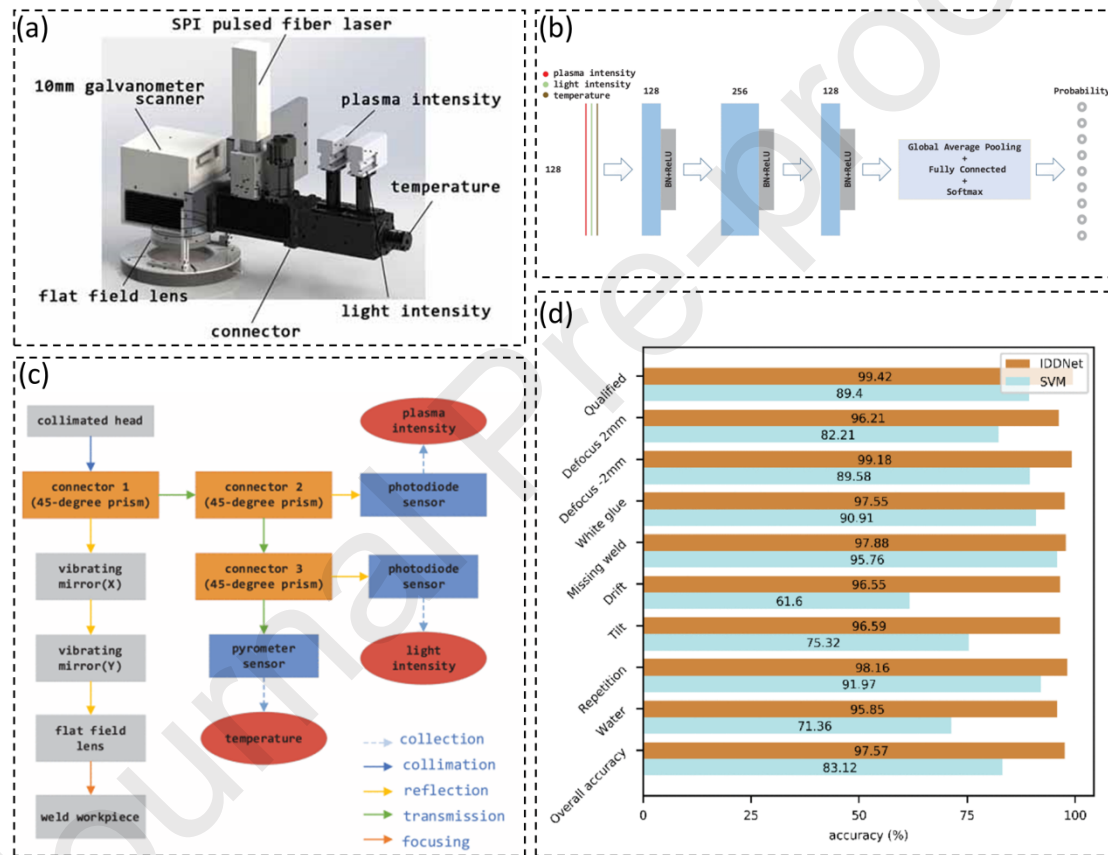


Fig. 24. A case study of inferring internal defects via multi-sensor fusion [45]: (a) a multi-sensor data fusion system for laser welding process monitoring; (b) the architecture of IDDNet; (c) overall schematic structure; (d) classification performance comparison between IDDNet and SVM.

Kim et al. [276] utilized transfer learning techniques based on pre-trained CNN architectures, including MobileNetV2, ResNet50V2, EfficientNetB3, and Xception. They developed a single-sensor model based on coaxial weld pool images and a multi-sensor model combining spectrometer data to predict keyhole depth in laser

welding. Experimental results indicated that the performance of the multi-sensor model outperformed that of the single-sensor CNN transfer learning model.

Given that welding process signals possess strong temporal dependence, meaning that current defect formation is often closely related to heat accumulation or keyhole status at previous moments, RNN and their variant, LSTM networks, have become the preferred architectures for processing such data. LSTM can effectively memorize long-range historical information through gating mechanisms, precisely capturing the dynamic patterns of signal evolution over time [277].

Addressing the issue that a single molten pool image contains insufficient information to predict weld penetration, Yu et al. [172] proposed a CNN-LSTM model combining CNNs and LSTM networks capable of extracting information from dynamic continuous molten pool images. The study found that the model could utilize continuous molten pool images to accurately predict the highly dynamic molten pool shape when the back weld width was 0.3 mm.

Targeting the randomness and uncertainty present in the dynamic characteristics of laser welding keyholes, Liu and Bao [150] proposed a periodic-sensitive long short-term memory (PLSTM) neural network. The method first extracts critical geometric features of the keyhole and utilizes the Fourier transform and autocorrelation coefficients to analyze periodic factors in the feature sequence. Results showed that compared to traditional methods, PLSTM possesses stronger long-term memory and temporal prediction capabilities, effectively enabling online inference of welding quality. Wang et al. [278] fully exploited the advantages of LSTM in processing long time-series data to propose an early event detection framework for the welding process. The study divided continuous process signals into finite time segments for network input. Leveraging LSTM's acute capability to capture signal evolution trends, the framework not only achieved high-precision classification of final welding quality but also precisely localized the abnormal starting point causing quality deviation by analyzing the cross-entropy convergence moment of classification probabilities.

To further enhance model performance on multi-modal data, attention mechanisms were introduced to address the problem of information overload. In multi-sensor fusion scenarios, signals from different modalities exhibit varying sensitivities to defects. For instance, acoustic signals may be more sensitive to spatter, whereas optical signals may be more sensitive to keyhole collapse. The attention

mechanism endows the model with adaptive weighting capabilities, enabling it to focus on the most discriminative sensor channels within specific time steps. Liang et al. [279] proposed a welding defect detection framework named WELD-DETR, which combines multi-scale feature fusion and multi-kernel perception cooperative optimization techniques. Based on the transformer architecture and its attention optimization mechanism, the model is capable of real-time and precise detection of micron-level defects during the welding process, such as cracks, porosity, and slag inclusions. Experimental results indicated that WELD-DETR achieved an mAP@0.5–0.95 of 98.2%, an accuracy of 96.8%, and an inference speed of 58 FPS on an RTX 2060 GPU. Furthermore, the model demonstrated outstanding detection accuracy and real-time performance in complex industrial scenarios characterized by high noise and strong reflections.

As shown in **Fig. 25**, Peng et al. [280] proposed a multi-sensor-based laser welding defect detection framework integrating the multiscale convolutional neural network, bidirectional long short-term memory, and attention mechanism to enable accurate in-situ defect identification. By capturing variations in physical signals during the welding process, the model effectively learns both spatial and temporal features from multi-modal data. Experimental results showed that the proposed fusion model achieved a detection accuracy of 99.38%, demonstrating its effectiveness for real-time welding quality monitoring.

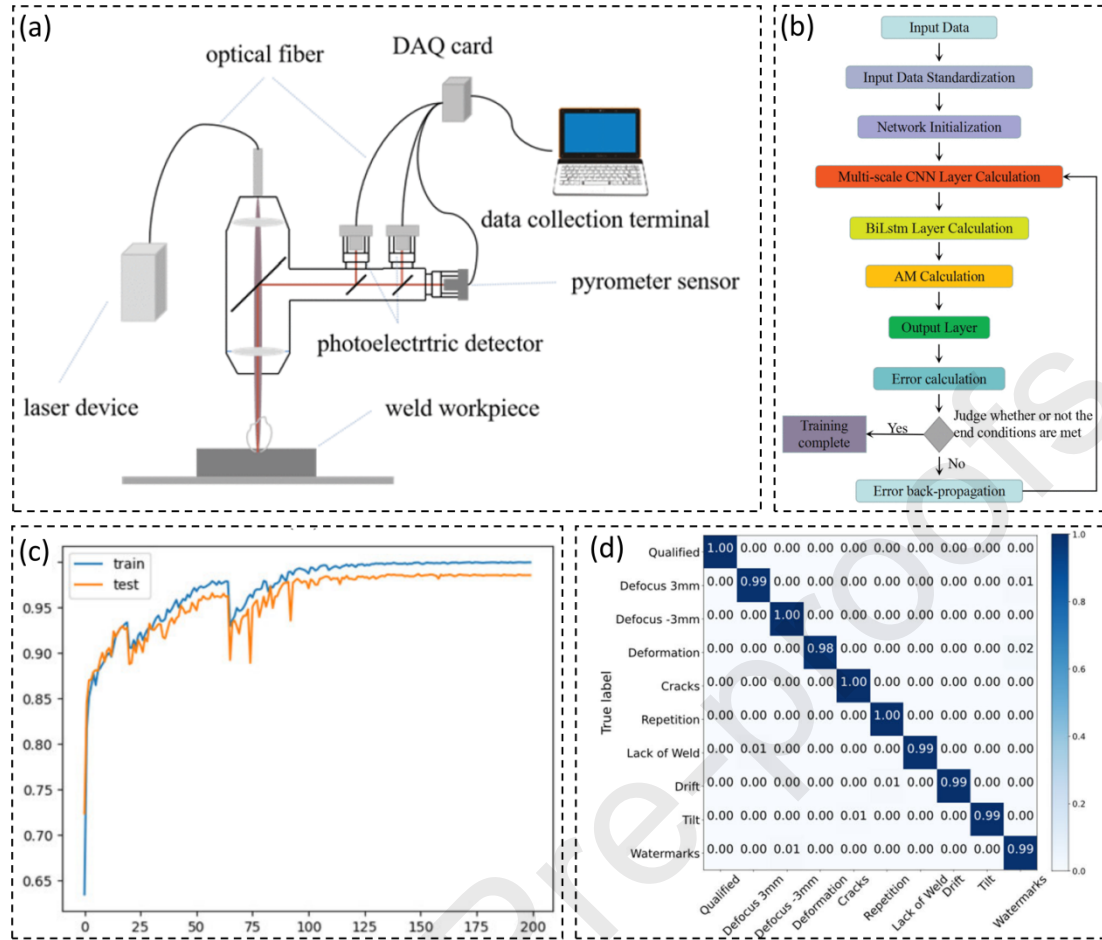


Fig. 25. Framework and performance of the MSCNN-BiLSTM-AM based defect detection system [280]: (a) multi-sensor data acquisition setup; (b) architecture of the MSCNN-BiLSTM-AM fusion model; (c) accuracy curves during training and testing; (d) confusion matrix of the identification results.

Although process signal-based internal defect inference has demonstrated immense application potential, it still faces severe challenges regarding data annotation during practical implementation. Training a high-precision supervised learning model necessitates massive datasets, whereas the ground truth labels of internal defects must be obtained through expensive X-ray or metallographic sectioning experiments. Consequently, future research is gradually shifting towards semi-supervised learning and unsupervised anomaly detection. These approaches aim to utilize massive amounts of unlabeled process data to learn the data distribution characteristics of normal welding, thereby achieving the identification of anomalous defects using only a small number of labeled samples. Nevertheless, the challenge lies not only in the scarcity of labels, but also in the reliability of the underlying ground truth used for model training and evaluation.

Despite these promising results, it should be emphasized that the reported accuracy of smart quality assurance models cannot be interpreted independently of the reliability of the underlying ground truth. In many reviewed studies, model performance is evaluated mainly by classification or regression metrics, such as accuracy, precision, recall, F1-score, or R^2 [117,281,282]. However, the scientific significance of these metrics depends fundamentally on whether the corresponding labels are rigorously validated. For laser welding of EV batteries, especially in tasks related to internal defect detection, microstructural evolution, or joint performance prediction, visually acceptable surface morphology does not necessarily guarantee defect-free internal quality. Therefore, labels derived solely from surface appearance, empirical thresholding, or simplified process windows may be insufficient to represent the true metallurgical state of the weld.

For quality inference tasks with strong physical and materials-science implications, rigorous validation methods are indispensable for establishing trustworthy ground truth. Techniques such as cross-sectional metallography, weld bead geometry analysis, X-ray or CT inspection, EBSD characterization, phase analysis, and mechanical or electrical testing are often required [283–286]. Without such validation, a model may achieve ostensibly high predictive accuracy while merely learning from weakly annotated, indirectly inferred, or even erroneously labeled datasets. In such scenarios, the reported performance holds limited scientific value and risks overstating the true capability of the AI model for industrial deployment.

4.2. Online quantitative prediction of key performance indicators

The advanced goal of the intelligent quality assurance system is to transcend the scope of defect classification and advance towards the quantitative prediction of key performance indicators. It requires models not only to identify specific anomaly categories but also to precisely infer numerical values of the mechanical strength and electrical performance of the weld seam via regression analysis based on multi-modal process data. The data-driven, non-destructive quantitative assessment technology provides a feasible technical pathway for realizing high-precision, full-coverage online monitoring of the production process.

4.2.1. Mechanical property prediction

Tensile strength and shear strength of solder joints are core indicators for assessing the structural integrity of battery modules under vehicle vibration, thermal expansion, and crash impact. Traditional quality control systems rely primarily on offline destructive tensile testing [287]. The method not only causes exorbitant material wastage and production capacity loss but, more critically, allows for the sampling of only an extremely low proportion of products, leaving significant quality blind spots. Furthermore, industrial sites frequently face the challenge of poor interfacial bonding. Although the surface morphology of the weld appears indistinguishable from normal welds, the actual load-bearing capacity is far below standards due to insufficient effective penetration depth or inadequate interfacial metallurgical reaction. Due to the lack of obvious external manifestations, such latent hazards are difficult to detect via conventional surface visual inspection and often serve as key inducements for structural fatigue failure of battery packs during long-term service. Addressing the dilemma, machine learning-based online prediction technology demonstrates broad application prospects. It makes full-coverage online assessment of the mechanical properties of every solder joint possible, thereby enabling the precise identification of latent weak joints that lack macroscopic defects but possess insufficient strength, effectively filling the quality control loopholes of traditional sampling modes.

The physical basis of machine learning-driven non-destructive prediction technology lies in the fact that dynamic heat input characteristics during the welding process, such as temporal fluctuations in voltage and current, and thermal cycle curves directly determine the microstructural evolution of the weld metal. It is particularly true for the thickness and distribution morphology of IMCs at the Cu-Al dissimilar metal interface, which are the fundamental internal causes determining macroscopic mechanical properties [288–290]. Based on this, the core implementation pathway of the technology involves constructing a high-quality database linking multi-modal sensor data with mechanical performance indicators. Specifically, it entails synchronously collecting high-frequency process signals and post-weld destructive tensile or shear test data via experiments. Subsequently, supervised learning is utilized to train regression models to learn the complex

mapping relationship from multi-modal signal features to specific strength values, such as tensile or shear strength [291,292].

Given that the exorbitant cost of destructive mechanical testing often results in a scarcity of labeled samples, algorithms such as SVR, GPR, and ensemble learning methods like RF and XGBoost have become classic choices for engineering applications. Zhang et al. [293] proposed an optical spectrum-based regression framework for predicting weld seam strength in laser welding using XGBoost. By performing parametric experiments with an in-situ spectral monitoring system, the relationship between spectral intensity and mechanical strength was established. The importance of key spectral lines was quantitatively analyzed, while RF and PCA were employed for feature selection and dimensionality reduction, showing strong complementarity. Experimental results demonstrated that the proposed RFPCA-XGBoost model outperformed other feature selection methods and ensemble learning models, achieving an R^2 value of 0.9383.

As shown in **Fig. 26**, Zhang et al. [294] proposed a machine learning-based framework for predicting the tensile strength and elongation of laser-welded joints to optimize welding parameters and improve joint performance. MLP, SVR, and four ensemble learning algorithms, including XGBoost, CatBoost, LightGBM, and RF, were developed to establish the mapping between process parameters and mechanical properties. Experimental results showed that XGBoost achieved the best performance in tensile strength prediction, while CatBoost outperformed other models in elongation prediction. SHAP analysis revealed that heat input was the most influential factor for tensile strength, whereas tensile strength itself played a dominant role in elongation prediction.

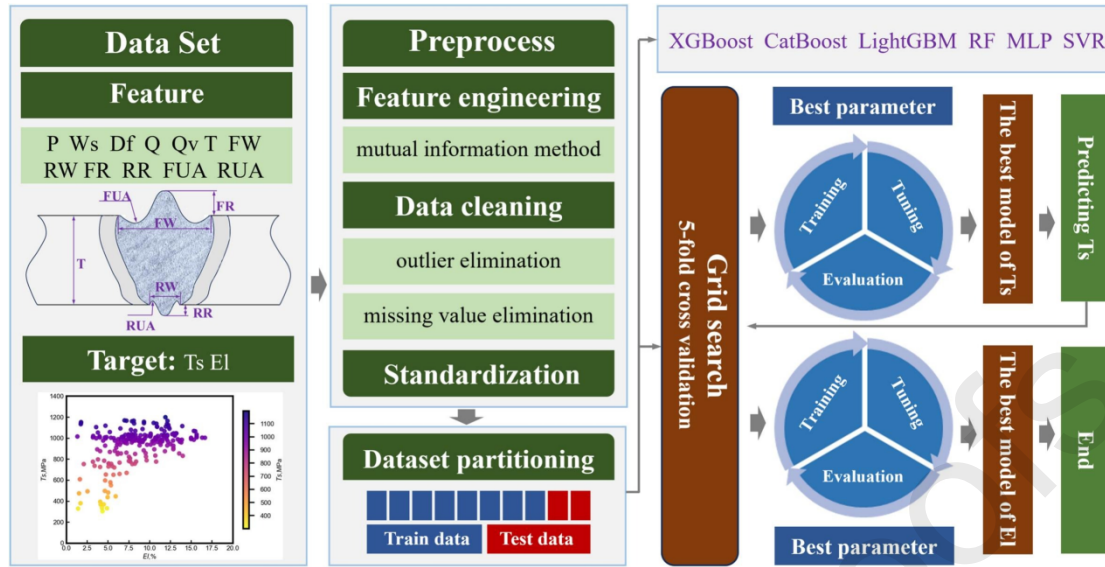


Fig. 26. Machine learning-based framework for predicting mechanical properties of laser-welded joints [294].

Currently, constrained by limited labeled datasets and the engineering necessity for model interpretability, traditional machine learning methods incorporating feature engineering remain the predominant approach for predicting mechanical properties of welded joints in industrial practice. Nevertheless, accelerated by the proliferation of high-frequency sensing technologies, expanding welding databases, and innovations in lightweight deep learning architectures, deep learning is progressively gaining traction for complex tasks such as end-to-end raw signal processing and dynamic multimodal fusion. The trajectory signals an evolving landscape where method selection will increasingly align with specific data availability, application requirements, and system integration needs rather than adhering to a single paradigm.

4.2.2. Electrical performance prediction

Joint contact resistance is a decisive indicator for measuring the reliability of electrical connections in battery modules during charge-discharge cycles. Under operating conditions of large-rate fast charging and high-power output in new energy vehicles, even minor resistance anomalies can induce severe Joule heating effects. It not only drastically increases the risk of thermal runaway but also undermines the energy efficiency of the entire module due to excessive voltage drop [295].

Traditional electrical performance detection relies primarily on offline four-probe micro-ohmmeter measurements [13]. Although the method offers high precision, it represents a single-point static test that struggles to adapt to the high-tempo

production requirements of large-scale automated production lines. Moreover, it cannot block the outflow of non-conforming products in real time during the welding process. Furthermore, the degradation of electrical contact performance often possesses a concealed nature. Certain solder joints may appear qualified in their initial state but rapidly age and fail under thermal cycling stress during service due to internal micro-cracks or insufficient effective bonding area [296]. Addressing the challenge, establishing an online contact resistance prediction model based on process data holds irreplaceable engineering value for ensuring the long-term electrical safety and power stability of battery systems.

The physical basis of machine learning-driven electrical performance prediction technology shares a fundamental origin with mechanical property prediction. During laser welding, the contact state of the interface, the spreading process of liquid metal, and the growth behavior of IMCs are all fed back in real time within the dynamic voltage and current waveforms. These process variables ultimately determine the effective conductive area and the integrity of the interfacial electron transport channel [18]. Based on this, the core implementation pathway of the technology involves constructing a database linking dynamic process signals with static resistance ground truths. Specifically, it entails real-time collection of welding voltage, current, and dynamic resistance waveforms, using contact resistance measured by high-precision micro-ohmmeters as labels, and utilizing regression algorithms to mine the numerical mapping laws from transient process features to steady-state electrical performance indicators [297].

Stavropoulos et al. [298] proposed a data-driven quality assurance approach for battery laser welding, where high-speed infrared process emissions were combined with machine learning to assess the electrical resistance and maximum tensile shear strength of tab-to-tab Al-Cu joints online. Their results demonstrated the feasibility of using in-process IR data for the classification of electrically and mechanically acceptable welds.

Ideally, a quality busbar weld must exhibit both sufficient mechanical strength and low electrical resistance. Given that high strength reflects a continuous and robust interfacial bond, it is reasonable to postulate that such structural continuity facilitates efficient electron transport and uniform current distribution, thereby inherently minimizing contact resistance. Liu et al. [35] noted that samples exhibiting superior mechanical strength also possessed favorable electrical conductivity, whereas the

reverse was not necessarily the case. This asymmetry implies that weld tensile strength can function as a standalone, comprehensive metric for quality evaluation, guaranteeing both robust mechanical integrity and electrical performance. Consequently, this significantly streamlines subsequent quality prediction, enabling feature extraction and the development of machine learning models to concentrate solely on the precise assessment of tensile strength.

4.3. Correlation between process signatures and lifecycle performance

The overarching objective of the intelligent quality assurance system is to bridge the gap between manufacturing and operational phases by utilizing multi-source data to precisely predict the long-term performance of the product. To realize the vision, efforts can be expanded from two dimensions. First, starting from monitoring data during the battery operation phase, to real-time assess the current health status and deduce the remaining lifespan. Second, starting from process characteristics during the battery manufacturing phase, to explore the decisive impact of welding quality on usage performance, thereby establishing a prospective screening mechanism based on the manufacturing source.

4.3.1. Lifecycle tracking via operational data

During the battery service cycle, voltage, current, and temperature data continuously collected by the battery management system (BMS) objectively record the physical trajectory of battery performance degradation [9,299]. The core task of the dimension is to utilize machine learning algorithms to extract performance evolution indicators from massive operational time-series data, thereby establishing a mapping model from the current state to the remaining useful life. Although the method falls under post-hoc evaluation, it provides indispensable data labels and verification benchmarks for establishing the correlation between manufacturing quality and long-term lifespan.

Yang et al. [300] proposed a gradient boosting regression tree (GBRT) model based on multi-dimensional features such as voltage, capacity, and temperature extracted from battery lifecycle data streams. Leveraging the powerful non-linear mapping capability of the GBRT algorithm, the study effectively captured the complex dynamic characteristics during the battery degradation process. It achieved

high-precision prediction of battery service life under conditions of limited data samples and multiple input features.

As shown in **Fig. 27**, Zhao et al. [301] introduced advanced architectures from the field of computer vision, proposing a multi-branch vision transformer model to address the problem of battery early life prediction. Utilizing a multi-input data structure, the model considered various battery parameters during charge and discharge phases. It employed a meticulously designed vision transformer network to extract high-dimensional global hidden features from inputs and integrated these features to predict battery lifespan. Experimental results indicated that the model possessed excellent performance and robustness, verifying its effectiveness and reliability in accurately predicting lifespan during the early stages of lithium-ion battery capacity decline.

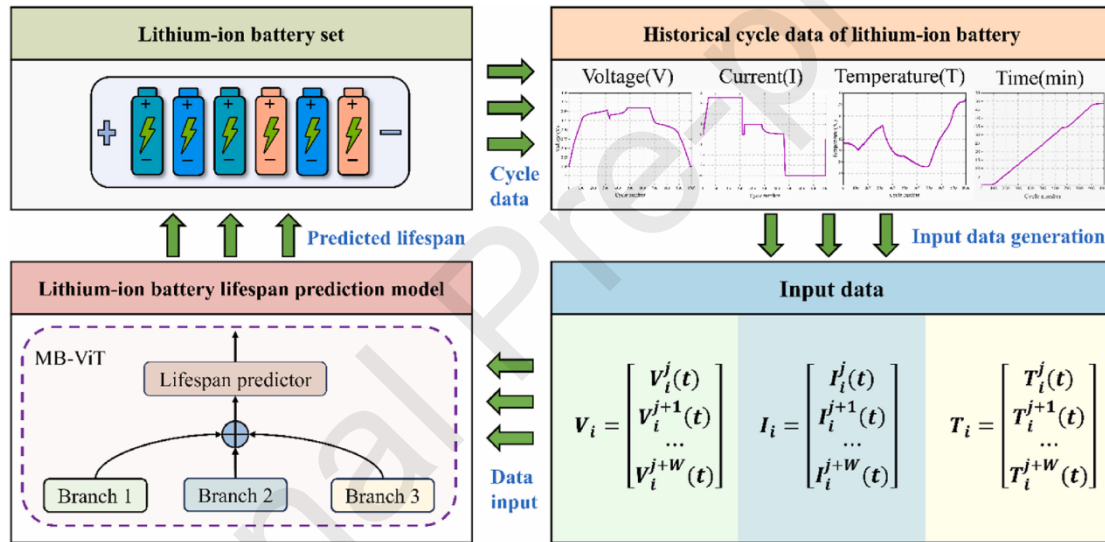


Fig. 27. Framework of lifespan early prediction method for Lithium-ion battery [301].

To overcome the challenge of limited early cycle historical data, Zhao et al. [302] proposed a deep learning model called CNN-AT, fusing attention mechanisms and transfer learning. The study used the capacity change curve of the battery's early discharge phase as input data. It utilized CNN to automatically extract features from input data. Additionally, by integrating attention layers into convolutional layers, it emphasized key information through weight recalibration, thereby enhancing the extraction of relevant features. Finally, utilizing transfer learning strategies, parameters from pre-trained models were selectively transferred to the target model, which was then partially retrained using limited target data. Experimental results

demonstrated that the model performed exceptionally well in terms of both prediction accuracy and generalization capability.

4.3.2. Process-based lifespan prediction

Although prediction based on operational data offers high accuracy, its inherent latency cannot meet the manufacturing demand for zero-defect delivery. Therefore, a more challenging yet valuable research direction is to take a step forward, specifically predicting the future structural durability and fatigue life of joints directly based on high-frequency process signals during welding, such as voltage and current waveforms. The theoretical basis of the dimension lies in the fact that transient energy input fluctuations during the welding process directly determine the micro-metallographic structure and effective bonding area of the joint. These micro-features serve as the fundamental internal causes determining whether the joint can withstand long-term vibration and thermal expansion stresses [303].

However, directly predicting the overall service life of battery systems based on welding process data remains highly challenging, primarily due to the scarcity of labeled lifetime data and the multifactorial nature of battery degradation, which involves complex electrochemical, thermal, and mechanical interactions. As a result, research in the direction remains relatively limited. In comparison, the fatigue performance of welded joints, as a critical determinant of structural integrity, is more directly governed by welding-induced microstructural features and thus provides a more tractable entry point for early-stage prediction. Particularly under new energy vehicle operating conditions, mechanical fatigue failure of welded joints may precede electrochemical aging, directly leading to module open circuits or sharp increases in internal resistance. Consequently, establishing a prediction model mapping welding process features to fatigue life has become key to ensuring the long-term mechanical reliability of battery systems under complex operating conditions [304].

Park and Kang [305] developed a BPNN to predict the fatigue life of spot welds under different geometric dimensions and loading conditions. The study utilized critical geometric indicators of the welded joint, such as nugget diameter, and complex loading conditions, such as R-ratio and loading angle, as multi-dimensional input variables, successfully establishing a non-linear mapping relationship between them and fatigue life. Experimental results indicated that the BPNN model could

predict the fatigue life of spot welds with high accuracy, verifying the practical value of utilizing data-driven methods to quantify long-term mechanical reliability.

Addressing the challenge of predicting the fatigue life of defective welded joints, Liu et al. [306] proposed a hybrid prediction model fusing damage tolerance theory and SVR. The model fully accounted for critical fatigue factors such as the geometric features and locations of welding defects. Experiments demonstrated that the method, featuring a deep fusion of physics and data, significantly outperformed single physical models or purely data-driven methods in terms of lifespan prediction accuracy. This provides a highly promising technical approach for resolving early fatigue failure caused by micro-defects in battery welding.

5. Challenges and future perspectives

Although machine learning has demonstrated tremendous potential for transitioning from offline assistance to online intelligence in the field of laser welding for EV batteries, deploying it robustly from controlled laboratory environments to large-scale, high-cycle industrial production lines still faces the daunting challenge of the final stage of industrial deployment. It is not merely an issue of algorithmic optimization but rather a convergence of multiple contradictions involving the uncertainty of physical metallurgical mechanisms, the complexity of industrial field data, and the computational constraints of edge computing.

5.1 Challenges in industrial deployment

5.1.1 Data scarcity and annotation costs

In the intelligent advancement of laser welding for EV batteries, data is considered a critical strategic asset. However, the industry currently encounters a significant paradox characterized by an abundance of raw data contrasted with a scarcity of actionable information. Despite modern welding workstations generating gigabytes of process monitoring data per second via photodiodes, high-speed cameras, and acoustic sensors, the raw data remains underutilized for supervised learning models absent precise quality labels known as ground truth. In contrast to standard image classification tasks in computer vision, the annotation of battery weld quality proves to be both prohibitively costly and destructive. As discussed in Section 2, critical defects in copper-aluminum dissimilar metal welding, including deep micro-

porosity, incomplete penetration, and excessively thick brittle IMC layers, are frequently located internally within the weld volume or at microscopic interfaces, rendering them undetectable through surface visual inspection. Obtaining accurate labels for these defects necessitates expensive and time-consuming procedures including industrial CT scans, destructive metallographic sectioning, and tensile peel tests. However, driven by the desire to circumvent these exorbitant costs, a prevailing yet alarming trend in current literature is the over-reliance on superficial visual inspections or indirect proxy signals to annotate datasets, bypassing rigorous physical validation. Many studies boast excessively high prediction accuracies for internal defects based on datasets where the ground truth is highly questionable [307–309]. It must be critically pointed out that high algorithmic precision on an improperly annotated or unvalidated dataset is fundamentally scientifically meaningless. To truly bridge the gap between AI algorithms and physical metallurgy, future research must shift away from evaluating models solely based on algorithmic metrics. It is imperative to mandate rigorous metallographic validation as the unequivocal gold standard for dataset annotation. This includes techniques such as destructive metallographic cross-sectioning for morphological verification, EBSD for microstructural evolution and phase transition analysis, or high-resolution industrial CT scanning. Consequently, training a high-precision regression model to predict joint strength may necessitate the destruction of thousands of high-cost cell samples and require weeks of physical testing. The resource-intensive and inefficient data acquisition paradigm leads to a severe shortage of high-quality annotated data. As a result, deep learning models demonstrating high performance in Section 4 often face the risk of overfitting during actual deployment. Furthermore, mature battery production lines typically achieve yield rates exceeding 99.9%, making defect samples rare events within the overall data stream. The extreme class imbalance biases models toward predicting all samples as qualified, thereby failing to identify critical yet rare long-tail defects. It fundamentally compromises the reliability of AI in critical quality control applications.

5.1.2 Multi-physics data synchronization and decoupling

When constructing multimodal fusion monitoring systems, achieving precise spatio-temporal alignment and feature decoupling of heterogeneous sensor data constitutes a significant physical barrier to implementing high-precision models.

Laser welding is a transient process involving the strong coupling of multiple physical fields including light, heat, flow, and force with its core physical phenomena occurring at microsecond or nanosecond scales. As discussed in Section 3, different sensors possess entirely distinct temporal resolutions and physical fields of view. Photodiodes achieve sampling rates of hundreds of kHz to capture instantaneous fluctuations in plasma light intensity. Conversely, high-speed cameras are typically constrained by transmission bandwidth, making it difficult to capture high-frequency oscillations in the molten pool, whereas acoustic signals are inherently limited by the propagation speed of sound waves in air which introduces natural physical delays. Achieving microsecond-level precise synchronization of these signals varying vastly in frequency and physical properties along the time axis presents an extremely challenging engineering problem. Even minute synchronization errors can cause the model to erroneously associate a sudden laser power change at one moment with a molten pool morphology change at another, thereby establishing incorrect causal mappings. More intractable is the problem of signal feature decoupling where fluctuations in sensor signals often result from multiple overlapping factors. For instance, a decrease in photoelectric signal intensity may stem from keyhole instability and collapse representing defect formation, or it could simply be due to disturbances in the shielding gas flow or spatter obscuring the sensor probe. Although current end-to-end deep learning models demonstrate strong fitting capabilities as shown in Section 4, most lack explicit modeling of physical mechanisms, making it difficult to distinguish between process noise and defect features. The black-box nature characterized by knowing the outcome without understanding the underlying cause leads to misjudgments when confronting complex coupled signals, preventing true precise deconstruction of the physical state of welding.

5.1.3 Real-time constraints and edge computing

As the production capacity demands for EV battery manufacturing continue to rise, cycle times have been compressed to their absolute limits, placing significant demands on the real-time inference capabilities of machine learning models. On a typical battery module production line, the welding process for a single tab or busbar is often completed within tens to hundreds of milliseconds. To achieve the adaptive closed-loop control envisioned in Section 3, intelligent systems must complete the entire process including data acquisition, preprocessing, model inference, and control

command issuance at the instant welding occurs. The total latency must be controlled within milliseconds or even lower. However, the real-time requirement inherently conflicts with the need for improved model accuracy. As reviewed in Section 4, researchers tend to employ deeper and more complex network architectures with larger parameter counts, including Transformer-based attention models or dual-stream convolutional neural networks, to enhance defect detection rates or joint performance prediction accuracy. While these massive models perform excellently on high-performance GPU servers, deploying them on resource-constrained edge computing devices in industrial settings such as embedded industrial control computers or smart cameras often results in inference delays that fail to meet the stringent standards of real-time control. The disparity between algorithmic complexity and edge computing power has resulted in many advanced AI models being confined to post-weld offline analysis. They cannot actively intervene in the welding process for real-time quality intervention and defect prevention, thereby limiting the practical value of intelligent welding systems in improving yield rates.

5.1.4 Model generalization and robustness

Moving from the idealized laboratory environment to the unpredictable industrial setting, the greatest challenge for machine learning models lies in their generalization capability and robustness. Existing research predominantly relies on clean data collected under strictly controlled experimental conditions where material surfaces exhibit high consistency, fixtures are assembled with precision, and environmental interference is minimal. However, the actual battery production line environment is far more complex than the laboratory. Minor contamination on laser protective lenses, variations in surface oxide layers due to material batch changes, fluctuations in assembly gaps caused by fixture wear, and even shifts in workshop temperature and humidity can induce domain shift in the data distribution captured by sensors.

More importantly, the reviewed studies often treat different welding tasks as if they belonged to a uniform data domain, although this assumption is rarely valid in industrial practice. In EV battery manufacturing, seemingly similar applications, such as prismatic-cell busbar welding and cylindrical-cell tab welding, may differ substantially in material combination, thickness stack-up, joint geometry, heat dissipation path, structural constraint, and acceptable penetration range. Likewise, even under the same nominal welding task, variations in optical setup, including

camera angle, magnification, illumination wavelength, filter configuration, and sensor position, can significantly alter the observed morphology and signal distribution. Therefore, a model trained on one specific material system or optical setup may experience severe degradation in predictive accuracy when deployed in a slightly modified manufacturing environment.

As analyzed in Section 2, the process window for copper-aluminum laser welding is inherently narrow and highly sensitive to external perturbations. Consequently, high accuracy achieved under a single experimental condition should not be directly interpreted as evidence of robust industrial applicability. Furthermore, industrial production involves numerous long-tail defects, defined as abnormalities with extremely low occurrence probabilities, diverse manifestations, and no presence in training datasets, including specific spatter trajectories or rare molten pool oscillation patterns. Current supervised learning models are fundamentally based on statistical induction from historical data and often lack the ability to reason about and respond to unseen anomaly patterns. This vulnerability under cross-domain and open-world conditions means that relying solely on AI for quality control still carries significant safety risks at the current stage, and remains a major concern for automakers when introducing intelligent welding technologies.

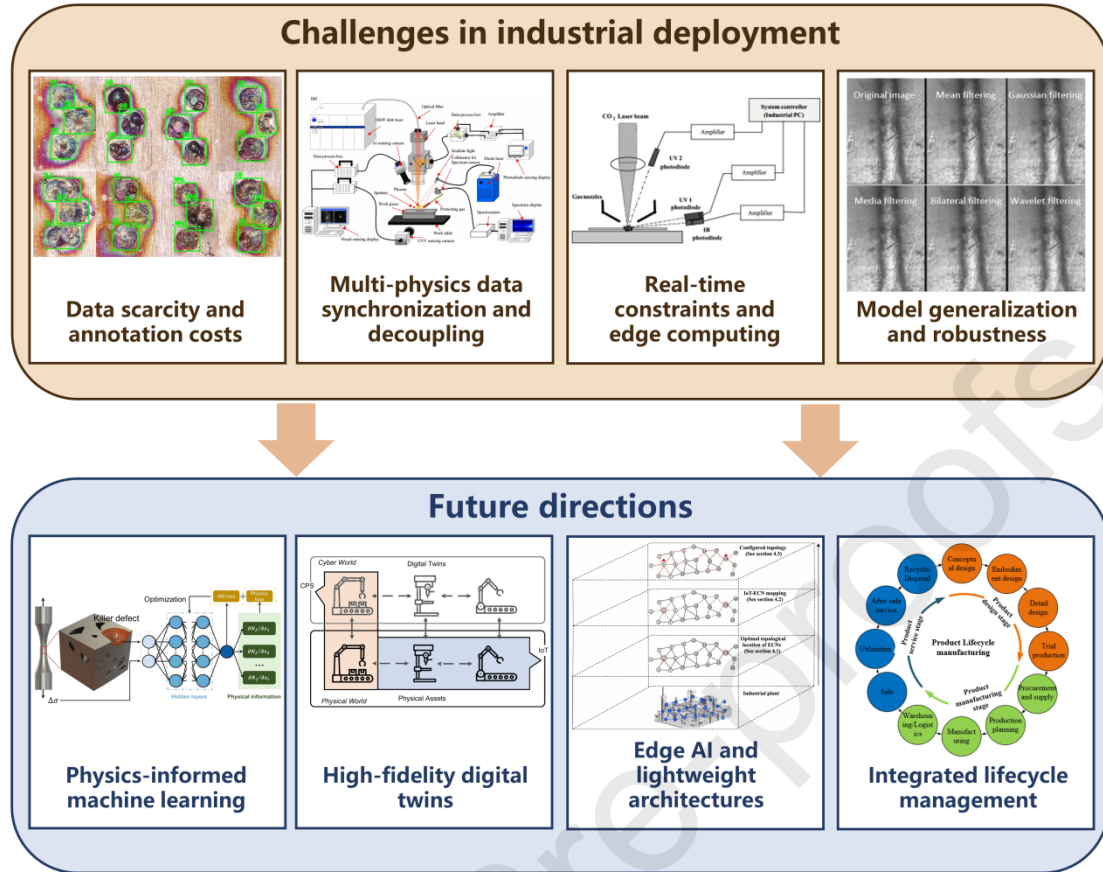


Fig. 28. Summary chart of challenges and future perspectives[147,310–316].

5.2 Future directions and technological pathways

5.2.1 Physics-informed machine learning

Addressing the inherent limitations of purely data-driven models, such as poor physical interpretability and weak generalization capabilities, future research will focus on breaking down the barriers between physical mechanisms and artificial intelligence to establish a new paradigm of physics-informed machine learning (PIML) [317,318]. As analyzed in Section 2, laser welding is not a simple input-output black box but a multi-physics coupled process strictly governed by heat conduction equations, Navier-Stokes equations, and phase-change dynamics. While traditional deep learning models demonstrated exceptional feature extraction capabilities in Section 4, they often disregard these fundamental physical laws. The oversight may lead to erroneous predictions under rare operating conditions, potentially violating energy conservation or thermodynamic principles. Physical-informative neural networks (PINNs) [319,320], explicitly embed partial differential

equations (PDEs) governing weld pool flow as regularization terms within the loss function, compelling the model to satisfy physical constraints while learning data distributions. The dual-driven data and mechanism approach revolutionizes the small sample challenge discussed earlier in Section 5. By incorporating universal physical prior knowledge, the model's reliance on expensive experimental data is significantly reduced, requiring only a small number of physical samples to calibrate boundary conditions. More importantly, PIML endows the model with logical reasoning capabilities for unseen operating conditions, enabling it to robustly predict Cu/Al dissimilar metal welding behavior across varying thicknesses and alloy compositions. The approach fundamentally resolves model failure issues in industrial scenarios caused by process window drift.

5.2.2 High-fidelity digital twins and sim-to-real transfer

To overcome the bottleneck of high training costs and safety concerns associated with reinforcement learning on physical entities, constructing high-fidelity digital twins [321–323] and achieving seamless sim-to-real transfer [38,324] will be the essential path toward Level 4 autonomous welding systems. Although the reinforcement learning algorithms mentioned in Section 3 possess powerful adaptive decision-making potential, their trial-and-error processes, often requiring millions of iterations, are unacceptable in actual battery production lines. Future research will focus on constructing high-fidelity virtual welding environments featuring full thermo-fluid-solid coupling. Leveraging high-performance computing, these environments will reproduce transient physical phenomena in laser-material interactions, including plasma eruptions, keyhole evolution, and spatter formation. On the virtual foundation, agents can undergo large-scale parallel training at zero cost and risk, exploring optimal control strategies under extreme process parameters. To address the reality gap where simulation models cannot perfectly replicate real-world physical details, domain randomization [325,326] will be widely applied. By introducing random perturbations such as laser power fluctuations, beam jitter, and variations in material absorption rates during simulation, agents are forced to learn universal features with strong robustness against environmental noise. Once these strategies, forged through rigorous testing in virtual simulation environments, are transferred to physical welding machines, robots can leverage their generalized

capabilities to achieve millisecond-level adaptive suppression of welding defects, even when confronted with unprecedented complex operating conditions.

5.2.3 Edge AI and lightweight architectures

As EV battery manufacturing strives for extreme efficiency, edge AI [327–329], which offloads AI processing to sensor terminals to mitigate cloud computing latency and bandwidth constraints, is poised to become an industry standard. Although the Transformers and two-stage detection networks mentioned in **Section 4** achieve exceptional accuracy, their massive parameter counts and computational overhead make deployment on embedded chips at the welding gun tip impractical, failing to meet the >1kHz real-time closed-loop control requirements. Future technological evolution will advance along two dimensions, specifically model lightweighting and dedicated hardware acceleration. At the algorithmic level, leveraging the sparsity and temporal characteristics of welding signals, research will deepen exploration into network pruning, weight quantization, and knowledge distillation techniques. The goal is to compress model size to one-tenth or less of its original volume without significant accuracy loss, enabling smooth operation on resource-constrained edge devices. On the hardware front, dedicated AI acceleration chips based on FPGA or ASIC will be integrated into smart cameras or laser controllers, enabling unified sensing-computing-control processing of molten pool images or optoelectronic signals. The extreme edge computing capability will endow welding equipment with reflex-like responsiveness, enabling it to detect precursors of keyhole collapse within microsecond-level time windows and instantly compensate laser power before defects form. It fundamentally transforms the current passive quality management model of post-detection.

5.2.4 Integrated lifecycle management via manufacturing-service data loops

The transformative potential of intelligent welding technology lies not only in improving manufacturing yield rates but also in eliminating data silos between production and operational phases, thereby establishing a closed-loop quality management system driven by full-lifecycle data. Current quality control systems largely stop at battery shipment, lacking forward-looking insights into the aging behavior of welded joints under long-term vibration and thermal cycling conditions in actual vehicles. Future research will focus on establishing unified data indexing

standards to correlate high-frequency waveform data from the welding process, representing the digital fingerprint of each weld point, with long-term in-service performance data from the battery management system (BMS), including internal resistance growth and capacity decay [330–332]. Extracting latent statistical patterns from these massive datasets enables not only the precise prediction of full-lifecycle battery reliability but also a systemic feedback loop of in-service failure data into R&D and production. It signifies that future welding process optimization will no longer pursue solely high tensile strength and porosity-free outcomes. Instead, it will proactively design microstructures with enhanced fatigue resistance and more stable electrical connections based on data feedback. It marks a paradigm shift from passive quality screening to active performance design, providing a core data engine for manufacturing next-generation high-safety, long-life EV batteries.

6. Conclusion

As the core component of the EV industry, the manufacturing quality of EV batteries directly impacts the safety and performance of end products. Laser welding, serving as an indispensable key process in battery manufacturing, has seen its level of intelligence become a crucial indicator of manufacturing advancement. This paper has systematically reviewed the current applications and core developments of machine learning technologies in the field of EV battery laser welding. Focusing on the two central themes of process control and quality assurance, the review provides an in-depth analysis of the current status, challenges, and future prospects of the cutting-edge interdisciplinary field, drawing the following key conclusions.

First, machine learning has driven a fundamental paradigm shift in welding process control from trial-and-error to intelligent adaptation. Traditional open-loop control relies on extensive and time-consuming offline experiments to determine static process windows, making it difficult to address dynamic disturbances during production. The introduction of machine learning technology has effectively addressed the limitation. During the offline phase, surrogate models based on GPR and SVR, combined with multi-objective optimization algorithms such as NSGA-II, have successfully achieved efficient optimization of welding process parameters. The approach demonstrates capabilities augmenting human experience in resolving the Pareto conflict between mechanical strength and electrical conductivity. During the

online phase, deep learning models achieve real-time perception of keyhole stability, molten pool dynamics, and spatter behavior by integrating multimodal sensor data from vision, acoustics, and photodiodes. Furthermore, model-free control strategies represented by deep reinforcement learning empower the welding system to autonomously adjust process parameters in milliseconds through closed-loop regulation. This effectively suppresses defects including porosity and incomplete penetration, marking a significant step toward autonomous and intelligent process control.

Second, machine learning has transformed quality assurance systems, enabling a transition from lagging spot checks to real-time comprehensive inspection. Traditional destructive sampling methods based on metallographic sections or tensile tests, along with offline CT scanning, fail to meet the stringent zero-defect requirements of large-scale EV battery manufacturing due to their inherent delays and high costs. Machine learning-driven quality assurance systems bridge the gap. In defect identification, the application of CNN and transformer architectures enables systems to automatically detect cracks, porosity, and spatter on weld surfaces with exceptional precision and speed. Their real-time capabilities now meet the cycle time demands of high-speed production lines. For performance prediction, deep learning regression models based on process signals, such as CNN-LSTM networks, establish strong nonlinear mappings between transient characteristics of the welding process and the final mechanical strength or contact resistance of the joint. It enables online quantitative prediction of critical performance metrics for each weld point without damaging the workpiece, providing robust technical support for achieving 100% comprehensive virtual quality inspection.

Despite significant progress, industrial applications in this field still face formidable challenges in data, modeling, and system integration. Key obstacles that future research must overcome include the scarcity of high-quality and fully annotated industrial datasets, the need to enhance model generalization, robustness, and interpretability, and the deep real-time integration of algorithms with production line hardware. Emerging technologies, exemplified by Physics-Informed Neural Networks, advanced learning paradigms tailored for industrial small-sample scenarios such as transfer learning and self-supervised learning, and laser welding digital twins, will provide powerful tools to address these challenges. The convergence of these

technologies is poised to enhance the transparency, precision, and adaptability of laser welding in EV battery manufacturing.

In summary, the integration of machine learning and laser welding represents not merely a technical refinement of a single process step but a profound evolution in the field of EV battery manufacturing. By transforming data into deep insights regarding physical processes and decision-making capabilities, it has successfully established a closed-loop data chain spanning process optimization, real-time monitoring, and quality assessment. Such intelligent manufacturing paradigm not only significantly enhances the yield and reliability of welded joints but also lays the foundation for building a future battery smart manufacturing ecosystem based on full lifecycle data. Closer collaboration between academia and industry remains essential to jointly tackle technical challenges, accelerate the transformation and implementation of research outcomes, and contribute to the high-quality development of the global new energy industry.

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Reference

- [1] J.Y. Yong, V.K. Ramachandramurthy, K.M. Tan, N. Mithulananthan, A review on the state-of-the-art technologies of electric vehicle, its impacts and prospects, *Renewable and Sustainable Energy Reviews* 49 (2015) 365–385. <https://doi.org/10.1016/j.rser.2015.04.130>.
- [2] M.A.I. Malik, M.A. Kalam, A. Ikram, S. Zeeshan, S.Q. Raza Zahidi, Energy transition towards electric vehicle technology: Recent advancements, *Energy Reports* 13 (2025) 2958–2996. <https://doi.org/10.1016/j.egy.2025.02.029>.
- [3] R. Zaino, V. Ahmed, A.M. Alhammadi, M. Alghoush, Electric Vehicle Adoption: A Comprehensive Systematic Review of Technological, Environmental, Organizational and Policy Impacts, *World Electric Vehicle Journal* 15 (2024) 375. <https://doi.org/10.3390/wevj15080375>.
- [4] F. Duffner, M. Wentker, M. Greenwood, J. Leker, Battery cost modeling: A review and directions for future research, *Renewable and Sustainable Energy Reviews* 127 (2020) 109872. <https://doi.org/10.1016/j.rser.2020.109872>.
- [5] B. Nykvist, M. Nilsson, Rapidly falling costs of battery packs for electric vehicles, *Nature Clim Change* 5 (2015) 329–332.

- <https://doi.org/10.1038/nclimate2564>.
- [6] J.B. Goodenough, K.-S. Park, The Li-Ion Rechargeable Battery: A Perspective, *J. Am. Chem. Soc.* 135 (2013) 1167–1176. <https://doi.org/10.1021/ja3091438>.
 - [7] K. Liu, K. Li, Q. Peng, C. Zhang, A brief review on key technologies in the battery management system of electric vehicles, *Front. Mech. Eng.* 14 (2019) 47–64. <https://doi.org/10.1007/s11465-018-0516-8>.
 - [8] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia, X. He, Thermal runaway mechanism of lithium ion battery for electric vehicles: A review, *Energy Storage Materials* 10 (2018) 246–267. <https://doi.org/10.1016/j.ensm.2017.05.013>.
 - [9] A. Turetskyy, S. Thiede, M. Thomitzek, N. von Drachenfels, T. Pape, C. Herrmann, Toward Data-Driven Applications in Lithium-Ion Battery Cell Manufacturing, *Energy Technology* 8 (2020) 1900136. <https://doi.org/10.1002/ente.201900136>.
 - [10] A. Kwade, W. Haselrieder, R. Leithoff, A. Modlinger, F. Dietrich, K. Droeder, Current status and challenges for automotive battery production technologies, *Nat Energy* 3 (2018) 290–300. <https://doi.org/10.1038/s41560-018-0130-3>.
 - [11] A. Das, D. Li, D. Williams, D. Greenwood, Joining Technologies for Automotive Battery Systems Manufacturing, *World Electric Vehicle Journal* 9 (2018) 22. <https://doi.org/10.3390/wevj9020022>.
 - [12] S.S. Lee, T.H. Kim, S.J. Hu, W.W. Cai, J.A. Abell, Joining Technologies for Automotive Lithium-Ion Battery Manufacturing: A Review, in: *American Society of Mechanical Engineers Digital Collection*, 2011: pp. 541–549. <https://doi.org/10.1115/MSEC2010-34168>.
 - [13] M.J. Brand, P.A. Schmidt, M.F. Zaeh, A. Jossen, Welding techniques for battery cells and resulting electrical contact resistances, *Journal of Energy Storage* 1 (2015) 7–14. <https://doi.org/10.1016/j.est.2015.04.001>.
 - [14] T. Solchenbach, P. Plapper, Mechanical characteristics of laser braze-welded aluminium–copper connections, *Optics & Laser Technology* 54 (2013) 249–256. <https://doi.org/10.1016/j.optlastec.2013.06.003>.
 - [15] V. Dimatteo, A. Ascari, A. Fortunato, Continuous laser welding with spatial beam oscillation of dissimilar thin sheet materials (Al-Cu and Cu-Al): Process optimization and characterization, *Journal of Manufacturing Processes* 44 (2019) 158–165. <https://doi.org/10.1016/j.jmapro.2019.06.002>.
 - [16] J. Feng, P. Zhang, H. Yan, H. Shi, Q. Lu, Z. Liu, D. Wu, T. Sun, R. Li, Q. Wang, Application of Laser Welding in Electric Vehicle Battery Manufacturing: A Review, *Coatings* 13 (2023) 1313. <https://doi.org/10.3390/coatings13081313>.
 - [17] P. Stavropoulos, K. Sabatakakis, H. Bikas, Welding Challenges and Quality Assurance in Electric Vehicle Battery Pack Manufacturing, *Batteries* 10 (2024) 146. <https://doi.org/10.3390/batteries10050146>.
 - [18] T. Solchenbach, P. Plapper, W. Cai, Electrical performance of laser braze-welded aluminum–copper interconnects, *Journal of Manufacturing Processes*

- 16 (2014) 183–189. <https://doi.org/10.1016/j.jmapro.2013.12.002>.
- [19] N. Kumar, I. Masters, A. Das, In-depth evaluation of laser-welded similar and dissimilar material tab-to-busbar electrical interconnects for electric vehicle battery pack, *Journal of Manufacturing Processes* 70 (2021) 78–96. <https://doi.org/10.1016/j.jmapro.2021.08.025>.
- [20] M. He, L.G. Hector, F. Dai, F. Xu, S. Kolluri, N. Hardin, M. Cai, Industry needs for practical lithium-metal battery designs in electric vehicles, *Nat Energy* 9 (2024) 1199–1205. <https://doi.org/10.1038/s41560-024-01624-5>.
- [21] A.F.H. Kaplan, J. Powell, Spatter in laser welding, *J. Laser Appl.* 23 (2011) 032005. <https://doi.org/10.2351/1.3597830>.
- [22] S. Yan, Z. Li, L. Song, Y. Zhang, S. Wei, Research and development status of laser micro-welding of aluminum-copper dissimilar metals: A review, *Optics and Lasers in Engineering* 161 (2023) 107312. <https://doi.org/10.1016/j.optlaseng.2022.107312>.
- [23] D. Zuo, S. Hu, J. Shen, Z. Xue, Intermediate layer characterization and fracture behavior of laser-welded copper/aluminum metal joints, *Materials & Design* 58 (2014) 357–362. <https://doi.org/10.1016/j.matdes.2014.02.004>.
- [24] Q. Wang, B. Mao, S.I. Stoliarov, J. Sun, A review of lithium ion battery failure mechanisms and fire prevention strategies, *Progress in Energy and Combustion Science* 73 (2019) 95–131. <https://doi.org/10.1016/j.pecs.2019.03.002>.
- [25] D.Y. You, X.D. Gao, S. Katayama, Review of laser welding monitoring, *Science and Technology of Welding and Joining* 19 (2014) 181–201. <https://doi.org/10.1179/1362171813Y.0000000180>.
- [26] W. Li, H. Dong, B. Zhang, S. Zou, W. Mu, Y. Cai, The influence of adjustable ring-mode laser on the formation of intermetallic compounds and mechanical properties of ultra-thin Al-Cu lap welded joints, *Journal of Materials Processing Technology* 332 (2024) 118537. <https://doi.org/10.1016/j.jmatprotec.2024.118537>.
- [27] T.N. Examilioti, P. Papanikos, N. Kashaev, B. Klusemann, N.D. Alexopoulos, Experimental and numerical investigation of laser beam-welded Al–Cu–Li joints using micro-mechanical characteristics, *Journal of Materials Research and Technology* 19 (2022) 2431–2446. <https://doi.org/10.1016/j.jmrt.2022.05.197>.
- [28] C. Zhao, N.D. Parab, X. Li, K. Fezzaa, W. Tan, A.D. Rollett, T. Sun, Critical instability at moving keyhole tip generates porosity in laser melting, *Science* 370 (2020) 1080–1086. <https://doi.org/10.1126/science.abd1587>.
- [29] G. Diot, A. Koudri-David, H. Walaszek, S. Guégan, J. Flifla, Non-destructive Testing of Porosity in Laser Welded Aluminium Alloy Plates: Laser Ultrasound and Frequency-Bandwidth Analysis, *J Nondestruct Eval* 32 (2013) 354–361. <https://doi.org/10.1007/s10921-013-0189-5>.
- [30] P.M. Attia, E. Moch, P.K. Herring, Challenges and opportunities for high-quality battery production at scale, *Nat Commun* 16 (2025) 611. <https://doi.org/10.1038/s41467-025-55861-7>.

- [31] N. Kumar, V.V. Pamarthi, C. Harris, E. Burbidge, I. Masters, Dual-mode laser beam welding of similar and dissimilar material tab-to-busbar for electric vehicle battery pack, *Journal of Advanced Joining Processes* 10 (2024) 100250. <https://doi.org/10.1016/j.jajp.2024.100250>.
- [32] S. Ali, J. Shin, Improvement in weld quality of Al and Cu joints using dual-beam laser welding, *Journal of Manufacturing Processes* 119 (2024) 499–510. <https://doi.org/10.1016/j.jmapro.2024.04.010>.
- [33] J. Li, J. Zhao, Y. Luo, J. Huang, Y. Ding, Fusing optical coherence tomography and coaxial image for contact width and penetration depth monitoring during laser lap welding, *Optics & Laser Technology* 183 (2025) 112294. <https://doi.org/10.1016/j.optlastec.2024.112294>.
- [34] S. Hwang, J. Lee, Classification of battery laser welding defects via enhanced image preprocessing methods and explainable artificial intelligence-based verification, *Engineering Applications of Artificial Intelligence* 133 (2024) 108311. <https://doi.org/10.1016/j.engappai.2024.108311>.
- [35] M. Liu, G. Li, H. Wang, Z. Wan, N. Bilinski, Y. Cai, F. Lu, Spectrum and vision based battery busbar weld quality inspection, *Journal of Manufacturing Processes* 152 (2025) 885–898. <https://doi.org/10.1016/j.jmapro.2025.08.054>.
- [36] M. Nilsen, F. Sikström, Integrated vision-based seam tracking system for robotic laser welding of curved closed square butt joints, *Int J Adv Manuf Technol* 137 (2025) 3387–3399. <https://doi.org/10.1007/s00170-025-15357-6>.
- [37] J. Gu, Z. Wang, J. Kuen, L. Ma, A. Shahroudy, B. Shuai, T. Liu, X. Wang, G. Wang, J. Cai, T. Chen, Recent advances in convolutional neural networks, *Pattern Recognition* 77 (2018) 354–377. <https://doi.org/10.1016/j.patcog.2017.10.013>.
- [38] P. Geng, Y. Xia, Z. Dong, B. Men, B. Pan, C. Shao, Y. Li, J. Li, Machine learning applications in welding processes: Progresses and opportunities, *International Journal of Machine Tools and Manufacture* 213 (2025) 104344. <https://doi.org/10.1016/j.ijmachtools.2025.104344>.
- [39] D. You, X. Gao, S. Katayama, Multisensor Fusion System for Monitoring High-Power Disk Laser Welding Using Support Vector Machine, *IEEE Transactions on Industrial Informatics* 10 (2014) 1285–1295. <https://doi.org/10.1109/TII.2014.2309482>.
- [40] S.A. Shevchik, C. Kenel, C. Leinenbach, K. Wasmer, Acoustic emission for in situ quality monitoring in additive manufacturing using spectral convolutional neural networks, *Additive Manufacturing* 21 (2018) 598–604. <https://doi.org/10.1016/j.addma.2017.11.012>.
- [41] B. Zhang, K.-M. Hong, Y.C. Shin, Deep-learning-based porosity monitoring of laser welding process, *Manufacturing Letters* 23 (2020) 62–66. <https://doi.org/10.1016/j.mfglet.2020.01.001>.
- [42] D. Tang, J. Zhang, P. Wang, Y. He, Weld TOFD defect classification method based on multi-scale CNN and cascaded focused attention, *Journal of Manufacturing Processes* 138 (2025) 157–168.

- <https://doi.org/10.1016/j.jmapro.2025.02.019>.
- [43] Z. Jin, H. Li, H. Gao, An intelligent weld control strategy based on reinforcement learning approach, *Int. J. Adv. Manuf. Technol.* 100 (2019) 2163–2175. <https://doi.org/10.1007/s00170-018-2864-2>.
 - [44] M.Z. Jamaludin, C.L.E. Swartz, Dynamic real-time optimization with closed-loop prediction, *AIChE Journal* 63 (2017) 3896–3911. <https://doi.org/10.1002/aic.15752>.
 - [45] F. Deng, Y. Huang, S. Lu, Y. Chen, J. Chen, H. Feng, J. Zhang, Y. Yang, J. Hu, T.L. Lam, F. Xia, A Multi-Sensor Data Fusion System for Laser Welding Process Monitoring, *IEEE Access* 8 (2020) 147349–147357. <https://doi.org/10.1109/ACCESS.2020.3015529>.
 - [46] Y. Zhang, D. You, X. Gao, N. Zhang, P.P. Gao, Welding defects detection based on deep learning with multiple optical sensors during disk laser welding of thick plates, *Journal of Manufacturing Systems* 51 (2019) 87–94. <https://doi.org/10.1016/j.jmsy.2019.02.004>.
 - [47] X. Wang, U. Zscherpel, P. Tripicchio, S. D’Avella, B. Zhang, J. Wu, Z. Liang, S. Zhou, X. Yu, A comprehensive review of welding defect recognition from X-ray images, *Journal of Manufacturing Processes* 140 (2025) 161–180. <https://doi.org/10.1016/j.jmapro.2025.02.039>.
 - [48] W. Cai, J. Wang, P. Jiang, L. Cao, G. Mi, Q. Zhou, Application of sensing techniques and artificial intelligence-based methods to laser welding real-time monitoring: A critical review of recent literature, *Journal of Manufacturing Systems* 57 (2020) 1–18. <https://doi.org/10.1016/j.jmsy.2020.07.021>.
 - [49] W. Shi, J. Huang, Y. Xie, Y. Li, F. An, Laser micro-welding technology for Cu–Al dissimilar metals and mechanisms of weld defect formation, *Int J Adv Manuf Technol* 93 (2017) 4197–4201. <https://doi.org/10.1007/s00170-017-0814-z>.
 - [50] S. Kohl, F. Kaufmann, M. Schmidt, Why Color Matters—Proposing a Quantitative Stability Criterion for Laser Beam Processing of Metals Based on Their Fundamental Optical Properties, *Metals* 12 (2022) 1118. <https://doi.org/10.3390/met12071118>.
 - [51] Awan I.Z., Khan A.Q., Precipitation from Solid Solutions., *Journal of the Chemical Society of Pakistan* 39 (2017) 319.
 - [52] M. Braunovic, Reliability of power connections, *J. Zhejiang Univ. - Sci. A* 8 (2007) 343–356. <https://doi.org/10.1631/jzus.2007.A0343>.
 - [53] S. Tavassoli, M. Abbasi, R. Tahavvori, Controlling of IMCs layers formation sequence, bond strength and electrical resistance in AlCu bimetal compound casting process, *Materials & Design* 108 (2016) 343–353. <https://doi.org/10.1016/j.matdes.2016.06.076>.
 - [54] A. Sadeghian, N. Iqbal, A review on dissimilar laser welding of steel-copper, steel-aluminum, aluminum-copper, and steel-nickel for electric vehicle battery manufacturing, *Optics & Laser Technology* 146 (2022) 107595. <https://doi.org/10.1016/j.optlastec.2021.107595>.

- [55] M. Abbasi, A. Karimi Taheri, M.T. Salehi, Growth rate of intermetallic compounds in Al/Cu bimetal produced by cold roll welding process, *Journal of Alloys and Compounds* 319 (2001) 233–241. [https://doi.org/10.1016/S0925-8388\(01\)00872-6](https://doi.org/10.1016/S0925-8388(01)00872-6).
- [56] W. Huang, W. Cai, T.J. Rinker, J. Bracey, W. Tan, Effects of laser oscillation on metal mixing, microstructure, and mechanical property of Aluminum–Copper welds, *International Journal of Machine Tools and Manufacture* 188 (2023) 104020. <https://doi.org/10.1016/j.ijmachtools.2023.104020>.
- [57] S. Ali, J. Shin, Improvement in weld quality of Al and Cu joints using dual-beam laser welding, *Journal of Manufacturing Processes* 119 (2024) 499–510. <https://doi.org/10.1016/j.jmapro.2024.04.010>.
- [58] P. Pan, J. Wang, S. Chen, W. Tao, Y. Wang, Q. Yin, Z. Lin, S. Yang, Effect of filler metal on microstructure and mechanical properties of welded SiCp/Al composites by dual beam laser welding, *Materials Letters* 372 (2024) 136984. <https://doi.org/10.1016/j.matlet.2024.136984>.
- [59] B. Ma, X. Gao, Y. Huang, Z. Zhou, Y. Zhang, Effect of pulse shape energy ratio on the laser welding of aluminum and copper, *Optics & Laser Technology* 183 (2025) 112360. <https://doi.org/10.1016/j.optlastec.2024.112360>.
- [60] Z. Wu, J. Wan, Y. Zhang, C. Li, Y. Liu, C. Yang, The influence of welding speed on nanosecond laser welding of AZ31B magnesium alloy and 304 stainless steel, *Optics & Laser Technology* 168 (2024) 109997. <https://doi.org/10.1016/j.optlastec.2023.109997>.
- [61] S. Katayama, *Fundamentals and details of laser welding*, Springer, 2020.
- [62] A. Sadeghian, N. Iqbal, Blue laser welding of low thickness Ni-coated copper and mild steel for electric vehicle (EV) battery manufacturing, *Optics & Laser Technology* 155 (2022) 108415. <https://doi.org/10.1016/j.optlastec.2022.108415>.
- [63] A. Das, R. Fritz, M. Finuf, I. Masters, Blue laser welding of multi-layered AISI 316L stainless steel micro-foils, *Optics & Laser Technology* 132 (2020) 106498. <https://doi.org/10.1016/j.optlastec.2020.106498>.
- [64] N. Iqbal, A. Sadeghian, Fundamental study of blue wavelength laser for welding low thickness dissimilar Cu and steel materials, *Materials Today Communications* 36 (2023) 106604. <https://doi.org/10.1016/j.mtcomm.2023.106604>.
- [65] Q.L. Zhu, J.M. Yu, C. Xu, X. Li, Z.H. Zhang, Z.R. Hu, X.N. Wang, H. Nagaumi, Advancing near-infrared and blue hybrid laser welding: Energy efficiency and microstructural refinement in Al-Mg-Si dissimilar joints, *Journal of Materials Processing Technology* 340 (2025) 118842. <https://doi.org/10.1016/j.jmatprotec.2025.118842>.
- [66] H. Yang, Z. Tang, L. Wan, Q. Wei, J. Wu, A. Wang, X. Jin, X. Li, Y. Wu, G. Lu, H. Wang, H. Wang, Achieving crack-free CuCrZr/AlSi7Mg interface by infrared-blue hybrid laser cladding with low power infrared laser, *Journal of Alloys and Compounds* 931 (2023) 167572.

- <https://doi.org/10.1016/j.jallcom.2022.167572>.
- [67] Y. Liu, N. Tan, Y. Li, G. Zhang, W. Yin, G. Li, Y. Cai, Y. Zhou, S. Han, C. Liu, Microstructure, hardness, and tribological properties of CoCrFeNiX (X = Mo, Ti, W) high entropy alloy coating by red-blue composite laser cladding on copper alloy, *Surface and Coatings Technology* 483 (2024) 130761. <https://doi.org/10.1016/j.surfcoat.2024.130761>.
- [68] T. Li, M. Zhang, W. Zeng, L. Dai, B. Cheng, Y. Niu, Q. Guo, K. Wang, H. Li, X. Wu, X. Liu, R. Huang, Achieving exceptional strength-toughness synergy in aluminum/copper welded joint by coaxial infrared-blue hybrid laser, *Optics & Laser Technology* 198 (2026) 114739. <https://doi.org/10.1016/j.optlastec.2026.114739>.
- [69] F. Kaufmann, R. Möttingdörfer, S. Roth, M. Schmidt, Investigation of coupling efficiency in laser beam welding of copper materials using brilliant infrared and green laser radiation, *Journal of Manufacturing Processes* 131 (2024) 2037–2050. <https://doi.org/10.1016/j.jmapro.2024.10.028>.
- [70] S. Yan, Y. Shi, Influence of Ni interlayer on microstructure and mechanical properties of laser welded joint of Al/Cu bimetal, *Journal of Manufacturing Processes* 59 (2020) 343–354. <https://doi.org/10.1016/j.jmapro.2020.09.078>.
- [71] J. Karlsson, P. Norman, A.F.H. Kaplan, P. Rubin, J. Lamas, A. Yañez, Observation of the mechanisms causing two kinds of undercut during laser hybrid arc welding, *Applied Surface Science* 257 (2011) 7501–7506. <https://doi.org/10.1016/j.apsusc.2011.03.068>.
- [72] W. Ke, X. Bu, J.P. Oliveira, W. Xu, Z. Wang, Z. Zeng, Modeling and numerical study of keyhole-induced porosity formation in laser beam oscillating welding of 5A06 aluminum alloy, *Optics & Laser Technology* 133 (2021) 106540. <https://doi.org/10.1016/j.optlastec.2020.106540>.
- [73] W. Ke, Z. Zeng, J.P. Oliveira, B. Peng, J. Shen, C. Tan, X. Song, W. Yan, Heat transfer and melt flow of keyhole, transition and conduction modes in laser beam oscillating welding, *International Journal of Heat and Mass Transfer* 203 (2023) 123821. <https://doi.org/10.1016/j.ijheatmasstransfer.2022.123821>.
- [74] B. Ma, X. Gao, Y. Huang, P.P. Gao, Y. Zhang, A review of laser welding for aluminium and copper dissimilar metals, *Optics & Laser Technology* 167 (2023) 109721. <https://doi.org/10.1016/j.optlastec.2023.109721>.
- [75] J. Li, S. Geng, Y. Wang, C. Wang, P. Jiang, Mitigation of porosity in adjustable-ring-mode laser welding of medium-thick aluminum alloy, *International Journal of Heat and Mass Transfer* 227 (2024) 125514. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125514>.
- [76] A.A. Lassila, D. Lönn, T. Andersson, W. Wang, R. Ghasemi, Effects of different laser welding parameters on the joint quality for dissimilar material joints for battery applications, *Optics & Laser Technology* 177 (2024) 111155. <https://doi.org/10.1016/j.optlastec.2024.111155>.
- [77] K. Lee, T.J. Rinker, M.M. Pour, W. Cai, W. Huang, W. Tan, J. Bracey, J. Li, A study on cracks and IMCs in laser welding of Al and Cu, *Manufacturing Letters*

- 35 (2023) 221–231. <https://doi.org/10.1016/j.mfglet.2023.08.026>.
- [78] T. Sun, P. Franciosa, C. Liu, F. Pierro, D. Ceglarek, Effect of Micro Solidification Crack on Mechanical Performance of Remote Laser Welded AA6063-T6 Fillet Lap Joint in Automotive Battery Tray Construction, *Applied Sciences* 11 (2021). <https://doi.org/10.3390/app11104522>.
- [79] F. Hugger, K. Hofmann, S. Kohl, M. Dobler, M. Schmidt, Spatter formation in laser beam welding using laser beam oscillation, *Weld World* 59 (2015) 165–172. <https://doi.org/10.1007/s40194-014-0189-9>.
- [80] W. Ke, Y. Liu, F.B. Teshome, Numerical investigation of thermal dynamics and local dry region evolution during shallow water laser welding, *International Journal of Thermal Sciences* 220 (2026) 110359. <https://doi.org/10.1016/j.ijthermalsci.2025.110359>.
- [81] W. Ke, Y. Liu, F.B. Teshome, Z. Zeng, Numerical study on multiphase evolution and molten pool dynamics of underwater wet laser welding in shallow water environment, *International Journal of Heat and Mass Transfer* 220 (2024) 124976. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124976>.
- [82] S. Grabmann, J. Kriegler, F. Harst, F.J. Günter, M.F. Zaeh, Laser welding of current collector foil stacks in battery production—mechanical properties of joints welded with a green high-power disk laser, *Int J Adv Manuf Technol* 118 (2022) 2571–2586. <https://doi.org/10.1007/s00170-021-07839-0>.
- [83] M. Zhang, M. Feng, C. Chen, X. Yu, G. Lian, Weld Defect Detection: Deep Learning-Based Image Processing and the Mechanisms of Defect Formation, *Arch Computat Methods Eng* (2025). <https://doi.org/10.1007/s11831-025-10407-4>.
- [84] S. Zhang, J. Sun, M. Zhu, L. Zhang, P. Nie, Z. Li, Fiber laser welding of HSLA steel by autogenous laser welding and autogenous laser welding with cold wire methods, *Journal of Materials Processing Technology* 275 (2020) 116353. <https://doi.org/10.1016/j.jmatprotec.2019.116353>.
- [85] N. Kumar, V.V. Pamarthi, J. Moffat, I. Masters, In-depth evaluation of laser welding of thick busbar to 21700 Li-ion cell terminal for electric supercar vehicle battery pack, *Journal of Materials Research and Technology* 33 (2024) 3058–3067. <https://doi.org/10.1016/j.jmrt.2024.10.012>.
- [86] Influencing the Weld Pool During Laser Welding, in: *Advances in Laser Materials Processing*, Woodhead Publishing, 2018: pp. 235–256. <https://doi.org/10.1016/B978-0-08-101252-9.00010-8>.
- [87] Defect formation mechanisms and preventive procedures in laser welding, in: *Handbook of Laser Welding Technologies*, Woodhead Publishing, 2013: pp. 332–373. <https://doi.org/10.1533/9780857098771.2.332>.
- [88] J.L. Huang, N. Warnken, J.-C. Gebelin, M. Strangwood, R.C. Reed, On the mechanism of porosity formation during welding of titanium alloys, *Acta Materialia* 60 (2012) 3215–3225. <https://doi.org/10.1016/j.actamat.2012.02.035>.
- [89] N. Coniglio, C.E. Cross, Initiation and growth mechanisms for weld

- solidification cracking, *International Materials Reviews* 58 (2013) 375–397. <https://doi.org/10.1179/1743280413Y.0000000020>.
- [90] M.J. Zhang, G.Y. Chen, Y. Zhou, S.C. Li, H. Deng, Observation of spatter formation mechanisms in high-power fiber laser welding of thick plate, *Applied Surface Science* 280 (2013) 868–875. <https://doi.org/10.1016/j.apsusc.2013.05.081>.
- [91] N. Kumar, M. Mukherjee, A. Bandyopadhyay, Comparative study of pulsed Nd:YAG laser welding of AISI 304 and AISI 316 stainless steels, *Optics & Laser Technology* 88 (2017) 24–39. <https://doi.org/10.1016/j.optlastec.2016.08.018>.
- [92] K.Y. Benyounis, A.G. Olabi, Optimization of different welding processes using statistical and numerical approaches – A reference guide, *Advances in Engineering Software* 39 (2008) 483–496. <https://doi.org/10.1016/j.advengsoft.2007.03.012>.
- [93] P. Sathiya, K. Panneerselvam, M.Y. Abdul Jaleel, Optimization of laser welding process parameters for super austenitic stainless steel using artificial neural networks and genetic algorithm, *Materials & Design* (1980-2015) 36 (2012) 490–498. <https://doi.org/10.1016/j.matdes.2011.11.028>.
- [94] A.G. Olabi, K.Y. Benyounis, M.S.J. Hashmi, Application of Response Surface Methodology in Describing the Residual Stress Distribution in CO₂ Laser Welding of AISI304, *Strain* 43 (2007) 37–46. <https://doi.org/10.1111/j.1475-1305.2007.00305.x>.
- [95] Y.-A. Tsai, Y.-L. Lo, M.M. Raza, A.N. Saleh, T.-C. Chuang, C.-Y. Chen, C.-P. Chiu, Optimization of lap-joint laser welding parameters using high-fidelity simulations and machine learning mode, *J. Mater. Res. Technol.-JMRT* 24 (2023) 6876–6892. <https://doi.org/10.1016/j.jmrt.2023.04.256>.
- [96] T.-C. Chuang, Y.-L. Lo, H.-C. Tran, Y.-A. Tsai, C.-Y. Chen, C.-P. Chiu, Optimization of Butt-joint laser welding parameters for elimination of angular distortion using High-fidelity simulations and Machine learning, *Optics & Laser Technology* 167 (2023) 109566. <https://doi.org/10.1016/j.optlastec.2023.109566>.
- [97] L. Ding, Q. Lu, S. Liu, R. Xu, X. Yan, X. Xu, M. Lu, Y. Chen, Quality inspection of micro solder joints in laser spot welding by laser ultrasonic method, *Ultrasonics* 118 (2022) 106567. <https://doi.org/10.1016/j.ultras.2021.106567>.
- [98] T. Sun, S. Jabar, N. Kumar, C. Liu, D. Ceglarek, P. Franciosa, The impact of ring-shaped laser beam on dissimilar welding of Al-Cu thin sheets for battery tab-to-busbar connection: Microstructural, mechanical and electrical characteristics, *Optics & Laser Technology* 179 (2024) 111312. <https://doi.org/10.1016/j.optlastec.2024.111312>.
- [99] D. Mery, V. Rizzo, U. Zscherpel, G. Mondragón, I. Lillo, I. Zuccar, H. Lobel, M. Carrasco, GDXray: The Database of X-ray Images for Nondestructive Testing, *J Nondestruct Eval* 34 (2015) 42. <https://doi.org/10.1007/s10921-015->

- 0315-7.
- [100] M. Hamed, M. Atashparva, A review of electrical contact resistance modeling in resistance spot welding, *Weld World* 61 (2017) 269–290.
<https://doi.org/10.1007/s40194-016-0419-4>.
 - [101] H. Zhao, H. Qi, Vision-based keyhole detection in laser full penetration welding process, *J. Laser Appl.* 28 (2016) 022412.
<https://doi.org/10.2351/1.4944003>.
 - [102] M. Luo, Y.C. Shin, Vision-based weld pool boundary extraction and width measurement during keyhole fiber laser welding, *Optics and Lasers in Engineering* 64 (2015) 59–70. <https://doi.org/10.1016/j.optlaseng.2014.07.004>.
 - [103] T. Sibillano, A. Ancona, D. Rizzi, V. Lupo, L. Tricarico, P.M. Lugarà, Plasma Plume Oscillations Monitoring during Laser Welding of Stainless Steel by Discrete Wavelet Transform Application, *Sensors* 10 (2010) 3549–3561.
<https://doi.org/10.3390/s100403549>.
 - [104] M. Schweier, M.W. Haubold, M.F. Zaeh, Analysis of spatters in laser welding with beam oscillation: A machine vision approach, *CIRP Journal of Manufacturing Science and Technology* 14 (2016) 35–42.
<https://doi.org/10.1016/j.cirpj.2016.05.004>.
 - [105] L. Zhang, A.C. Basantes-Defaz, D. Ozevin, E. Indacochea, Real-time monitoring of welding process using air-coupled ultrasonics and acoustic emission, *Int J Adv Manuf Technol* 101 (2019) 1623–1634.
<https://doi.org/10.1007/s00170-018-3042-2>.
 - [106] X. Gao, Y. Zhang, Monitoring of welding status by molten pool morphology during high-power disk laser welding, *Optik - International Journal for Light and Electron Optics* 126 (2015) 1797–1802.
<https://doi.org/10.1016/j.ijleo.2015.04.060>.
 - [107] G. Zhang, C.S. Wu, X. Liu, Single vision system for simultaneous observation of keyhole and weld pool in plasma arc welding, *Journal of Materials Processing Technology* 215 (2015) 71–78.
<https://doi.org/10.1016/j.jmatprotec.2014.07.033>.
 - [108] L.J. Zhang, J.X. Zhang, A. Gumenyuk, M. Rethmeier, S.J. Na, Numerical simulation of full penetration laser welding of thick steel plate with high power high brightness laser, *Journal of Materials Processing Technology* 214 (2014) 1710–1720. <https://doi.org/10.1016/j.jmatprotec.2014.03.016>.
 - [109] D. You, X. Gao, S. Katayama, Visual-based spatter detection during high-power disk laser welding, *Optics and Lasers in Engineering* 54 (2014) 1–7.
<https://doi.org/10.1016/j.optlaseng.2013.09.010>.
 - [110] F. Chen, X. Chen, X. Xie, X. Feng, L. Yang, Full-field 3D measurement using multi-camera digital image correlation system, *Optics and Lasers in Engineering* 51 (2013) 1044–1052.
<https://doi.org/10.1016/j.optlaseng.2013.03.001>.
 - [111] Z. Ye, G. Fang, S. Chen, J.J. Zou, Passive vision based seam tracking system for pulse-MAG welding, *Int J Adv Manuf Technol* 67 (2013) 1987–1996.

- <https://doi.org/10.1007/s00170-012-4625-y>.
- [112] Y. Zhang, C. Zhang, L. Tan, S. Li, Coaxial monitoring of the fibre laser lap welding of Zn-coated steel sheets using an auxiliary illuminant, *Optics & Laser Technology* 50 (2013) 167–175.
<https://doi.org/10.1016/j.optlastec.2013.03.001>.
- [113] F. Kong, J. Ma, B. Carlson, R. Kovacevic, Real-time monitoring of laser welding of galvanized high strength steel in lap joint configuration, *Optics & Laser Technology* 44 (2012) 2186–2196.
<https://doi.org/10.1016/j.optlastec.2012.03.003>.
- [114] B. Xue, B. Chang, G. Peng, Y. Gao, Z. Tian, D. Du, G. Wang, A Vision Based Detection Method for Narrow Butt Joints and a Robotic Seam Tracking System, *Sensors* 19 (2019) 1144. <https://doi.org/10.3390/s19051144>.
- [115] J. Xiong, S. Zou, Active vision sensing and feedback control of back penetration for thin sheet aluminum alloy in pulsed MIG suspension welding, *J. Process Control* 77 (2019) 89–96.
<https://doi.org/10.1016/j.jprocont.2019.03.013>.
- [116] X. Chen, Q. Ma, Z. He, X. Sun, Y. Ren, Real-time detection and localization method for weld seam of narrow butt joint based on semantic segmentation, *Meas. Sci. Technol.* 35 (2024) 035205. <https://doi.org/10.1088/1361-6501/ad16b9>.
- [117] B. Xue, B. Chang, D. Du, Multi-Output Monitoring of High-Speed Laser Welding State Based on Deep Learning, *Sensors* 21 (2021) 1626.
<https://doi.org/10.3390/s21051626>.
- [118] K. Nam, H. Ki, Vision-to-code framework for automated weld seam tracking in laser welding, *Journal of Manufacturing Processes* 156 (2025) 63–78.
<https://doi.org/10.1016/j.jmapro.2025.10.111>.
- [119] F. Tenner, B. Berg, C. Brock, F. Klämpfl, M. Schmidt, Experimental approach for quantification of fluid dynamics in laser metal welding, *J. Laser Appl.* 27 (2015) S29003. <https://doi.org/10.2351/1.4906302>.
- [120] Visual sensing of the weld pool geometry from the topside view in keyhole plasma arc welding, *Journal of Manufacturing Processes* 26 (2017) 74–83.
<https://doi.org/10.1016/j.jmapro.2017.01.011>.
- [121] Weld seam profile detection and feature point extraction for multi-pass route planning based on visual attention model, *Robotics and Computer-Integrated Manufacturing* 37 (2016) 251–261. <https://doi.org/10.1016/j.rcim.2015.04.005>.
- [122] Y. Ma, J. Fan, H. Yang, L. Yang, Z. Ji, F. Jing, M. Tan, A Fast and Robust Seam Tracking Method for Spatial Circular Weld Based on Laser Visual Sensor, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 1–11. <https://doi.org/10.1109/TIM.2021.3106685>.
- [123] Y. Zou, T. Chen, X. Chen, J. Li, Robotic seam tracking system combining convolution filter and deep reinforcement learning, *Mechanical Systems and Signal Processing* 165 (2022) 108372.
<https://doi.org/10.1016/j.ymssp.2021.108372>.

- [124] A.R. Konuk, R.G.K.M. Aarts, A.J.H. in't Veld, T. Sibillano, D. Rizzi, A. Ancona, Process Control of Stainless Steel Laser Welding using an Optical Spectroscopic Sensor, *Physics Procedia* 12 (2011) 744–751. <https://doi.org/10.1016/j.phpro.2011.03.093>.
- [125] D. Rizzi, T. Sibillano, P.P. Calabrese, A. Ancona, P.M. Lugara, Spectroscopic, energetic and metallographic investigations of the laser lap welding of AISI 304 using the response surface methodology, *Opt. Lasers Eng.* 49 (2011) 892–898. <https://doi.org/10.1016/j.optlaseng.2011.02.014>.
- [126] T. Sibillano, D. Rizzi, A. Ancona, S. Saludes-Rodil, J. Rodriguez-Nieto, H. Chmelickova, H. Sebestova, Spectroscopic monitoring of penetration depth in CO₂ Nd:YAG and fiber laser welding processes, *J. Mater. Process. Technol.* 212 (2012) 910–916. <https://doi.org/10.1016/j.jmatprotec.2011.11.016>.
- [127] T. Sibillano, A. Ancona, D. Rizzi, S.S. Rodil, J.R. Nieto, A.R. Konuk, R. Aarts, A.J. Huis in't Veld, Study on the correlation between plasma electron temperature and penetration depth in laser welding processes, *Physics Procedia* 5 (2010) 429–436. <https://doi.org/10.1016/j.phpro.2010.08.070>.
- [128] S. Kang, M. Kang, Y.H. Jang, C. Kim, Deep learning-based penetration depth prediction in Al/Cu laser welding using spectrometer signal and CCD image, *J. Laser Appl.* 34 (2022) 042035. <https://doi.org/10.2351/7.0000767>.
- [129] S. Kang, M. Kang, Y. Hoon Jang, C. Kim, Spectrometer as a quantitative sensor for predicting the weld depth in laser welding, *Optics & Laser Technology* 175 (2024) 110855. <https://doi.org/10.1016/j.optlastec.2024.110855>.
- [130] I. Eriksson, J. Powell, A.F.H. Kaplan, Signal overlap in the monitoring of laser welding, *Meas. Sci. Technol.* 21 (2010) 105705. <https://doi.org/10.1088/0957-0233/21/10/105705>.
- [131] G. Chianese, P. Franciosa, J. Nolte, D. Ceglarek, S. Patalano, Characterization of Photodiodes for Detection of Variations in Part-to-Part Gap and Weld Penetration Depth During Remote Laser Welding of Copper-to-Steel Battery Tab Connectors, *J. Manuf. Sci. Eng.-Trans. ASME* 144 (2022) 071004. <https://doi.org/10.1115/1.4052725>.
- [132] L. Shu, Y. Zhou, W. Wu, P. Jiang, Y. Wang, Online diagnosis of weak welds in busbar laser welding based on photoelectric signals and machine learning, *Engineering Applications of Artificial Intelligence* 156 (2025) 111277. <https://doi.org/10.1016/j.engappai.2025.111277>.
- [133] K. Lee, S. Kang, M. Kang, S. Yi, C. Kim, Estimation of Al/Cu laser weld penetration in photodiode signals using deep neural network classification, *J. Laser Appl.* 33 (2021) 042009. <https://doi.org/10.2351/7.0000506>.
- [134] Y. Luo, L. Zhu, J. Han, X. Xie, R. Wan, Y. Zhu, Study on the acoustic emission effect of plasma plume in pulsed laser welding, *Mechanical Systems and Signal Processing* 124 (2019) 715–723. <https://doi.org/10.1016/j.ymssp.2019.01.045>.
- [135] M.F.M. Yusof, M. Ishak, M.F. Ghazali, Feasibility of using acoustic method in

- monitoring the penetration status during the Pulse Mode Laser Welding process, *IOP Conf. Ser.: Mater. Sci. Eng.* 238 (2017) 012006. <https://doi.org/10.1088/1757-899X/238/1/012006>.
- [136] N. Lv, Y.L. Xu, G. Fang, X.W. Yu, S.B. Chen, Research on welding penetration state recognition based on BP-Adaboost model for pulse GTAW welding dynamic process, in: *2016 IEEE Workshop on Advanced Robotics and Its Social Impacts (ARSO)*, 2016: pp. 100–105. <https://doi.org/10.1109/ARSO.2016.7736264>.
- [137] S.-S. Ao, Z. Luo, C.-F. Zhao, M.-N. Feng, W.-D. Liu, Numerical and Experimental Study of the Acoustic Signal Generated by Vapour Flow in Laser Welding, *Laser Eng.* 34 (2016) 145–165.
- [138] S. Ao, Z. Luo, M. Feng, F. Yan, Simulation and experimental analysis of acoustic signal characteristics in laser welding, *Int. J. Adv. Manuf. Technol.* 81 (2015) 277–287. <https://doi.org/10.1007/s00170-015-7164-5>.
- [139] W. Huang, R. Kovacevic, A neural network and multiple regression method for the characterization of the depth of weld penetration in laser welding based on acoustic signatures, *J. Intell. Manuf.* 22 (2011) 131–143. <https://doi.org/10.1007/s10845-009-0267-9>.
- [140] N. Lv, J. Zhong, H. Chen, T. Lin, S. Chen, Real-time control of welding penetration during robotic GTAW dynamical process by audio sensing of arc length, *Int J Adv Manuf Technol* 74 (2014) 235–249. <https://doi.org/10.1007/s00170-014-5875-7>.
- [141] Z. Luo, W. Liu, Z. Wang, S. Ao, Monitoring of laser welding using source localization and tracking processing by microphone array, *Int J Adv Manuf Technol* 86 (2016) 21–28. <https://doi.org/10.1007/s00170-015-8095-x>.
- [142] K. He, J. Li, Z. Li, W. Tao, J. Huang, R. Zhou, Detection method of austenitic stainless steel laser welding quality based on the AE signal, *Nondestruct. Test. Eval.* (2025). <https://doi.org/10.1080/10589759.2025.2572396>.
- [143] L. Tomcic, A. Ederer, S. Grabmann, M. Kick, J. Kriegler, M.F. Zaeh, Interpreting acoustic emissions to determine the weld depth during laser beam welding, *J. Laser Appl.* 34 (2022) 042052. <https://doi.org/10.2351/7.0000796>.
- [144] H. Yu, Y. Xu, N. Lv, H. Chen, S. Chen, Arc spectral processing technique with its application to wire feed monitoring in Al–Mg alloy pulsed gas tungsten arc welding, *Journal of Materials Processing Technology* 213 (2013) 707–716. <https://doi.org/10.1016/j.jmatprotec.2012.11.016>.
- [145] Y. Zhang, X. Gao, S. Katayama, Weld appearance prediction with BP neural network improved by genetic algorithm during disk laser welding, *Journal of Manufacturing Systems* 34 (2015) 53–59. <https://doi.org/10.1016/j.jmsy.2014.10.005>.
- [146] H.J. Zhang, F.J. Wang, W.G. Gao, Y.Y. Hou, Quality assessment for resistance spot welding based on binary image of electrode displacement signal and probabilistic neural network, *Science and Technology of Welding and Joining* 19 (2014) 242–249. <https://doi.org/10.1179/1362171813Y.0000000187>.

- [147] G. Liu, X. Gao, D. You, N. Zhang, Prediction of high power laser welding status based on PCA and SVM classification of multiple sensors, *J. Intell. Manuf.* 30 (2019) 821–832. <https://doi.org/10.1007/s10845-016-1286-y>.
- [148] Z. Lei, J. Shen, Q. Wang, Y. Chen, Real-time weld geometry prediction based on multi-information using neural network optimized by PCA and GA during thin-plate laser welding, *J. Manuf. Process.* 43 (2019) 207–217. <https://doi.org/10.1016/j.jmapro.2019.05.013>.
- [149] L. Cao, Y. Yang, P. Jiang, Q. Zhou, G. Mi, Z. Gao, Y. Rong, C. Wang, Optimization of processing parameters of AISI 316L laser welding influenced by external magnetic field combining RBFNN and GA, *Results Phys.* 7 (2017) 1329–1338. <https://doi.org/10.1016/j.rinp.2017.03.029>.
- [150] T. Liu, J. Bao, A Novel Period-Sensitive LSTM for Laser Welding Quality Prediction, *IEEE Transactions on Industrial Informatics* 21 (2025) 830–838. <https://doi.org/10.1109/TII.2024.3463706>.
- [151] Z. Luo, D. Wu, P. Zhang, X. Ye, H. Shi, X. Cai, Y. Tian, Laser Welding Penetration Monitoring Based on Time-Frequency Characterization of Acoustic Emission and CNN-LSTM Hybrid Network, *Materials* 16 (2023) 1614. <https://doi.org/10.3390/ma16041614>.
- [152] M. Luo, Y.C. Shin, Estimation of keyhole geometry and prediction of welding defects during laser welding based on a vision system and a radial basis function neural network, *Int J Adv Manuf Technol* 81 (2015) 263–276. <https://doi.org/10.1007/s00170-015-7079-1>.
- [153] Y. Li, Y. Gao, H. Pan, D. Tao, H. Zhang, Keyhole Depth Prediction in Laser Deep Penetration Welding of Titanium Alloy Based on Spectral Information, *Metals* 15 (2025) 527. <https://doi.org/10.3390/met15050527>.
- [154] H.H.L. Núñez, L.-W. Hsu, K.S.B. Ribeiro, A. Salminen, W.M. Bessa, In-situ monitoring and online prediction of keyhole depth in laser welding by coaxial imaging, *Procedia CIRP* 124 (2024) 793–796. <https://doi.org/10.1016/j.procir.2024.08.227>.
- [155] X. Gao, G. Liu, Elucidation of Metallic Plume and Spatter Characteristics Based on SVM During High-Power Disk Laser Welding, *Plasma Sci. Technol.* 17 (2015) 32–36. <https://doi.org/10.1088/1009-0630/17/1/07>.
- [156] W. Cai, L. Shu, S. Geng, Q. Zhou, L. Cao, Real-time tracking method for motion spatter in high-power laser welding of stainless steel plate based on a lightweight deep learning model, *Expert Syst. Appl.* 254 (2024) 124386. <https://doi.org/10.1016/j.eswa.2024.124386>.
- [157] T. Will, K. Schwarzkopf, C. Hölbling, L. Müller, M. Schmidt, Feature extraction based on scalable hypothesis tests from photodiode data in laser welding processes, *Procedia CIRP* 111 (2022) 527–531. <https://doi.org/10.1016/j.procir.2022.08.084>.
- [158] L. Aucott, H. Dong, W. Mirihanage, R. Atwood, A. Kidess, S. Gao, S. Wen, J. Marsden, S. Feng, M. Tong, T. Connolley, M. Drakopoulos, C.R. Kleijn, I.M. Richardson, D.J. Browne, R.H. Mathiesen, H.V. Atkinson, Revealing internal

- flow behaviour in arc welding and additive manufacturing of metals, *Nat Commun* 9 (2018) 5414. <https://doi.org/10.1038/s41467-018-07900-9>.
- [159] C. Jiang, W. Qiao, G. Yang, F. Liu, Efficient Signal Alignment Algorithm for Undersampling Digital Predistortion, *IEEE Microwave and Wireless Components Letters* 32 (2022) 1487–1490. <https://doi.org/10.1109/LMWC.2022.3192925>.
- [160] R. Alizadeh, J.K. Allen, F. Mistree, Managing computational complexity using surrogate models: a critical review, *Res Eng Design* 31 (2020) 275–298. <https://doi.org/10.1007/s00163-020-00336-7>.
- [161] Response surface methodology (RSM) as a tool for optimization in analytical chemistry, *Talanta* 76 (2008) 965–977. <https://doi.org/10.1016/j.talanta.2008.05.019>.
- [162] J. Yang, H. Zhang, Y. Li, Optimization of Nd:YAG Laser Welding of Aluminum Alloy to Stainless Steel Thin Sheets Via Taguchi Method and Response Surface Methodology (RSM), *Lasers in Engineering* 31 (2015) 141–159.
- [163] K. He, W. Tao, J. Li, J. Yong, Optimization of Excitation Coil Parameters for PECT of Defects in Laser Welding of Power Battery Pack Based on Kriging Surrogate Model, *Journal of Sensors* 2025 (2025) 1301327. <https://doi.org/10.1155/js/1301327>.
- [164] J.P.C. Kleijnen, Kriging metamodeling in simulation: A review, *European Journal of Operational Research* 192 (2009) 707–716. <https://doi.org/10.1016/j.ejor.2007.10.013>.
- [165] H.S. Dhiman, D. Deb, J.M. Guerrero, Hybrid machine intelligent SVR variants for wind forecasting and ramp events, *Renewable and Sustainable Energy Reviews* 108 (2019) 369–379. <https://doi.org/10.1016/j.rser.2019.04.002>.
- [166] Z. Gomolka, Backpropagation algorithm with fractional derivatives, *ITM Web Conf.* 21 (2018) 00004. <https://doi.org/10.1051/itmconf/20182100004>.
- [167] K. Pavlíček, V. Kotlan, I. Doležel, Estimation of laser weld parameters using surrogate modelling technique, *Journal of Electrical Engineering* 69 (2018) 170–176. <https://doi.org/10.2478/jee-2018-0021>.
- [168] H. Dong, M. Cong, Y. Zhang, Y. Liu, H. Chen, Modeling and real-time prediction for complex welding process based on weld pool, *Int J Adv Manuf Technol* 96 (2018) 2495–2508. <https://doi.org/10.1007/s00170-018-1685-7>.
- [169] M.F.A. Zaharuddin, D. Kim, S. Rhee, An ANFIS based approach for predicting the weld strength of resistance spot welding in artificial intelligence development, *J. Mech. Sci. Technol.* 31 (2017) 5467–5476. <https://doi.org/10.1007/s12206-017-1041-0>.
- [170] M.J.H. Rawa, M.H. Razavi Dehkordi, M.J. Kholoud, N.H. Abu-Hamdeh, H. Azimy, Using the numerical simulation and artificial neural network (ANN) to evaluate temperature distribution in pulsed laser welding of different alloys, *Engineering Applications of Artificial Intelligence* 126 (2023) 107025. <https://doi.org/10.1016/j.engappai.2023.107025>.

- [171] S. Kumaresan, K.S.J. Aultrin, S.S. Kumar, M.D. Anand, Transfer Learning With CNN for Classification of Weld Defect, *IEEE Access* 9 (2021) 95097–95108. <https://doi.org/10.1109/ACCESS.2021.3093487>.
- [172] R. Yu, J. Kershaw, P. Wang, Y. Zhang, How to Accurately Monitor the Weld Penetration From Dynamic Weld Pool Serial Images Using CNN-LSTM Deep Learning Model?, *IEEE Robotics and Automation Letters* 7 (2022) 6519–6525. <https://doi.org/10.1109/LRA.2022.3173659>.
- [173] L. Li, F. Cheng, S. Wu, An LSTM-based measurement method of 3D weld pool surface in GTAW, *Measurement* 171 (2021) 108809. <https://doi.org/10.1016/j.measurement.2020.108809>.
- [174] J. Edwin Raja Dhas, S. Kumanan, Evolutionary fuzzy SVR modeling of weld residual stress, *Applied Soft Computing* 42 (2016) 423–430. <https://doi.org/10.1016/j.asoc.2016.01.050>.
- [175] R. Liang, R. Yu, Y. Luo, Y. Zhang, Machine learning of weld joint penetration from weld pool surface using support vector regression, *Journal of Manufacturing Processes* 41 (2019) 23–28. <https://doi.org/10.1016/j.jmapro.2019.01.039>.
- [176] B. Xing, Y. Xiao, Q.H. Qin, H. Cui, Quality assessment of resistance spot welding process based on dynamic resistance signal and random forest based, *Int J Adv Manuf Technol* 94 (2018) 327–339. <https://doi.org/10.1007/s00170-017-0889-6>.
- [177] P. Podrżaj, A. Čebular, The Application of LVQ Neural Network for Weld Strength Evaluation of RF-Welded Plastic Materials, *IEEE/ASME Transactions on Mechatronics* 21 (2016) 1063–1071. <https://doi.org/10.1109/TMECH.2015.2498278>.
- [178] K. Chen, H. Chen, L. Liu, S. Chen, Prediction of weld bead geometry of MAG welding based on XGBoost algorithm, *Int J Adv Manuf Technol* 101 (2019) 2283–2295. <https://doi.org/10.1007/s00170-018-3083-6>.
- [179] A. Avcı, M. Kocakulak, N. Acır, E. Gunes, S. Turan, A study on the monitoring of weld quality using XGBoost with Particle Swarm Optimization, *Ain Shams Engineering Journal* 15 (2024) 102651. <https://doi.org/10.1016/j.asej.2024.102651>.
- [180] S.S. Karganroudi, M. Moradi, M.A. Attar, S.A. Rasouli, M. Ghoreishi, J. Lawrence, H. Ibrahim, Experimental and Numerical Analysis on TIG Arc Welding of Stainless Steel Using RSM Approach, *Metals* 11 (2021). <https://doi.org/10.3390/met11101659>.
- [181] M. Ragavendran, N. Chandrasekhar, R. Ravikumar, R. Saxena, M. Vasudevan, A.K. Bhaduri, Optimization of hybrid laser–TIG welding of 316LN steel using response surface methodology (RSM), *Optics and Lasers in Engineering* 94 (2017) 27–36.
- [182] D. Sterling, T. Sterling, Y. Zhang, H. Chen, Welding parameter optimization based on Gaussian process regression Bayesian optimization algorithm, in: 2015 IEEE International Conference on Automation Science and Engineering

- (CASE), 2015: pp. 1490–1496. <https://doi.org/10.1109/CoASE.2015.7294310>.
- [183] Z. Wang, L. Duan, L. Kuang, H. Zhou, J. Duan, Physics-Informed Gaussian Process Regression with Bayesian Optimization for Laser Welding Quality Control in Coaxial Laser Diodes, *CMC-Comput. Mat. Contin.* 84 (2025) 2587–2604. <https://doi.org/10.32604/cmc.2025.065648>.
- [184] C. Li, S. Li, Y. Xie, J. Wu, C. Zhang, Optimization of laser welding process parameters considering carbon emissions and weld quality based on DBO-BP and NSGA-II, *INT. J. PRECIS. ENG. MAN-GT.* 12 (2025) 1429–1444. <https://doi.org/10.1007/s40684-025-00707-2>.
- [185] X. Zhang, X. Hu, H. Li, Z. Zhang, H. Chen, H. Sun, Research on Predicting Welding Deformation in Automated Laser Welding Processes with an Enhanced DEWOA-BP Algorithm, *Machines* 12 (2024) 307. <https://doi.org/10.3390/machines12050307>.
- [186] F. Zhang, T. Zhou, Process parameter optimization for laser-magnetic welding based on a sample-sorted support vector regression, *J. Intell. Manuf.* 30 (2019) 2217–2230. <https://doi.org/10.1007/s10845-017-1378-3>.
- [187] X. Zhang, Y. Guo, X. Fan, J. Zhao, B. Yan, L. Zhang, Multi-objective optimization of laser welding process parameters of steel/Al based on BO-RF and MOJS, *J. Mech. Sci. Technol.* 38 (2024) 861–871. <https://doi.org/10.1007/s12206-024-0133-x>.
- [188] Z. Zhang, Y. Huang, R. Qin, W. Ren, G. Wen, XGBoost-based on-line prediction of seam tensile strength for Al-Li alloy in laser welding: Experiment study and modelling, *J. Manuf. Process.* 64 (2021) 30–44. <https://doi.org/10.1016/j.jmapro.2020.12.004>.
- [189] Y. Ai, X. Shao, P. Jiang, P. Li, Y. Liu, W. Liu, Welded joints integrity analysis and optimization for fiber laser welding of dissimilar materials, *Opt. Lasers Eng.* 86 (2016) 62–74. <https://doi.org/10.1016/j.optlaseng.2016.05.011>.
- [190] R. Palanivel, A Contemporary Review of the Advancements in Joining Technologies for Battery Applications, *Mater. Tehnol.* 57 (2023) 275–281. <https://doi.org/10.17222/mit.2023.797>.
- [191] H. Aghaei, H.A. Larki, M. Taqizadeh, M. Aoudia, W. Aich, L. Kolsi, M.H.R. Dehkordi, Multi- objective modeling and optimization of dissimilar laser welding by integrating an artificial intelligence predictive model with NSGA-II algorithm, *Mater. Today Commun.* 40 (2024) 109765. <https://doi.org/10.1016/j.mtcomm.2024.109765>.
- [192] Y. Tan, G. Zhu, F. Tian, Z. Zhao, B. Chai, Multi-objective optimization of T-shaped bilateral laser welding parameters based on NSGA-II and MOPSO, *J. Mater. Sci.* 59 (2024) 9547–9573. <https://doi.org/10.1007/s10853-024-09727-w>.
- [193] A.K. Perka, M. John, U.B. Kuruveri, P.L. Menezes, Advanced High-Strength Steels for Automotive Applications: Arc and Laser Welding Process, Properties, and Challenges, *Metals* 12 (2022) 1051. <https://doi.org/10.3390/met12061051>.

- [194] K. Pal, S.K. Pal, Effect of Pulse Parameters on Weld Quality in Pulsed Gas Metal Arc Welding: A Review, *J. of Materi Eng and Perform* 20 (2011) 918–931. <https://doi.org/10.1007/s11665-010-9717-y>.
- [195] F. Di Fiore, M. Nardelli, L. Mainini, Active Learning and Bayesian Optimization: A Unified Perspective to Learn with a Goal, *Arch. Comput. Method Eng.* 31 (2024) 2985–3013. <https://doi.org/10.1007/s11831-024-10064-z>.
- [196] C. Dang, M. Beer, Semi-Bayesian active learning quadrature for estimating extremely low failure probabilities, *Reliab. Eng. Syst. Saf.* 246 (2024) 110052. <https://doi.org/10.1016/j.ress.2024.110052>.
- [197] M. Haas, V. Onuseit, J. Powell, F. Zaiß, J. Wahl, T. Menold, C. Hagenlocher, A. Michalowski, Improving the weld seam quality in laser welding processes by means of Bayesian optimization, *Procedia CIRP* 124 (2024) 772–775. <https://doi.org/10.1016/j.procir.2024.08.222>.
- [198] Z. Zhang, Y. Zhang, F. Luo, J. Li, C. Lu, Y. Zhao, H. Zhang, A. Lu, Convolutional Neural Network Using Bayesian Optimization for Laser Welding Tailor Rolled Blanks Penetration Detection, in: n.d. <https://doi.org/10.1115/MSEC2019-2836>.
- [199] J. Kershaw, R. Yu, Y. Zhang, P. Wang, Hybrid machine learning-enabled adaptive welding speed control, *Journal of Manufacturing Processes* 71 (2021) 374–383. <https://doi.org/10.1016/j.jmapro.2021.09.023>.
- [200] W. Cai, P. Jiang, L. Shu, S. Geng, Q. Zhou, Real-time identification of molten pool and keyhole using a deep learning-based semantic segmentation approach in penetration status monitoring, *J. Manuf. Process.* 76 (2022) 695–707. <https://doi.org/10.1016/j.jmapro.2022.02.058>.
- [201] A. Bollig, D. Abel, Ch. Kratzsch, S. Kaierle, Identification and predictive control of laser beam welding using neural networks, in: 2003 European Control Conference (ECC), 2003: pp. 2457–2462. <https://doi.org/10.23919/ECC.2003.7085334>.
- [202] K. Zhang, Y. Chen, J. Zheng, J. Huang, X. Tang, Adaptive filling modeling of butt joints using genetic algorithm and neural network for laser welding with filler wire, *Journal of Manufacturing Processes* 30 (2017) 553–561. <https://doi.org/10.1016/j.jmapro.2017.10.025>.
- [203] M. Safari, R.J. Alves de Sousa, A.H. Rabiee, V. Tahmasbi, Investigation of Dissimilar Resistance Spot Welding Process of AISI 304 and AISI 1060 Steels with TLBO-ANFIS and Sensitivity Analysis, *Metals* 11 (2021) 1324. <https://doi.org/10.3390/met11081324>.
- [204] D. Wu, J. Chen, H. Liu, P. Zhang, Z. Yu, H. Chen, S. Chen, Weld penetration in situ prediction from keyhole dynamic behavior under time-varying VPPAW pools via the OS-ELM model, *Int. J. Adv. Manuf. Technol.* 104 (2019) 3929–3941. <https://doi.org/10.1007/s00170-019-04142-x>.
- [205] N. Nievas, A. Pagès-Bernaus, F. Bonada, L. Echeverria, X. Domingo, Reinforcement Learning for Autonomous Process Control in Industry 4.0:

- Advantages and Challenges, *Applied Artificial Intelligence* 38 (2024) 2383101. <https://doi.org/10.1080/08839514.2024.2383101>.
- [206] J. Romero-Hdz, B.N. Saha, S. Tstutsumi, R. Fincato, Incorporating domain knowledge into reinforcement learning to expedite welding sequence optimization, *Engineering Applications of Artificial Intelligence* 91 (2020) 103612. <https://doi.org/10.1016/j.engappai.2020.103612>.
- [207] Y. Guo, J. Gao, C. Wu, X. Gui, Correlation between fusion hole morphology and weld penetration in TIG welding, *Int. J. Adv. Manuf. Technol.* 101 (2019) 2991–3000. <https://doi.org/10.1007/s00170-018-3063-x>.
- [208] Q.-C. Nguyen, H.Q.B. Hua, P.-T. Pham, Development of a vision system integrated with industrial robots for online weld seam tracking, *J. Manuf. Process.* 119 (2024) 414–424. <https://doi.org/10.1016/j.jmapro.2024.03.090>.
- [209] G. Gokmen, O. Akgun, T.C. Akinci, S. Seker, Fast fourier transformation of emitted noises from welding machines and their classification with acoustic method, *Mechanika* 23 (2017) 588–593. <https://doi.org/10.5755/j01.mech.23.4.14876>.
- [210] M. Sheng, W. Shi, C. Lu, Z. Lei, Laser-generated guided wave imaging using the PSO-SSWT-FFT method, *Nondestruct. Test. Eval.* (2025). <https://doi.org/10.1080/10589759.2025.2574339>.
- [211] G. Masinelli, T. Le-Quang, S. Zanolli, K. Wasmer, S.A. Shevchik, Adaptive Laser Welding Control: A Reinforcement Learning Approach, *IEEE Access* 8 (2020) 103803–103814. <https://doi.org/10.1109/ACCESS.2020.2998052>.
- [212] Y. Cai, L. Zhao, X. Chen, Z. Li, Deep Reinforcement Learning-Based Motion Control Optimization for Defect Detection System, *Actuators* 14 (2025) 180. <https://doi.org/10.3390/act14040180>.
- [213] R. Yu, Y.-Y. Chen, Coordinated Ship Welding with Optimal Lazy Robot Ratio and Energy Consumption via Reinforcement Learning, *J. Mar. Sci. Eng.* 12 (2024) 1765. <https://doi.org/10.3390/jmse12101765>.
- [214] J. Günther, P.M. Pilarski, G. Helfrich, H. Shen, K. Diepold, Intelligent laser welding through representation, prediction, and control learning: An architecture with deep neural networks and reinforcement learning, *Mechatronics* 34 (2016) 1–11. <https://doi.org/10.1016/j.mechatronics.2015.09.004>.
- [215] K. Arulkumaran, M.P. Deisenroth, M. Brundage, A.A. Bharath, Deep Reinforcement Learning: A Brief Survey, *IEEE Signal Processing Magazine* 34 (2017) 26–38. <https://doi.org/10.1109/MSP.2017.2743240>.
- [216] T.L. Quang, B. Meylan, G. Masinelli, F. Saeidi, S.A. Shevchik, F.V. Farahani, K. Wasmer, Smart closed-loop control of laser welding using reinforcement learning, *Procedia CIRP* 111 (2022) 479–483. <https://doi.org/10.1016/j.procir.2022.08.074>.
- [217] W. Tian, P. Hu, C. Zhang, Optimization framework of laser oscillation welding based on a deep predictive reward reinforcement learning net, *J Intell Manuf* 36 (2025) 4331–4350. <https://doi.org/10.1007/s10845-024-02465-5>.

- [218] H. Alzorgan, A. Razi, A.J. Moshayedi, Invited Paper: Actuator Trajectory Planning for UAVs with Overhead Manipulator using Reinforcement Learning, in: 2023 IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications, Pimrc, IEEE, New York, 2023. <https://doi.org/10.1109/PIMRC56721.2023.10293891>.
- [219] J. Romero-Hdz, B. Saha, G. Toledo-Ramirez, I. Lopez-Juarez, A Reinforcement Learning Based Approach for Welding Sequence Optimization, in: S. Chen, Y. Zhang, Z. Feng (Eds.), Transactions on Intelligent Welding Manufacturing, Springer Singapore, Singapore, 2018: pp. 33–45. https://doi.org/10.1007/978-981-10-7043-3_2.
- [220] F. Fuica, A Posteriori Error Estimates for a Bang-Bang Optimal Control Problem, Appl. Math. Optim. 91 (2025) 74. <https://doi.org/10.1007/s00245-025-10276-0>.
- [221] X. Chen, Bang-Bang and Linear Quadratic Zero-Sum Game for Noncausal and Causal Systems, Mathematical Methods in the Applied Sciences 48 (2025) 10470–10486. <https://doi.org/10.1002/mma.10892>.
- [222] K. Wang, Z. Chen, Z. Wei, F. Lu, J. Li, A New Smoothing Technique for Bang-Bang Optimal Control Problems, in: AIAA SCITECH 2024 Forum, American Institute of Aeronautics and Astronautics, n.d. <https://doi.org/10.2514/6.2024-0727>.
- [223] T. Gao, Y. Wang, X. Wang, Multi-objective optimization of rail welded joint grinding in railroad tracks via reinforcement learning, Eng. Appl. Artif. Intell. 164 (2026) 113386. <https://doi.org/10.1016/j.engappai.2025.113386>.
- [224] Y. He, X. Wang, Path planning of multiple spot-welding digital twin robots based on reinforcement pointer network, Int J Adv Manuf Technol 139 (2025) 587–602. <https://doi.org/10.1007/s00170-025-15840-0>.
- [225] J. Zheng, Y. Zhang, Y. Gao, Z. Chen, Y. Gao, C. Zhou, X. Zhou, Optimization of Multi-Agent Scheduling Based on MA-ID3QN for the Riveting and Welding Work Cell, IEEE Access 13 (2025) 153171–153188. <https://doi.org/10.1109/ACCESS.2025.3604143>.
- [226] Y. He, J. Zhang, M. Wei, Y. Guo, Y. Wang, Y. Huang, Reinforcement Learning Based Motion Planning for Robotic Arm of Welding Robot, in: J. Gu, F. Hu, H. Zhou, Z. Fei, E. Yang (Eds.), Robotics and Autonomous Systems and Engineering Applications of Computational Intelligence, Springer Nature, Singapore, 2024: pp. 59–70. https://doi.org/10.1007/978-981-96-0313-8_5.
- [227] W. Chen, L. Hua, L. Xu, B. Zhang, M. Li, T. Ma, Y.-Y. Chen, MADDPG Algorithm for Coordinated Welding of Multiple Robots, in: 2021 6th International Conference on Automation, Control and Robotics Engineering (CACRE), 2021: pp. 1–5. <https://doi.org/10.1109/CACRE52464.2021.9501327>.
- [228] B.C. Routara, S.D. Mohanty, S. Datta, A. Bandyopadhyay, S.S. Mahapatra, Combined quality loss (CQL) concept in WPCA-based Taguchi philosophy for optimization of multiple surface quality characteristics of UNS C34000 brass in

- cylindrical grinding, *Int J Adv Manuf Technol* 51 (2010) 135–143.
<https://doi.org/10.1007/s00170-010-2599-1>.
- [229] J. Roy, B.D. Barma, J.D. Barma, S.C. Saha, Optimization of Process Parameters for Submerged Arc Welding by Weighted Principal Component Analysis Based Taguchi Method, *Advanced Materials Research* 622–623 (2013) 45–50. <https://doi.org/10.4028/www.scientific.net/AMR.622-623.45>.
- [230] S. Liu, G. Mi, F. Yan, C. Wang, P. Jiang, Correlation of high power laser welding parameters with real weld geometry and microstructure, *Optics & Laser Technology* 94 (2017) 59–67.
<https://doi.org/10.1016/j.optlastec.2017.03.004>.
- [231] N. Chandrasekhar, M. Vasudevan, A.K. Bhaduri, T. Jayakumar, Intelligent modeling for estimating weld bead width and depth of penetration from infra-red thermal images of the weld pool, *J Intell Manuf* 26 (2015) 59–71.
<https://doi.org/10.1007/s10845-013-0762-x>.
- [232] J. Zhang, P.-H. Li, M.-X. Zhang, K.-H. Shen, Y.-Z. Xing, A review of algorithms for surface defect detection in welds based on machine learning, *Expert Systems with Applications* 299 (2026) 130188.
<https://doi.org/10.1016/j.eswa.2025.130188>.
- [233] L. Zhang, H. Pan, B. Jia, L. Li, M. Pan, L. Chen, Lightweight DCGAN and MobileNet based model for detecting X-ray welding defects under unbalanced samples, *Sci Rep* 15 (2025) 6221. <https://doi.org/10.1038/s41598-025-89558-0>.
- [234] L. Xu, R. Xiao, H. Chen, A novel few-shot learning based feature relation model for robotic welding states monitoring, *Journal of Manufacturing Processes* 138 (2025) 203–213. <https://doi.org/10.1016/j.jmapro.2025.02.018>.
- [235] Z. Zhang, G. Wen, S. Chen, Weld image deep learning-based on-line defects detection using convolutional neural networks for Al alloy in robotic arc welding, *Journal of Manufacturing Processes* 45 (2019) 208–216.
<https://doi.org/10.1016/j.jmapro.2019.06.023>.
- [236] B.W. Seo, S.B. Han, J.Y. Heo, S. Kim, Y.T. Cho, A Study on the Development of Deep Learning Algorithm for Determining External Quality of Welded Parts Using Transfer Learning, *J Weld Join* 41 (2023) 132–139.
<https://doi.org/10.5781/JWJ.2023.41.2.7>.
- [237] W. Jiao, Q. Wang, Y. Cheng, Y. Zhang, End-to-end prediction of weld penetration: A deep learning and transfer learning based method, *Journal of Manufacturing Processes* 63 (2021) 191–197.
<https://doi.org/10.1016/j.jmapro.2020.01.044>.
- [238] J. Wang, Z. Zhang, S. Zhang, Z. Bai, R. Qin, J. Huang, Y. Su, G. Wen, X. Chen, Hybrid transfer learning and GAN-driven approach for online detection of welding defects, *Journal of Manufacturing Processes* 135 (2025) 82–99.
<https://doi.org/10.1016/j.jmapro.2024.12.039>.
- [239] K. Han, Y. Wang, H. Chen, X. Chen, J. Guo, Z. Liu, Y. Tang, A. Xiao, C. Xu, Y. Xu, Z. Yang, Y. Zhang, D. Tao, A Survey on Vision Transformer, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (2023) 87–110.

- <https://doi.org/10.1109/TPAMI.2022.3152247>.
- [240] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention Is All You Need, (2023).
<https://doi.org/10.48550/arXiv.1706.03762>.
- [241] L. Liu, C. Hu, Z. Liu, Y. Xiong, L. Wu, B. Ao, Multi-scale feature fusion-enhanced intelligent defect detection for X-ray weld images using CNN-Transformer, *Measurement* 258 (2026) 119182.
<https://doi.org/10.1016/j.measurement.2025.119182>.
- [242] Y. Zhang, Y. Yang, J. Sun, R. Ji, P. Zhang, H. Shan, Surface defect detection of wind turbine based on lightweight YOLOv5s model, *Measurement* 220 (2023) 113222. <https://doi.org/10.1016/j.measurement.2023.113222>.
- [243] R. Liu, L. Zhao, Y. Ren, Z. Shen, L. Li, J. Luo, Z. Abbas, A lightweight model based on multi-scale feature fusion for ultrasonic welding surface defect detection, *Engineering Applications of Artificial Intelligence* 161 (2025) 112208. <https://doi.org/10.1016/j.engappai.2025.112208>.
- [244] Y. Yang, R. Yang, L. Pan, J. Ma, Y. Zhu, T. Diao, L. Zhang, A lightweight deep learning algorithm for inspection of laser welding defects on safety vent of power battery, *Computers in Industry* 123 (2020) 103306.
<https://doi.org/10.1016/j.compind.2020.103306>.
- [245] L. Zhang, Y. He, T. Diao, Z. Liu, A real-time lightweight laser welding defect inspection algorithm based on deep learning, *Computers in Industry* 174 (2026) 104413. <https://doi.org/10.1016/j.compind.2025.104413>.
- [246] A. Chattopadhyay, A. Sarkar, P. Howlader, V.N. Balasubramanian, Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks, in: 2018 IEEE Winter Conference on Applications of Computer Vision (WACV), 2018: pp. 839–847.
<https://doi.org/10.1109/WACV.2018.00097>.
- [247] G. López, P.D. Ramírez, E. Vega, F. Pizarro, J. Toro, C. Parra, WeldVGG: A VGG-Inspired Deep Learning Model for Weld Defect Classification from Radiographic Images with Visual Interpretability, *Sensors* 25 (2025).
<https://doi.org/10.3390/s25196183>.
- [248] S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39 (2017) 1137–1149.
<https://doi.org/10.1109/TPAMI.2016.2577031>.
- [249] C. Ajmi, J. Zapata, S. Elferchichi, K. Laabidi, Advanced Faster-RCNN Model for Automated Recognition and Detection of Weld Defects on Limited X-Ray Image Dataset, *J Nondestruct Eval* 43 (2023) 14.
<https://doi.org/10.1007/s10921-023-01032-x>.
- [250] W. Yang, Y. Xiao, H. Shen, Z. Wang, Generalized weld bead region of interest localization and improved faster R-CNN for weld defect recognition, *Measurement* 222 (2023) 113619.
<https://doi.org/10.1016/j.measurement.2023.113619>.

- [251] S. Ruan, C. Zhan, B. Liu, Q. Wan, K. Song, A high precision YOLO model for surface defect detection based on PyConv and CISBA, *Sci Rep* 15 (2025) 15841. <https://doi.org/10.1038/s41598-025-91930-z>.
- [252] B. Eren, Deep learning approaches for weld defect detection: A comprehensive review of models, applications, and future directions, *Computers & Industrial Engineering* 212 (2026) 111725. <https://doi.org/10.1016/j.cie.2025.111725>.
- [253] Y. Luo, J. Ling, J. Wang, H. Zhang, F. Chen, X. Xiao, N. Lu, SFW-YOLO: A lightweight multi-scale dynamic attention network for weld defect detection in steel bridge inspection, *Measurement* 253 (2025) 117608. <https://doi.org/10.1016/j.measurement.2025.117608>.
- [254] L. Liu, M. Zhang, X. Zhou, R. Zhai, D. Wang, S. Zhu, A weld defect detection algorithm designed based on YOLOv9, *Measurement* 257 (2026) 118630. <https://doi.org/10.1016/j.measurement.2025.118630>.
- [255] Y. Yang, Y. Zhou, N.U. Din, J. Li, Y. He, L. Zhang, An Improved YOLOv5 Model for Detecting Laser Welding Defects of Lithium Battery Pole, *Applied Sciences* 13 (2023). <https://doi.org/10.3390/app13042402>.
- [256] S. Deshpande, V. Venugopal, M. Kumar, S. Anand, Deep learning-based image segmentation for defect detection in additive manufacturing: an overview, *Int J Adv Manuf Technol* 134 (2024) 2081–2105. <https://doi.org/10.1007/s00170-024-14191-6>.
- [257] J. Pei, J. Wang, F. Su, X-ray weld image segmentation based on improved U-net model, *Structures* 80 (2025) 110058. <https://doi.org/10.1016/j.istruc.2025.110058>.
- [258] X. Wu, D. Hong, J. Chanussot, UIU-Net: U-Net in U-Net for Infrared Small Object Detection, *IEEE Transactions on Image Processing* 32 (2023) 364–376. <https://doi.org/10.1109/TIP.2022.3228497>.
- [259] Y. Yang, Y. He, H. Guo, Z. Chen, L. Zhang, Semantic segmentation supervised deep-learning algorithm for welding-defect detection of new energy batteries, *Neural Comput & Applic* 34 (2022) 19471–19484. <https://doi.org/10.1007/s00521-022-07474-0>.
- [260] D. Wang, Y. Zheng, W. Dai, D. Tang, Y. Peng, Deep Network-Assisted Quality Inspection of Laser Welding on Power Battery, *Sensors* 23 (2023). <https://doi.org/10.3390/s23218894>.
- [261] K. He, G. Gkioxari, P. Dollar, R. Girshick, Mask R-CNN, in: 2017: pp. 2961–2969. https://openaccess.thecvf.com/content_iccv_2017/html/He_Mask_R-CNN_ICCV_2017_paper.html (accessed January 28, 2026).
- [262] C. Kim, S. Hwang, H. Sohn, Weld crack detection and quantification using laser thermography, mask R-CNN, and CycleGAN, *Automation in Construction* 143 (2022) 104568. <https://doi.org/10.1016/j.autcon.2022.104568>.
- [263] Z.-H. Zhou, A brief introduction to weakly supervised learning, *Natl Sci Rev* 5 (2018) 44–53. <https://doi.org/10.1093/nsr/nwx106>.
- [264] X. Wu, T. Wang, Y. Li, P. Li, Y. Liu, A CAM-Based Weakly Supervised Method for Surface Defect Inspection, *IEEE Transactions on Instrumentation*

- and Measurement 71 (2022) 1–10. <https://doi.org/10.1109/TIM.2022.3168895>.
- [265] S. Zhang, Y. Zhang, Automated weld defect segmentation from phased array ultrasonic data based on U-net architecture, *NDT & E International* 146 (2024) 103165. <https://doi.org/10.1016/j.ndteint.2024.103165>.
- [266] Y. Zhu, R. Yang, Y. He, J. Ma, H. Guo, Y. Yang, L. Zhang, A Lightweight Multiscale Attention Semantic Segmentation Algorithm for Detecting Laser Welding Defects on Safety Vent of Power Battery, *IEEE Access* 9 (2021) 39245–39254. <https://doi.org/10.1109/ACCESS.2021.3064180>.
- [267] N. Munir, H.-J. Kim, S.-J. Song, S.-S. Kang, Investigation of deep neural network with drop out for ultrasonic flaw classification in weldments, *J Mech Sci Technol* 32 (2018) 3073–3080. <https://doi.org/10.1007/s12206-018-0610-1>.
- [268] J.D. Cullen, N. Athi, M. Al-Jader, P. Johnson, A.I. Al-Shamma'a, A. Shaw, A.M.A. El-Rasheed, Multisensor fusion for on line monitoring of the quality of spot welding in automotive industry, *Measurement* 41 (2008) 412–423. <https://doi.org/10.1016/j.measurement.2007.01.006>.
- [269] A. Mohammed, M. Hussain, Advances and Challenges in Deep Learning for Automated Welding Defect Detection: A Technical Survey, *IEEE Access* 13 (2025) 94553–94569. <https://doi.org/10.1109/ACCESS.2025.3574083>.
- [270] B. Das, S. Pal, S. Bag, Torque based defect detection and weld quality modelling in friction stir welding process, *Journal of Manufacturing Processes* 27 (2017) 8–17. <https://doi.org/10.1016/j.jmapro.2017.03.012>.
- [271] A. Darwish, M. Persson, S. Ericson, R. Ghasemi, K. Salomonsson, Weld Defect Detection in Laser Beam Welding Using Multispectral Emission Sensor Features and Machine Learning, *Sensors* 25 (2025). <https://doi.org/10.3390/s25165120>.
- [272] G. Chianese, P. Franciosa, T. Sun, D. Ceglarek, S. Patalano, Using photodiodes and supervised machine learning for automatic classification of weld defects in laser welding of thin foils copper-to-steel battery tabs, *J. Laser Appl.* 34 (2022) 042040. <https://doi.org/10.2351/7.0000800>.
- [273] T. Wang, J. Chen, X. Gao, Y. Qin, Real-time Monitoring for Disk Laser Welding Based on Feature Selection and SVM, *Applied Sciences* 7 (2017). <https://doi.org/10.3390/app7090884>.
- [274] Z. Zhang, G. Wen, S. Chen, Audible Sound-Based Intelligent Evaluation for Aluminum Alloy in Robotic Pulsed GTAW: Mechanism, Feature Selection, and Defect Detection, *IEEE Transactions on Industrial Informatics* 14 (2018) 2973–2983. <https://doi.org/10.1109/TII.2017.2775218>.
- [275] L. Caprio, B. Previtali, A.G. Demir, Sensor Selection and Defect Classification via Machine Learning During the Laser Welding of Busbar Connections for High-Performance Battery Pack Production, *Lasers Manuf. Mater. Process.* 11 (2024) 329–352. <https://doi.org/10.1007/s40516-023-00238-1>.
- [276] B.-J. Kim, Y.-M. Kim, C. Kim, Transfer Learning-Based Multi-Sensor Approach for Predicting Keyhole Depth in Laser Welding of 780DP Steel, *Materials* 18 (2025) 3961. <https://doi.org/10.3390/ma18173961>.

- [277] S. Hochreiter, J. Schmidhuber, Long Short-Term Memory, *Neural Computation* 9 (1997) 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>.
- [278] B. Wang, Y. Li, Y. Luo, X. Li, T. Freiheit, Early event detection in a deep-learning driven quality prediction model for ultrasonic welding, *Journal of Manufacturing Systems* 60 (2021) 325–336. <https://doi.org/10.1016/j.jmsy.2021.06.009>.
- [279] Y. Liang, B. Yu, M. Ding, W. Hu, Y. Jin, Y. Yuan, WELD-DETR: A Real-Time Welding Defect Detection Framework with Multi-Scale Feature Fusion and Multi-Kernel Perception Optimization, *Sensors* 25 (2025) 7024. <https://doi.org/10.3390/s25227024>.
- [280] P. Peng, K. Fan, X. Fan, H. Zhou, Z. Guo, Real-Time Defect Detection Scheme Based on Deep Learning for Laser Welding System, *IEEE Sensors Journal* 23 (2023) 17301–17309. <https://doi.org/10.1109/JSEN.2023.3277732>.
- [281] S.V. Sotubadi, A. Hosseinzadeh, H. Ghassemi-Armaki, M.M. Pour, J. Ma, X. Zhao, J.T. Bracey, B.E. Carlson, P.E. Wang, Harnessing unlabeled plant data and labeled lab data for enhanced quality prediction in laser welding, *Journal of Manufacturing Processes* 157 (2026) 1015–1034. <https://doi.org/10.1016/j.jmapro.2025.12.023>.
- [282] S. Kang, K. Lee, M. Kang, Y.H. Jang, C. Kim, Weld-penetration-depth estimation using deep learning models and multisensor signals in Al/Cu laser overlap welding, *Optics & Laser Technology* 161 (2023) 109179. <https://doi.org/10.1016/j.optlastec.2023.109179>.
- [283] Z.-H. Lai, S. Xu, S.J. Clark, K. Fezzaa, J. Li, Unveiling mechanisms and onset threshold of humping in high-speed laser welding, *Nat Commun* 15 (2024) 9546. <https://doi.org/10.1038/s41467-024-53888-w>.
- [284] N.B. Pugacheva, S.M. Zadvorkin, N.S. Michurov, EBSD Analysis of an Austenitic Cr–Ni Steel Laser Weld, *Phys. Metals Metallogr.* 123 (2022) 791–796. <https://doi.org/10.1134/S0031918X22080087>.
- [285] P. Lehto, H. Remes, EBSD characterisation of grain size distribution and grain sub-structures for ferritic steel weld metals, *Weld World* 66 (2022) 363–377. <https://doi.org/10.1007/s40194-021-01225-w>.
- [286] W. Sun, Y. Xia, M. Chen, G. Fan, A study on the role of local residual stress on the local strength and defect growth in laser-welded Ti-6Al-4V, *Journal of Alloys and Compounds* 1018 (2025) 179174. <https://doi.org/10.1016/j.jallcom.2025.179174>.
- [287] C. Zhang, B. Gong, C. Deng, D. Wang, Computational prediction of mechanical properties of a C-Mn weld metal based on the microstructures and micromechanical properties, *Materials Science and Engineering: A* 685 (2017) 310–316. <https://doi.org/10.1016/j.msea.2017.01.023>.
- [288] A. Heidarzadeh, T. Saeid, Prediction of mechanical properties in friction stir welds of pure copper, *Materials & Design* (1980-2015) 52 (2013) 1077–1087. <https://doi.org/10.1016/j.matdes.2013.06.068>.
- [289] E.B. Farahani, A. Sarhadi, M. Alizadeh-Sh, S. Fæster, H.K. Danielsen, M.A.

- Eder, Thermomechanical modeling and experimental study of a multi-layer cast iron repair welding for weld-induced crack prediction, *Journal of Manufacturing Processes* 104 (2023) 443–459. <https://doi.org/10.1016/j.jmapro.2023.08.059>.
- [290] K.-M. Hong, Y.C. Shin, Analysis of microstructure and mechanical properties change in laser welding of Ti6Al4V with a multiphysics prediction model, *Journal of Materials Processing Technology* 237 (2016) 420–429. <https://doi.org/10.1016/j.jmatprotec.2016.06.034>.
- [291] Y. Meng, C. Shao, Physics-informed ensemble learning for online joint strength prediction in ultrasonic metal welding, *Mechanical Systems and Signal Processing* 181 (2022) 109473. <https://doi.org/10.1016/j.ymssp.2022.109473>.
- [292] M. Russell, J. Kershaw, Y. Xia, T. Lv, Y. Li, H. Ghassemi-Armaki, B.E. Carlson, P. Wang, Comparison and explanation of data-driven modeling for weld quality prediction in resistance spot welding, *J Intell Manuf* 35 (2024) 1305–1319. <https://doi.org/10.1007/s10845-023-02108-1>.
- [293] Z. Zhang, Y. Huang, R. Qin, W. Ren, G. Wen, XGBoost-based on-line prediction of seam tensile strength for Al-Li alloy in laser welding: Experiment study and modelling, *Journal of Manufacturing Processes* 64 (2021) 30–44. <https://doi.org/10.1016/j.jmapro.2020.12.004>.
- [294] S. Zhang, R. Cao, M.K. Ghalati, H. Dong, Y. Wei, Integrated machine-learning modelling for mechanical property prediction – A case study on laser-welded TC4 titanium alloy, *Optics & Laser Technology* 199 (2026) 115019. <https://doi.org/10.1016/j.optlastec.2026.115019>.
- [295] P. Taheri, S. Hsieh, M. Bahrami, Investigating electrical contact resistance losses in lithium-ion battery assemblies for hybrid and electric vehicles, *Journal of Power Sources* 196 (2011) 6525–6533. <https://doi.org/10.1016/j.jpowsour.2011.03.056>.
- [296] D. Zhao, Y. Bezgans, Y. Wang, W. Du, D. Lodkov, Performances of dimension reduction techniques for welding quality prediction based on the dynamic resistance signal, *Journal of Manufacturing Processes* 58 (2020) 335–343. <https://doi.org/10.1016/j.jmapro.2020.08.037>.
- [297] H. Zhang, H. Fu, X. He, C. Wang, L. Jiang, L.-Q. Chen, J. Xie, Dramatically Enhanced Combination of Ultimate Tensile Strength and Electric Conductivity of Alloys via Machine Learning Screening, *Acta Materialia* 200 (2020) 803–810. <https://doi.org/10.1016/j.actamat.2020.09.068>.
- [298] P. Stavropoulos, H. Bikas, K. Sabatakakis, C. Theoharatos, S. Grossi, Quality assurance of battery laser welding: A data-driven approach, *Procedia CIRP* 111 (2022) 784–789. <https://doi.org/10.1016/j.procir.2022.08.129>.
- [299] S. Wang, M. Verbrugge, J.S. Wang, P. Liu, Power prediction from a battery state estimator that incorporates diffusion resistance, *Journal of Power Sources* 214 (2012) 399–406. <https://doi.org/10.1016/j.jpowsour.2012.04.070>.
- [300] F. Yang, D. Wang, F. Xu, Z. Huang, K.-L. Tsui, Lifespan prediction of lithium-ion batteries based on various extracted features and gradient boosting

- regression tree model, *Journal of Power Sources* 476 (2020) 228654.
<https://doi.org/10.1016/j.jpowsour.2020.228654>.
- [301] W. Zhao, W. Ding, S. Zhang, Z. Zhang, Enhancing lithium-ion battery lifespan early prediction using a multi-branch vision transformer model, *Energy* 302 (2024) 131816. <https://doi.org/10.1016/j.energy.2024.131816>.
- [302] W. Zhao, W. Ding, S. Zhang, Z. Zhang, A deep learning approach incorporating attention mechanism and transfer learning for lithium-ion battery lifespan prediction, *Journal of Energy Storage* 75 (2024) 109647.
<https://doi.org/10.1016/j.est.2023.109647>.
- [303] G. Kang, H. Luo, Review on fatigue life prediction models of welded joint, *Acta Mech. Sin.* 36 (2020) 701–726. <https://doi.org/10.1007/s10409-020-00957-0>.
- [304] M.D. Chapetti, L.F. Jaureguizar, Fatigue behavior prediction of welded joints by using an integrated fracture mechanics approach, *International Journal of Fatigue* 43 (2012) 43–53. <https://doi.org/10.1016/j.ijfatigue.2012.02.004>.
- [305] J.M. Park, H.T. Kang, Prediction of fatigue life for spot welds using back-propagation neural networks, *Materials & Design* 28 (2007) 2577–2584.
<https://doi.org/10.1016/j.matdes.2006.10.014>.
- [306] M. Liu, X. Wang, D. Mi, Z. Liu, X. Long, S. Wen, C. Jiang, A new fatigue life prediction method for welded joints based on machine learning incorporating defect information and physics of failure, *International Journal of Fatigue* 202 (2026) 109234. <https://doi.org/10.1016/j.ijfatigue.2025.109234>.
- [307] J. Liu, B. Xu, Y. Feng, P. Chen, C. Yan, Z. Li, K. Yang, K. She, Y. Huang, Study on Porosity Defect Detection in Narrow Gap Laser Welding Based on Spectral Diagnosis, *Materials* 16 (2023). <https://doi.org/10.3390/ma16144989>.
- [308] D. Fan, C. Yu, L. Sha, H. Zhang, X. Liu, Failure Detection of Laser Welding Seam for Electric Automotive Brake Joints Based on Image Feature Extraction, *Machines* 13 (2025). <https://doi.org/10.3390/machines13070616>.
- [309] A. Raj, U. Chadha, A. Chadha, R.R. Mahadevan, B.R. Sai, D. Chaudhary, S.K. Selvaraj, R. Lokeshkumar, S. Das, B. Karthikeyan, R. Nagalakshmi, V. Chandramohan, H. Hadidi, Weld quality monitoring via machine learning-enabled approaches, *Int J Interact Des Manuf* (2023).
<https://doi.org/10.1007/s12008-022-01165-9>.
- [310] H. Deng, Y. Cheng, Y. Feng, J. Xiang, Industrial Laser Welding Defect Detection and Image Defect Recognition Based on Deep Learning Model Developed, *Symmetry* 13 (2021). <https://doi.org/10.3390/sym13091731>.
- [311] J. Feng, X. Zhao, J. Wang, Z. Liu, Y. Ding, Laser weld spot detection based on MMG-YOLO, *J Supercomput* 81 (2025) 1478. <https://doi.org/10.1007/s11227-025-07958-5>.
- [312] H. Gauttam, K.K. Pattanaik, S. Bhadauria, D. Saxena, Sapna, A cost aware topology formation scheme for latency sensitive applications in edge infrastructure-as-a-service paradigm, *Journal of Network and Computer Applications* 199 (2022) 103303. <https://doi.org/10.1016/j.jnca.2021.103303>.

- [313] Y. Lu, C. Liu, K.I.-K. Wang, H. Huang, X. Xu, Digital Twin-driven smart manufacturing: Connotation, reference model, applications and research issues, *Robotics and Computer-Integrated Manufacturing* 61 (2020) 101837. <https://doi.org/10.1016/j.rcim.2019.101837>.
- [314] Y.W. Park, H. Park, S. Rhee, M. Kang, Real time estimation of CO2 laser weld quality for automotive industry, *Optics & Laser Technology* 34 (2002) 135–142. [https://doi.org/10.1016/S0030-3992\(01\)00103-7](https://doi.org/10.1016/S0030-3992(01)00103-7).
- [315] L. Wang, S.-P. Zhu, B. Wu, Z. Xu, C. Luo, Q. Wang, Multi-fidelity physics-informed machine learning framework for fatigue life prediction of additive manufactured materials, *Computer Methods in Applied Mechanics and Engineering* 439 (2025) 117924. <https://doi.org/10.1016/j.cma.2025.117924>.
- [316] L. Wang, Z. Liu, A. Liu, F. Tao, Artificial intelligence in product lifecycle management, *Int J Adv Manuf Technol* 114 (2021) 771–796. <https://doi.org/10.1007/s00170-021-06882-1>.
- [317] W. Deng, K.T.P. Nguyen, C. Gogu, K. Mejaher, J. Morio, H. Le, D. Wu, A Novel PIML Architecture with Innovative Learning Paradigm Applied in Battery Prognostics, in: *2024 10th International Conference on Control, Decision and Information Technologies (CoDIT)*, 2024: pp. 248–253. <https://doi.org/10.1109/CoDIT62066.2024.10708278>.
- [318] A. Sudjianto, A. Zhang, Z. Yang, Y. Su, N. Zeng, PiML Toolbox for Interpretable Machine Learning Model Development and Diagnostics, (2023). <https://doi.org/10.48550/arXiv.2305.04214>.
- [319] Z.K. Lawal, H. Yassin, D.T.C. Lai, A.C. Idris, Physics-Informed Neural Network (PINN) Evolution and Beyond: A Systematic Literature Review and Bibliometric Analysis, *Big Data and Cognitive Computing* 6 (2022). <https://doi.org/10.3390/bdcc6040140>.
- [320] P.-H. Chiu, J.C. Wong, C. Ooi, M.H. Dao, Y.-S. Ong, CAN-PINN: A fast physics-informed neural network based on coupled-automatic-numerical differentiation method, *Computer Methods in Applied Mechanics and Engineering* 395 (2022) 114909. <https://doi.org/10.1016/j.cma.2022.114909>.
- [321] F. Tao, H. Zhan, A. Liu, A.Y.C. Nee, Digital Twin in Industry: State-of-the-Art, *IEEE Trans. Ind. Inform.* 15 (2019) 2405–2415. <https://doi.org/10.1109/TII.2018.2873186>.
- [322] O. Mata, P. Ponce, C. Perez, M. Ramirez, B. Anthony, B. Russel, P. Apte, B. Maccleery, A. Molina, Digital twin designs with generative AI: crafting a comprehensive framework for manufacturing systems, *J. Intell. Manuf.* (2025). <https://doi.org/10.1007/s10845-025-02583-8>.
- [323] M.B. Al Zami, S. Shaon, V.K. Quy, D.C. Nguyen, Digital Twin in Industries: A Comprehensive Survey, *IEEE Access* 13 (2025) 47291–47336. <https://doi.org/10.1109/ACCESS.2025.3551532>.
- [324] X. Sun, F. Zhang, J. Wang, Z. Yang, Z. Huang, R. Xue, Digital twin for smart manufacturing equipment: modeling and applications, *Int. J. Adv. Manuf. Technol.* 137 (2025) 4929–4946. <https://doi.org/10.1007/s00170-025-15468-0>.

- [325] S.-Y. Park, C. Lee, H. Kim, S.-H. Ahn, Enhancement of Control Performance for Degraded Robot Manipulators Using Digital Twin and Proximal Policy Optimization, *IEEE Access* 12 (2024) 19569–19583.
<https://doi.org/10.1109/ACCESS.2024.3359268>.
- [326] J.P. Allamaa, P. Patrinos, H. Van der Auweraer, T.D. Son, Learning-Based NMPC Adaptation for Autonomous Driving Using Parallelized Digital Twin, *IEEE Trans. Control Syst. Technol.* 33 (2025) 1317–1332.
<https://doi.org/10.1109/TCST.2024.3437163>.
- [327] B. Kim, S. Li, B. Taylor, Y. Chen, Efficient and Robust Edge AI: Software, Hardware, and the Co-design, *ACM Trans. Embed. Comput. Syst.* 24 (2025) 43. <https://doi.org/10.1145/3724396>.
- [328] T.L. Kaltenbach, J.C. Mosley, L.B. McNew, J.T. Beaver, Can edge AI mitigate environmental effects on camera trap performance?, *Wildl. Soc. Bull.* 49 (2025). <https://doi.org/10.1002/wsb.1598>.
- [329] S.S. Gill, M. Golec, J. Hu, M. Xu, J. Du, H. Wu, G.K. Walia, S.S. Murugesan, B. Ali, M. Kumar, K. Ye, P. Verma, S. Kumar, F. Cuadrado, S. Uhlig, Edge AI: A Taxonomy, Systematic Review and Future Directions, *Cluster Comput.* 28 (2024) 18. <https://doi.org/10.1007/s10586-024-04686-y>.
- [330] H.A. Gabbar, A.M. Othman, M.R. Abdussami, Review of Battery Management Systems (BMS) Development and Industrial Standards, *Technologies* 9 (2021). <https://doi.org/10.3390/technologies9020028>.
- [331] K.W.E. Cheng, B.P. Divakar, H. Wu, K. Ding, H.F. Ho, Battery-Management System (BMS) and SOC Development for Electrical Vehicles, *IEEE Transactions on Vehicular Technology* 60 (2011) 76–88.
<https://doi.org/10.1109/TVT.2010.2089647>.
- [332] M.T. Lawder, B. Suthar, P.W.C. Northrop, S. De, C.M. Hoff, O. Leitermann, M.L. Crow, S. Santhanagopalan, V.R. Subramanian, Battery Energy Storage System (BESS) and Battery Management System (BMS) for Grid-Scale Applications, *Proceedings of the IEEE* 102 (2014) 1014–1030.
<https://doi.org/10.1109/JPROC.2014.2317451>.